

PROJECT CONTROLS INSTITUTE GLOBAL

PCI Project Management Leader - AI

PCI PML-AI

Body of Knowledge & Official Study Text



Finance intelligently. Control predictively. Deliver successfully.

PCI AI Project Leadership Certification Suite

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Foreword

Every organisation that matters delivers its strategy through projects, and every project stands or falls on the quality of its leadership. The Project Controls Institute (PCI) created the PCI Project Management Leader – AI (PCI PML-AI) credential, and this Body of Knowledge that underpins it, for one purpose: to define what a competent project delivery leader must know and be able to do in an era in which artificial intelligence has become part of the everyday machinery of professional work.

The Institute's position on that machinery is deliberate and unambiguous: **AI proposes; the professional disposes**. AI can draft a schedule, surface a risk, forecast a cost outcome or summarise a contract — but every such output must be explainable, validated and owned by a competent human before it informs a decision. This volume therefore treats AI neither as a threat nor as a novelty, but as an instrument whose value depends entirely on the judgement, accountability and professionalism of the person wielding it.

This Body of Knowledge is the authoritative reference for the PCI PML-AI examination and, we intend, a working companion for delivery professionals long after the examination is passed.

— The Project Controls Institute

About this credential

The PCI Project Management Leader – AI (PCI PML-AI) is an **examination-based** professional credential: it is earned solely by demonstrating competence in the PCI PML-AI examination, which is built directly on this Body of Knowledge. It is awarded by the Project Controls Institute on its own authority as a professional certification body, and forms part of the PCI AI Project Leadership Certification Suite.

Its scope is comprehensive project management, leadership and delivery: initiation and business justification, governance, planning and execution, scope and stakeholder leadership, procurement and commercial delivery, integrated cost, schedule and risk management, predictive, agile and hybrid delivery, quality and change, sustainability, benefits realisation and closure — together with the governed, responsible use of AI in project management. It is intended for project managers, delivery leads, PMO professionals and programme practitioners who lead delivery, and for those preparing to step into such roles.

How to use this book

The Body of Knowledge comprises twenty-four chapters, and every chapter is examinable. They are best read as five movements:

- **Foundations and leadership** (Chapters 1–5): principles of project management, project leadership, initiation, business cases and governance — the footings on which everything else rests.
- **Managing the work** (Chapters 6–11): scope and requirements, planning and execution, stakeholder leadership and engagement, procurement, contracts and commercial delivery, and integrated cost, schedule and risk management.
- **Delivery approaches** (Chapters 12–14): predictive, agile and hybrid delivery — presented not as rival ideologies but as choices to be made deliberately, project by project.
- **Outcomes and stewardship** (Chapters 15–18): quality and change management, benefits realisation, sustainability in project delivery and project closure — because a project is judged by what it leaves behind.
- **The AI-enabled professional** (Chapters 19–24): AI-enabled project management, decision intelligence, digital delivery, automation, responsible AI and human accountability.

Candidates should read sequentially on a first pass, since later chapters assume earlier vocabulary, then revisit chapters selectively against their own experience. Practitioners may equally use the volume as a desk reference. Throughout, tools and technologies are described **generically, by capability**: nothing in this book endorses any vendor or product, and competence is defined in terms that outlast any particular software. British English spelling is used throughout.

Examination overview

The PCI PML-AI examination is a **competency-based, multiple-choice examination**. Items are drawn from across all twenty-four chapters of this Body of Knowledge and are predominantly scenario-based: rather than rewarding memorisation alone, they present realistic delivery situations and ask the candidate to apply principles, analyse trade-offs and select the most defensible professional course of action. Items span a range of cognitive levels, from recall of essential definitions through application and analysis.

Questions addressing AI-enabled practice are examined to the same doctrine that governs this book: candidates are expected to know when to rely on AI-generated outputs, how to validate and explain them, and where human accountability can never be delegated.

The examination is administered under controlled, invigilated conditions, and the pass standard is criterion-referenced — set against the competence the credential attests, not against the performance of other candidates. Practical parameters of the current examination form — including the number of questions, permitted duration, delivery arrangements and pass mark — are published by the Institute in the current examination specification and candidate handbook, which candidates should consult at registration, as these parameters may be revised between editions of this volume.

Read this book to pass the examination, by all means. But read it, above all, to deserve to.

1. Principles of Project Management

This chapter establishes the foundations of the discipline: what a project is, how it relates to programmes, portfolios and operations, how delivery life cycles are selected, where accountability sits and how the profession's standards fit together. It also introduces the credential's governing doctrine for artificial intelligence, because foundational decisions are where unexamined machine output does the greatest downstream damage.

Projects, programmes and portfolios

A **project** is a temporary endeavour undertaken to deliver a defined output — a product, service, facility or other result — against agreed objectives for time, cost, quality and benefit. Two properties are definitional. A project is temporary: it is formally authorised to start, is planned to finish, and its organisation is disbanded when its objectives are achieved or it is terminated. A project is unique: even where similar work has been done before, the specific combination of scope, stakeholders, constraints and context has not, so the work carries uncertainty that must be actively managed. From these flow the characteristic features of project working: transient teams, progressive elaboration, baseline-based control and a distinctive risk profile.

Business-as-usual (BAU) operations are the opposite case: ongoing, repetitive activity conducted through stable functional structures. The distinction matters because the governance regimes differ. Projects are justified by a business case, funded by staged authorisation and controlled against baselines through gates and tolerances; operations are funded through recurring budgets and controlled through performance indicators. Work crosses the boundary at handover, when outputs enter operational use and benefits begin to accrue — a transition examined in the benefits realisation chapter.

A **programme** is a coordinated group of related projects, together with associated business-change activity, managed as a unity to deliver outcomes and benefits that could not be obtained by managing the components independently. Programmes are outcome-led where projects are output-led. A **portfolio** is the totality — or a defined segment — of an organisation's investment in change: projects, programmes and related work selected, prioritised and balanced against strategic objectives, constrained resources and risk appetite. The three levels answer different questions: the portfolio asks are we doing the right things; the programme, are we combining them to achieve the intended outcomes; the project, are we delivering this thing well. The delivery leader must know which level a given decision belongs to: escalating it to the wrong level is itself a governance failure.

Classifying the work before managing it

Classification determines the governance that follows — who owns the business case, where tolerances are set, to whom the delivery leader escalates — so it must be settled before any plan is drafted. Apply this test in order:

1. **Is there a unique output and a defined end point?** If yes, it is a candidate project; if the work is repetitive and continuous, it is business-as-usual and belongs with operational management.
2. **Are the intended benefits achievable only by coordinating several related projects and change activities?** If yes, it is a programme, and benefits ownership sits above any single project.
3. **Is the question one of selecting and balancing investments against strategy and constrained resources?** If yes, the decision belongs at portfolio level.
4. **Record the classification and its consequence** for governance and escalation.

Project life cycles and delivery contexts

A **life cycle** is the framework of phases through which a project passes from concept to closure and, in its extended form, onward through adoption and benefits realisation. The recognised archetypes are: the **linear** (sequential or predictive) life cycle, in which scope is defined up front and phases proceed in order, suited to stable requirements and high integration cost; the **iterative** life cycle, which revisits and refines the solution as knowledge grows; the **incremental** life cycle, which delivers usable output in successive slices; the **evolutionary or adaptive (agile)** life cycle, which fixes time and cost in short cycles and flexes scope by prioritisation; and **hybrid** arrangements combining these within one project. No archetype is inherently superior. Selection is a professional judgement governed by the delivery context — stability of requirements, novelty of the technology, safety and regulatory criticality, contractual environment and the organisation's capacity to absorb change. Whatever the archetype, phase-gate governance can be applied over it: gates are decision points owned by the sponsoring organisation, not features of any single life cycle.

A structured method for selecting a delivery life cycle

Life cycle selection is a justified decision, not a preference. The examination expects candidates to perform and defend it as follows:

1. **Characterise the delivery context.** Assess requirements stability, regulatory or contractual constraint, urgency of benefit, stakeholder availability, team and supplier capability, and technical interdependency.
2. **Convert the context into weighted criteria** that sum to 1.00, agreed with the sponsor before scoring, so the weights cannot be reverse-engineered to fit a preferred answer.

3. **Score each candidate life cycle** — typically predictive, agile and hybrid — from 1 (poor fit) to 5 (strong fit) against each criterion, with a one-line rationale per score.
4. **Compute the weighted score** for each candidate: multiply each score by its weight and sum.
5. **Sense-check against absolute constraints.** A statutory deadline or a safety-critical approval regime is a constraint, not a criterion; it can veto an otherwise winning option.
6. **Record the selection and its justification** in the charter or delivery approach document, obtain sponsor endorsement, and revisit at each stage gate.

An AI assistant may draft the criteria, propose weights from comparable projects or pre-populate scores from tender and requirements documents; the delivery leader must be able to explain every figure, correct the machine's inferences, and sign the result.

Worked example: life cycle selection for a billing-platform replacement

Setup. An organisation must replace its customer-billing platform: budget USD 3,600,000, duration 14 months, with a statutory cut-over date that cannot move. Core regulatory requirements are fixed; user-experience requirements are expected to evolve. The supplier has moderate experience of iterative delivery.

Criterion	Weight	Predictive	Agile	Hybrid
Accommodates evolving requirements	0.25	2	4	4
Fits fixed regulatory scope and date	0.20	5	2	4
Delivers benefit early	0.25	1	5	4
Matches team and supplier capability	0.15	4	2	3
Handles technical interdependency	0.15	4	2	4

Substitution and result (weighted score = Σ weight $\times$ score):

- Predictive: $(0.25 \times 2) + (0.20 \times 5) + (0.25 \times 1) + (0.15 \times 4) + (0.15 \times 4) = 0.50 + 1.00 + 0.25 + 0.60 + 0.60 = \mathbf{2.95}$
- Agile: $(0.25 \times 4) + (0.20 \times 2) + (0.25 \times 5) + (0.15 \times 2) + (0.15 \times 2) = 1.00 + 0.40 + 1.25 + 0.30 + 0.30 = \mathbf{3.25}$
- Hybrid: $(0.25 \times 4) + (0.20 \times 4) + (0.25 \times 4) + (0.15 \times 3) + (0.15 \times 4) = 1.00 + 0.80 + 1.00 + 0.45 + 0.60 = \mathbf{3.85}$

Interpretation. The hybrid life cycle scores highest (3.85 against 3.25 and 2.95): iterative build of the evolving user-facing scope inside a predictive governance

wrapper anchored to the statutory cut-over. The sense-check confirms the choice — the fixed date is an absolute constraint that a purely agile approach, despite its strong early-benefit score, could not guarantee. The leader records this reasoning; the score justifies the decision, it does not replace it.

Roles and responsibilities of the project delivery leader

The Institute uses **project delivery leader** as the umbrella term for the accountable managerial role classically titled project manager. The delivery leader is accountable to the **sponsor** for day-to-day leadership of the project: shaping the definition, planning and integrating scope, schedule, cost, risk, quality and resources into a single coherent baseline, building and directing the team, managing stakeholders, controlling change, reporting performance honestly and closing the project in an orderly way. The sponsor, by contrast, owns the business case, secures funding, chairs governance and remains accountable for benefits; the team owns the technical work; the wider organisation supplies assurance, functional resources and the portfolio context. The delivery leader's distinct contribution is integration: no other role is positioned to reconcile these parts into one deliverable whole, and the responsibility cannot be delegated — to a subordinate, a contractor or a tool.

In practice, responsibilities are made visible stage by stage: list the key decisions and management products, assign responsibility using a RACI convention, and publish the matrix with the stage plan so the organisation can see who decides what. Verify two boundaries every time: the sponsor remains accountable for the business case and benefits; the delivery leader for delivery within agreed tolerances, escalating promptly when a forecast breaches them.

Standards, methods and the project profession

Candidates must be able to situate and compare the profession's published frameworks: the ISO 21500 family, including ISO 21502 on project management and its companions for programme and portfolio management; PMI's PMBOK Guide; the APM Body of Knowledge; PRINCE2 as a structured method; and the Scrum, Kanban and Lean traditions within agile delivery, with ISO 31000 informing risk principles. A standard codifies agreed good practice; a method prescribes a repeatable way of applying it; a framework leaves the practitioner to tailor. The PCI PML-AI credential is examination-based and framework-neutral: it assesses the candidate's ability to apply these sources with judgement, not allegiance to any one of them.

AI and the foundations of the discipline

Artificial intelligence now reaches into the most conceptual layer of the discipline — the layer this chapter defines. Predictive tooling can mine delivery history for reference-class comparators, flag the contextual characteristics that bear on life cycle

selection and score candidate investments for portfolio prioritisation; generative tooling can assemble first drafts of charters, business cases and stage-gate documentation in minutes rather than days.

Yet the judgements at the heart of this chapter are definitional acts that set the terms on which every later analytic operates. Whether a piece of work is a project, a programme tranche or business-as-usual determines how it is governed, funded and measured; which life cycle is adopted determines what "on track" even means. If an AI-assisted classification is wrong, or a predictive life cycle is recommended for genuinely emergent scope, every downstream forecast, dashboard and early-warning signal inherits the error with mechanical confidence. Foundational decisions are therefore the last place in the discipline where AI output may pass unexamined.

The competent leader accordingly automates the clerical substrate of initiation and governance — drafting of charters and responsibility matrices, retrieval and summarisation of lessons learned, consistency checks between business case, plan and governance documentation, assembly of portfolio reporting — tasks where AI compresses effort and errors are visible and cheap to correct. The leader validates, personally and demonstrably, anything that shapes accountability or approach: the classification of the endeavour, the recommended life cycle and its reasoning, the allocation of roles, and any AI-generated content placed before a sponsor, board or gate review. Validation is not a glance: it means the leader can reconstruct the recommendation's logic, has tested it against context no model can see — organisational politics, contractual posture, team capability, stakeholder appetite — and will defend the decision as their own.

Three caveats temper enthusiasm. Historic project data is sparse, inconsistently coded and survivor-biased, so models trained on it reproduce an organisation's past habits — including its optimism bias — rather than good practice. Generative tools occasionally fabricate plausible-sounding standards clauses or governance requirements, so every citation must be checked at source. And commercially sensitive material must be processed only within tooling approved under the organisation's data-governance arrangements.

None of this dilutes accountability. Boards hold the sponsor accountable for the business case and the delivery leader for the delivery approach — not the tooling either of them used. "The model recommended it" is never a defence, at a gate review or in this Institute's examinations. Candidates must be able to state, for any AI-assisted foundational decision, who owns it, how it was validated and how it would be explained to a governance authority.

AI proposes; the professional disposes. AI may draft the charter, rank the portfolio and recommend the life cycle; only a competent, accountable professional may decide what the endeavour is, how it will be governed and who answers for it. Every AI output used in founding a project must be explainable,

validated against the real delivery context and owned by a named human — because governance can delegate work to a machine, but never accountability.

Common pitfalls

- **Relabelling operations as projects**, burdening routine work with project governance while genuine projects are starved of it.
- **Ideological life cycle choice** — defaulting to agile or predictive by habit rather than deriving the choice from the delivery context.
- **Letting the matrix override an absolute constraint**, such as a statutory date or mandated approval regime.
- **Setting weights after scoring**, or accepting AI-generated weights and scores unexamined, to rationalise a predetermined answer.
- **Running a programme with project-level tools**, losing sight of cross-project dependencies and benefits.
- **Blurring the sponsor-leader boundary**, with the delivery leader effectively owning the business case.

Key points

- A project is a temporary, unique endeavour delivering a defined output; these properties drive its governance by business case, baselines, gates and tolerances, and distinguish it from business-as-usual operations.
- Portfolios select the right investments, programmes combine projects and business change to achieve outcomes, projects deliver outputs well; every significant decision belongs to exactly one of these levels.
- Life cycle archetypes — linear, iterative, incremental, evolutionary (agile) and hybrid — must be selected from the delivery context through justified, sponsor-endorsed reasoning, revisited at stage gates.
- A weighted-criteria matrix disciplines the selection, but absolute constraints such as statutory dates sit outside the scoring and can veto the highest-scoring option.
- The delivery leader's defining contribution is integration; the sponsor owns the business case and benefits; neither accountability can be delegated to a subordinate, a contractor or a tool.
- The profession's standards, methods and frameworks — the ISO 21500 family, the PMBOK Guide, the APM Body of Knowledge, PRINCE2, the agile traditions — are sources to apply with judgement; the credential is framework-neutral and examination-based.

- AI proposes; the professional disposes: AI may draft, classify, score and recommend, but every output used must be explainable, validated against the real context and owned by a named, competent human.

Practice questions

1. An organisation plans a new contact-centre system, revised processes, a retrained workforce and a rebranding initiative; the intended benefits are achievable only if these are coordinated with one another and with sustained business change. How should the endeavour be classified?

A. A single large project with multiple work packages B. A programme C. A portfolio D. Business-as-usual operations

Correct answer: B. Benefits obtainable only by coordinating several related projects together with business-change activity are the defining characteristic of a programme.

2. A delivery leader's weighted-criteria matrix ranks a fully agile life cycle highest, but the project has a statutory compliance date that cannot move. What should the leader do?

A. Adopt the agile life cycle, because the matrix score is the objective result B. Increase the weight of the regulatory criterion until agile no longer wins, then adopt the new winner C. Treat the statutory date as an absolute constraint that sense-checks the result, select an option that can guarantee the date, and record the justification D. Escalate the decision to portfolio level because the matrix and the constraint conflict

Correct answer: C. An absolute constraint is not a weighted criterion but a veto applied after scoring — and adjusting weights to force a preferred outcome is itself a recognised pitfall.

3. An AI assistant analyses an organisation's delivery history and recommends a predictive life cycle for a new project. Before acting on the recommendation, the delivery leader must:

A. Adopt it, because the model has analysed more projects than any individual could B. Reconstruct and test its reasoning against the real delivery context, adjust it where the inference is wrong, and take personal ownership of the decision C. Ask the sponsor to approve the recommendation so that accountability transfers to the governance board D. Obtain a second recommendation from a different model and adopt the majority view

Correct answer: B. Every AI output shaping the delivery approach must be explainable, validated and owned by the accountable human; accountability can be neither delegated to a tool nor transferred by approval.

2. Project Leadership

2.1 Leadership in the delivery context: core definitions

Leadership, as this Body of Knowledge uses the term, is the practice of setting direction, aligning people and securing willing commitment to project objectives. It is distinct from management, which is the exercise of positional authority to plan, organise and control work within an established structure. The project delivery leader must do both, but the distinction matters because the two rest on different foundations: management rests on **authority** — the legitimate right to direct, conferred by the project mandate, charter or contract — while leadership rests on **power**, the broader capacity to shape behaviour. Recognised analyses of power (notably French and Raven's bases: legitimate, reward, coercive, expert, referent and informational) show that formal authority is only one such base, and often the weakest one available. Most of the people on whom delivery depends — matrix-assigned specialists, supplier staff, business stakeholders, sponsors — do not report to the delivery leader at all. Influence, not instruction, is therefore the operative currency of project leadership, and that premise underpins this entire chapter.

2.2 The governing frameworks and standards context

Project leadership is not folk wisdom; it is a defined professional competence within the recognised bodies of guidance. ISO 21502 situates the project manager's people responsibilities within a governance structure of sponsor, board and team. The IPMA Individual Competence Baseline treats "people" competences — leadership, teamwork, conflict and crisis, negotiation, communication — as one of its three competence dimensions, alongside perspective and practice. The PMBOK's principles likewise place leadership behaviour, team performance and stakeholder engagement at the centre of delivery, and the Scrum Guide's self-managing team implies a facilitative, servant-style stance for adaptive work. Codes of professional conduct — including the PCI Code of Professional Conduct, which binds every credential holder — convert leadership values into enforceable obligations. The PCI PML-AI examination draws on these published sources: candidates are assessed on their ability to apply recognised models to delivery situations and to justify the choice, not to recite taxonomies.

2.3 Leadership theories and styles

Leadership theory has progressed from **trait** approaches (leaders possess distinguishing characteristics), through **behavioural** approaches (leadership is observable, learnable conduct, typically mapped on task-focus and people-focus axes), to **contingency and situational** approaches, which hold that effectiveness depends on matching behaviour to circumstance. The situational model associated with Hersey and Blanchard matches four styles — directing, coaching, supporting and delegating — to the competence and commitment of those being led. The Tannenbaum-Schmidt

continuum describes the same terrain as a graduated transfer of decision-making freedom from leader to team. Complementing these, the transactional-transformational distinction contrasts leadership through exchange and monitoring with leadership through vision, stimulation and individual consideration; servant leadership inverts the hierarchy so that the leader's role is to remove impediments; and the literature on emotional intelligence links a repertoire of styles to self-awareness and self-regulation. The doctrinal point the candidate must hold is that **no style is universally superior**: the examinable skill is diagnosis — reading team maturity, task criticality, risk exposure and delivery approach — and then selecting, and justifying, the style that fits.

2.4 Teams, development and motivation

A project team is a group with complementary skills, a common purpose and mutual accountability for delivery outcomes — assembled temporarily, often part-time and cross-organisational, which is precisely why deliberate team development matters. Tuckman's sequence — forming, storming, norming, performing, adjourning — models how groups mature and warns that changes in composition or mandate can return a team to earlier stages. Role-balance models such as Belbin's explain performance in terms of complementary behavioural roles rather than technical skills alone. Motivation theory divides into **content** theories, which identify what people need (Maslow's hierarchy; Herzberg's separation of hygiene factors from motivators; McClelland's achievement, affiliation and power needs), and **process** theories, which explain how effort decisions are made (expectancy and equity reasoning), augmented by findings on intrinsic motivation — autonomy, mastery and purpose. Within this syllabus these are diagnostic instruments: a performance issue is located within a development stage or a motivational deficit, and the intervention follows from the diagnosis.

2.5 Communication, influence, conflict and the ethical boundary

Communication is modelled as sender, encoding, channel, decoding and feedback, degraded by noise — a model that explains most project misunderstandings. Conflict is a divergence of interests or views; it can be functional (surfacing risk, testing options) or dysfunctional, and the Thomas-Kilmann framework maps five handling modes — competing, collaborating, compromising, avoiding and accommodating — against axes of assertiveness and cooperativeness. Principled negotiation directs attention to interests rather than positions and to objective criteria rather than contests of will. Ethics is the boundary condition on all of this: influence must never shade into misrepresentation, and codes of conduct make integrity, responsibility, respect and fairness enforceable duties. The same boundary governs AI-enabled tooling, examined in depth in section 2.7.

These elements form one operating loop: the leader diagnoses the team's state using development and motivation models; selects and justifies a style; enacts it through

communication, influence and conflict-resolution technique — above all where no formal authority exists; and keeps every action, including every use of AI, within the discipline's ethical frame.

2.6 Applying leadership methods in practice

Leadership in a delivery setting is examinable as method, not personality. The practitioner follows a repeatable cycle: gather evidence about the team and its stakeholders, diagnose against a recognised model, select an intervention that matches the diagnosis, and verify the result against measurable criteria. The three methods below apply that cycle to the chapter's core practice areas.

Method 1 — Selecting a leadership style

1. **Establish the team's development stage** using observable evidence — delivery trends, conflict frequency, decision latency — mapped to Tuckman's stages. Impressions are not evidence; two independent indicators are the minimum.
2. **Assess task-specific readiness** of individuals using the situational dimensions of competence and commitment. Readiness is task-specific: a senior engineer may be at the highest development level on design work and at the lowest on regulatory documentation.
3. **Match the style**: directing for low competence with high commitment; coaching for some competence but low commitment; supporting for high competence with variable commitment; delegating for high competence with high commitment.
4. **Record the justification** in the delivery log and re-assess at defined trigger events: membership change, phase transition, or a sustained performance shift. Style is a decision to be revisited, not a fixed trait.

Method 2 — Diagnosing team performance: worked example

Scenario. A nine-person hybrid team on a claims-platform programme. Two engineers were replaced at the start of sprint 12; performance has since deteriorated.

Step 1 — quantify the delivery signal. Velocity for sprints 12-14 was 46, 38 and 31 story points against a previously sustained 45.

- Mean velocity = $(46 + 38 + 31) \div 3 = 115 \div 3 = \mathbf{38.3 \text{ points}}$, an average shortfall of $(45 - 38.3) \div 45 = \mathbf{14.9\%}$ against plan.
- Peak-to-latest decline = $(46 - 31) \div 46 = 15 \div 46 = \mathbf{32.6\%}$.
- Defect escapes rose from 8 to 14 per sprint = $(14 - 8) \div 8 = \mathbf{+75\%}$.

Step 2 — run a structured team health pulse. All nine members score five dimensions on a 1-5 scale. Mean results: goal clarity 4.1, role clarity 2.3, trust 2.6, process fit 3.8, recognition 3.9.

- Composite = $(4.1 + 2.3 + 2.6 + 3.8 + 3.9) \div 5 = 16.7 \div 5 = \mathbf{3.34}$, below the 3.5 action threshold, with two dimensions below the 3.0 red line.

Step 3 — diagnose against the models. The pattern — declining output, rising defects, low role clarity and trust, intact goal clarity and process fit — indicates **regression from performing to storming**, triggered by the composition change. In Herzberg's terms, the deficit lies in hygiene factors such as role definition and interpersonal relations, not in motivators such as the work itself; adding incentives would treat the wrong variable.

Step 4 — intervene to match the diagnosis. Shift from a delegating to a coaching style for the affected workstreams; run a role-clarification workshop producing a refreshed responsibility assignment; pair the two new engineers with established members; reinstate explicit recognition of completed work at sprint reviews.

Step 5 — set exit criteria and verify. Criteria: velocity of at least 40 points within three sprints; role clarity and trust of at least 3.5 at the next pulse. Actual sprints 15-17: 35, 41, 44 points; pulse results: role clarity 3.7, trust 3.6. The intervention is confirmed effective and the leader steps back towards a supporting, then delegating, style.

Method 3 — Influencing and resolving conflict without authority

1. **Map commitment:** for each stakeholder, record current versus required commitment to the outcome and prioritise the largest gaps held by the most influential parties.
2. **Identify interests and currencies:** what each party values — risk reduction, resource relief, visibility — and what the delivery leader can legitimately exchange.
3. **Select the conflict mode deliberately** (Thomas-Kilmann): collaborate where stakes and relationships both matter; compromise under genuine time pressure; accommodate on trivial issues; compete only where a non-negotiable standard is at risk.
4. **Frame the proposal in the other party's terms**, confirm agreements in writing, and escalate through governance only after direct, interest-based resolution has demonstrably failed.

2.7 AI and the leadership competency

Of all the competencies in this Body of Knowledge, leadership is the one AI changes least in substance and most in context. Predictive and generative tooling now sits inside the everyday fabric of team leadership: meeting transcription and

summarisation, drafting of status narratives and stakeholder communications, collation of survey results, sentiment signals from collaboration platforms, and workload analytics that flag overload or attrition risk before they surface in behaviour. Used well, these tools extend a delivery leader's attention — they widen what the leader can notice across a large or distributed team. What they cannot extend is the leader's judgement: the choice of style, the diagnosis of underperformance and the resolution of conflict remain human acts, informed by models of the kind examined in this chapter, not delegated to a model of the statistical kind.

The practical discipline is a clean division of labour. A delivery leader may legitimately **automate** the mechanical layer of leadership work: minutes and action capture, first drafts of routine communications, consolidation of feedback and survey data, scheduling, and the assembly of team-health dashboards. The leader must personally **validate** — and, where necessary, override — anything that carries an inference about a person or a decision that shapes one: any AI-generated reading of individual performance, sentiment or "flight risk"; any summary attributing positions to named people; any drafted message on a sensitive matter. Difficult conversations, feedback, recognition and conflict resolution are delivered by the leader in person, whatever tooling helped prepare them. A team that discovers its leader's empathy was machine-drafted loses trust at precisely the moment trust is the leader's only instrument.

Leadership analytics carry model risk of a distinctive kind, because the data is people. Sentiment models trained on populations that do not match a multicultural, multilingual team may score the same message very differently across members — a direct bias risk. Proxy metrics mislead: message volume is not engagement, and quietness is not disengagement. Generative summaries can misattribute or invent statements, which in a personnel context is not a nuisance but a potential grievance. Collaboration-metadata analytics can shade into surveillance, chilling the candour that psychological safety depends on, and personal data fed to external tools may breach data-protection law and the confidentiality obligations of the PCI Code of Professional Conduct. Pulse and survey data is personal data: obtain consent, report only aggregates, and never disaggregate groups small enough to identify individuals. The delivery leader is expected to know what data a tool ingests, whether the team has been told, and whether consent and lawful basis exist — before the first dashboard is opened.

Accountability for people decisions is not transferable to a tool, a vendor or an algorithm. Where an AI signal contributes to a decision affecting an individual — workload, role, development, dispute — the leader must be able to explain the signal, evidence its validation, and own the outcome as their own professional judgement. In the examination, candidates are assessed on exactly this discrimination: identifying which leadership tasks may be safely automated, which outputs demand validation, and where acting on an unvalidated inference about a person would constitute a failure of professional conduct.

AI proposes; the professional disposes. An algorithm may surface the signal — a sentiment dip, a workload spike, a drafted difficult message. Only a competent, accountable delivery leader may decide what that signal means, whether the data behind it is fair and lawfully held, and how to act on it with the people concerned. Leadership judgement, and the accountability that goes with it, is never delegated to a model.

2.8 Common pitfalls

- **Diagnosing from a single data point.** One bad sprint is noise; a triangulated trend is a signal.
- **Choosing a style by personal preference** rather than team maturity, then blaming the team for the mismatch.
- **Turning measures into targets.** A velocity target invites point inflation and destroys the diagnostic value of the metric.
- **Skipping re-contracting after membership change** — the most common cause of storming regression.
- **Defaulting to forcing or avoiding** in every conflict instead of selecting the mode consciously; and **escalating first**, which spends political capital and signals inability to influence.
- **Treating hygiene factors as motivators.** Pay and conditions remove grievances; they do not create engagement.
- **Accepting AI-generated team diagnoses unvalidated**, or using sentiment analytics as covert surveillance — both breach professional conduct expectations.

Key points

- Leadership secures willing commitment through influence; management directs through positional authority — and because most contributors do not report to the delivery leader, influence is the operative currency.
- No leadership style is universally superior: the examinable skill is diagnosing team maturity and task-specific readiness, then selecting and justifying the style that fits.
- Team performance is diagnosed with recognised models — Tuckman for development stage, Belbin for role balance, content and process theories for motivation — and interventions must match the diagnosis.
- Practice follows a verifiable cycle: gather triangulated evidence, diagnose against a model, intervene, and confirm the result against pre-set exit criteria.

- Conflict modes are chosen deliberately: interests before positions, collaboration where stakes and relationships matter, escalation only after direct resolution has failed.
- AI may automate the mechanical layer of leadership work, but every inference about a person must be validated by the accountable leader: AI proposes; the professional disposes.
- People analytics are personal data: consent, lawful basis, aggregate-only reporting and freedom from covert surveillance are professional-conduct obligations, not options.

Practice questions

1. A delivery leader inherits a workstream whose members are highly competent but whose commitment has become inconsistent following a reorganisation. Under the situational leadership model, which style is most appropriate?

A. Directing B. Coaching C. Supporting D. Delegating

Correct answer: C. High competence with variable commitment calls for a supporting style — facilitation and encouragement rather than close task direction, which the team's competence does not require.

2. Three sprints after two new engineers join, velocity has fallen and a team pulse shows low role clarity and trust but high goal clarity. The delivery leader's best first intervention is to:

A. Introduce a performance bonus linked to velocity recovery B. Run a role-clarification workshop and pair new members with established ones C. Escalate the underperformance to the project board D. Replace the two new engineers

Correct answer: B. The evidence indicates storming regression driven by hygiene-factor deficits in role definition and relationships, so the intervention must rebuild clarity and trust; incentives, escalation or removal treat the wrong variable.

3. An AI workload-analytics tool flags a named team member as a "high attrition risk". What must the delivery leader do before acting on this signal?

A. Forward the report to human resources for action B. Validate the inference against direct observation and conversation, confirm the data was fairly and lawfully processed, and own any resulting decision C. Act immediately, since early intervention is the purpose of predictive tooling D. Ignore the output, since AI has no legitimate role in people leadership

Correct answer: B. Under the doctrine "AI proposes; the professional disposes", an inference about an individual must be explainable, validated and owned by the accountable leader before any action is taken.

3. Project Initiation

A project is a temporary endeavour undertaken to deliver defined outputs that enable an organisation to achieve outcomes and realise benefits. Because every project consumes scarce resources and carries risk, no project should exist by accident: a governing authority must have decided that this investment, at this time, serves strategy better than the alternatives. Project initiation is the disciplined process by which that decision is prepared, evidenced and formally made. This chapter sets out the concepts that govern initiation, the method by which the professional executes it, a worked example, and the proper uses of AI-enabled tooling at this stage.

3.1 From strategy to mandate: the governed start of a project

The starting instrument is the **project mandate** — the instruction, issued by an authority above the project (an executive, a portfolio board or a programme), that commissions initiation. A mandate is permission to prepare, not permission to deliver: it names the strategic driver, the intended contribution, the sponsor and the boundaries within which initiation effort may be spent. The **sponsor** is the accountable owner of the investment — the individual who owns the business case, secures funding and will ultimately answer for benefits. The **business case** is the documented justification: the options considered, the expected benefits, the estimated costs and timescales, and the principal risks. Initiation converts the mandate's intention into evidence on which the sponsor can authorise, redirect or decline the project.

The principal frameworks of professional practice converge on this architecture even where their vocabulary differs. ISO 21502 situates initiation within organisational governance, requiring a defined authority to approve the start of a project; the PMBOK Guide treats development of the project charter as the integrating act that links strategy to delivery; PRINCE2 separates pre-project start-up from initiation proper and insists that the business case remain continually justified; the APM Body of Knowledge places this work in the concept and definition phases of the lifecycle. The PML-AI examination privileges no single framework: it examines the underlying logic — traceability from strategy to mandate to authorised project — that is common to all of them.

These elements interlock as a single golden thread: strategy explains why, the mandate commissions, the business case argues whether, the charter authorises what and who, and the stakeholder and risk appraisals condition on what terms. Any element that cannot be traced along this thread is a candidate for challenge before the sponsor signs anything.

3.2 Objectives, success criteria, constraints and assumptions

An **objective** is a statement of what the project must achieve, expressed so that achievement can be verified: a measure, a baseline value, a target value and a date. **Success criteria** are the agreed tests by which the project will be judged successful; they operationalise the objectives and must be fixed at initiation, because criteria invented after the fact can justify anything. The professional distinguishes **outputs** (the deliverables the project produces), **outcomes** (the changed state those outputs enable) and **benefits** (the measurable improvement a stakeholder values), and traces every objective through this chain back to the strategy named in the mandate.

Constraints are externally imposed limits — on time, cost, scope, quality, resources or regulatory compliance — within which the project must be delivered. **Assumptions** are conditions taken as true without proof for planning purposes; each is a latent risk and must be recorded as such. Because the classic constraints trade off against one another, initiation should also record the sponsor's priority among them: knowing whether the deadline or the budget yields first is a decision to be made deliberately, not discovered in a crisis.

3.3 The project charter and initiation documentation

The **project charter** is the instrument of authorisation: the document by which the sponsor formally brings the project into existence, appoints the project manager and delegates defined authority. A competent charter records the purpose and strategic alignment; the objectives and success criteria; the scope boundaries, stated as explicit inclusions and exclusions; the constraints and assumptions; a milestone-level view of the timeline; the initial funding released and the gate at which further funding will be considered; and the project manager's authority levels and tolerances — the thresholds of cost, time or risk beyond which escalation to the sponsor is mandatory.

The charter is deliberately concise: it authorises and bounds, while the detailed management plans that follow describe how delivery will be conducted. Where an organisation uses a project initiation document, that fuller artefact assembles the charter's authorisation together with the first integrated plans for approval at the close of initiation.

3.4 Early stakeholder identification and initial risk appraisal

A **stakeholder** is any individual, group or organisation that can affect, or be affected by, the project. Initiation requires a first systematic identification — recorded in a stakeholder register with each party's interest, influence and attitude — because unidentified stakeholders resurface later as opposition, rework or missed requirements. In parallel, an **initial risk appraisal** identifies the principal threats and opportunities at the level of definition then available, tests them against the sponsor's risk appetite (ISO 31000 supplies the general vocabulary), and informs both the

contingency held within the initial funding and the credibility of the business case. Neither exercise is the full analysis developed in later chapters; each exists so that the decision to proceed is taken with eyes open.

3.5 The initiation method, step by step

Initiation is examinable as a method, not a mood. The candidate should be able to execute — and reproduce under examination conditions — the following sequence.

1. **Confirm the mandate.** Obtain the mandate in writing and identify the specific strategic objective it serves. If the link cannot be stated in one sentence — "this project advances objective X by achieving Y" — return the mandate to its originator before committing effort.
2. **Quantify the intended contribution.** Convert strategic intent into measurable objectives, each with a baseline value, a target value, a target date and a named measurement owner.
3. **Set success criteria, constraints and assumptions.** Separate acceptance criteria (the deliverable works) from business success criteria (the benefit arrives). Record constraints — funding ceilings, fixed dates, regulatory obligations — and log every assumption that, if false, would invalidate the case.
4. **Draw the scope boundary.** Write explicit in-scope and out-of-scope lists; a candid out-of-scope list prevents more later disputes than the in-scope list ever will.
5. **Estimate and fund.** Produce an order-of-magnitude estimate with a stated accuracy range, never a bare point figure, and distinguish the indicative whole-life envelope from the funding actually released for the next stage.
6. **Identify stakeholders.** Scan systematically — funders, users, operators, regulators, suppliers, affected third parties — and grade each by interest and influence, noting who can stop the project and who must change how they work.
7. **Appraise initial risks.** Identify the principal risks, assign each a probability and a monetary impact, and compute expected monetary value ($EMV = \text{probability} \times \text{impact}$) as a first-cut contingency signal.
8. **Assemble the charter and secure approval.** Consolidate steps 1-7 into the charter, add authority levels and tolerances, name the project manager, and present the document for the sponsor's recorded decision to proceed, amend or stop.

3.6 Worked example: from mandate to a fundable charter

Northbridge Insurance's strategy commits it to reducing the cost of claims operations. The mandate instructs a project manager to initiate a claims-handling modernisation. She works the method, showing her arithmetic at each step.

Step 2 — quantify the contribution. The baseline: 380,000 claims a year at USD 42 processing cost per claim, so the annual cost is $380,000 \times 42 = \text{USD } 15,960,000$. The strategic target is USD 31 per claim within 18 months of go-live. The projected annual saving is $380,000 \times (42 - 31) = 380,000 \times 11 = \text{USD } 4,180,000$. The objective is now measurable: "reduce unit processing cost from USD 42 to USD 31 by the stated date, measured quarterly by the finance director."

Step 5 — estimate and funding. The order-of-magnitude estimate is USD 5,200,000 with an initiation-stage accuracy range of -25% / $+75\%$: lower bound $5,200,000 \times 0.75 = \text{USD } 3,900,000$; upper bound $5,200,000 \times 1.75 = \text{USD } 9,100,000$. Simple payback at the point estimate is $5,200,000 \div 4,180,000 = 1.24$ years; even at the upper bound it is $9,100,000 \div 4,180,000 = 2.18$ years. Both sit inside the sponsor's three-year payback threshold, so the case survives a pessimistic cost outcome — a far stronger statement than a single-figure payback.

Step 7 — initial risk appraisal. Three principal risks are identified and costed:

Risk	Probability	Impact (USD)	EMV (USD)
Legacy data migration fails quality checks	0.30	1,500,000	450,000
Regulatory approval delays go-live	0.20	800,000	160,000
Key supplier under-resources the build	0.25	600,000	150,000
Total			760,000

The EMV total is $450,000 + 160,000 + 150,000 = \text{USD } 760,000$, which is $760,000 \div 5,200,000 = 14.6\%$ of the point estimate — a defensible first-cut contingency provision for this stage of maturity.

Step 8 — the charter. The indicative funding envelope is $5,200,000 + 760,000 = \text{USD } 5,960,000$; the funding released now is USD 400,000, sufficient only to complete detailed planning to the next gate. The project manager's authority is set at $\pm 10\%$ of the released stage budget, with escalation to the sponsor beyond that tolerance. The sponsor sees traceability to strategy (a USD 4,180,000 annual saving), a payback that holds across the whole estimate range, and a risk-informed envelope — and approves progression to planning.

3.7 Common pitfalls

- **Outputs masquerading as objectives.** "Implement the new system" is a deliverable, not an objective; without a baseline, target and date it cannot govern anything.
- **The point-estimate illusion.** Quoting USD 5,200,000 without its -25% / $+75\%$ range invites the sponsor to anchor on false precision the estimate cannot support.

- **The retrospective charter.** Drafting the charter after work has begun, so it records what is happening rather than authorising what may happen.
- **The convenient stakeholder list.** Listing only supportive insiders and omitting regulators, operational staff and those who bear the disruption.
- **Treating EMV as a forecast.** The USD 760,000 is a screening signal for contingency, not a prediction of what will be spent.
- **Unowned AI output.** Pasting an AI-generated risk register or stakeholder list into the charter unvalidated; every figure and claim must be explainable and owned by the professional who signs.

3.8 AI in project initiation: use, limits and accountability

Initiation is the stage at which the least is known and the most is decided, and it is precisely this asymmetry that AI-enabled tooling addresses. Predictive models trained on an organisation's delivery history can test a proposed mandate against the record of projects of comparable size, sector and complexity — reference-class techniques that confront the optimism bias distorting many mandates before a charter is ever drafted. Language models can assemble a first-draft charter from the mandate and the sponsor's brief; classification tools can surface analogous past projects and their close-out lessons; pattern analysis can propose an initial stakeholder long-list and a candidate risk register far faster than a workshop alone.

The delivery leader may properly automate the mechanical work of initiation: assembling draft charter text, checking objectives for internal consistency and traceability, retrieving comparators, and generating candidate lists of stakeholders, risks and assumptions. What the leader must personally validate is everything on which the decision to proceed rests: the strategic alignment argument; the measurability and ownership of each success criterion; the scope boundaries, constraints, assumptions, authority levels and funding recorded in the charter; the salience assigned to each stakeholder; and the initial risk appraisal presented to the sponsor. A generated stakeholder map is a hypothesis, not a finding; a machine-drafted charter is a starting text, not an approved document. The proceed, hold or stop recommendation is composed by the leader and decided by the sponsor — it is never delegated to a model.

Initiation-stage AI carries model risk that the leader must be able to name. Historical delivery data reflects the portfolio the organisation happened to run — survivorship bias, thin or inconsistently recorded lessons, contexts that no longer apply; a reference class of three projects is an anecdote, not a distribution. Generative tools can produce plausible but unfounded content — invented constraints, unsupported figures, assumptions presented as facts — and may render genuinely ranged estimates with false precision. Mining correspondence or organisational records for stakeholder analysis raises data-protection and confidentiality obligations that apply before any insight is used. Every AI-derived statement destined for initiation documentation must

carry its provenance: what data produced it, what the model assumed, and who checked it.

Accountability is not diluted by tooling. The sponsor remains accountable for the decision to initiate and the charter approved; the delivery leader for the completeness and honesty of the analysis placed before the sponsor, including every element a model contributed. Candidates are examined on their ability to state where AI assisted an initiation product, what was verified and by whom — a charter that cannot answer those questions is not ready for signature.

AI proposes; the professional disposes. At initiation, AI may draft the charter, mine the history, rank the stakeholders and seed the risk register — but the alignment of the project to strategy, the realism of its success criteria and the recommendation to proceed are judgements a competent professional must make, explain and own. The model informs the mandate; the sponsor's signature, and the accountability behind it, remain irreducibly human.

Key points

- A mandate commissions initiation only — permission to prepare, not to deliver; the charter is the instrument by which the sponsor authorises the project, appoints the project manager and delegates bounded authority.
- Objectives must be verifiable (measure, baseline, target, date, owner); the professional traces outputs → outcomes → benefits back to the mandate's strategy.
- Success criteria, constraints, assumptions and the sponsor's priority among competing constraints are fixed at initiation, not reconstructed afterwards.
- Estimates at initiation are ranged (e.g. -25 % / +75 %); funding is released in stages to the next gate, distinct from the whole-life envelope.
- Early stakeholder identification and an EMV-based initial risk appraisal exist so the decision to proceed is taken with eyes open; EMV is a contingency screening signal, not a forecast.
- AI can draft, mine, rank and seed initiation products at speed, but every AI output must be explainable, validated and owned by a named, competent human before it enters an authorising document.
- Accountability is never delegated: the leader answers for the analysis, the sponsor for the decision.

Practice questions

1. A project mandate has been issued by the portfolio board. What does the mandate itself authorise?

A. Delivery of the full project scope within the stated budget B. The expenditure of initiation effort within defined boundaries C. Release of the whole-life funding envelope to the project manager D. Commitment of suppliers under contract to the delivery schedule

Answer: B. A mandate is permission to prepare, not permission to deliver; delivery is authorised only by the sponsor's approval of the charter.

2. During initial risk appraisal, a risk is assessed at a probability of 0.4 with a monetary impact of USD 250,000 if it occurs. What is its expected monetary value (EMV)?

A. USD 62,500 B. USD 100,000 C. USD 250,000 D. USD 625,000

Answer: B. $EMV = \text{probability} \times \text{impact} = 0.4 \times 250,000 = \text{USD } 100,000$, used as a first-cut contingency signal rather than a spending forecast.

3. An AI tool generates a candidate risk register for inclusion in the project charter. Under the Institute's doctrine, what must occur before it may be used?

A. The register must be reformatted to the organisation's document template B. The model's confidence score must exceed an agreed threshold C. A competent, named professional must validate, be able to explain, and take ownership of its content D. The sponsor must sign the register in place of the project manager

Answer: C. AI proposes; the professional disposes — nothing enters an authorising document until a competent, accountable human has explained, validated and owned it.

Chapter 4: Business Cases

4.1 The business case as the instrument of justified investment

A **business case** is the documented justification for undertaking a project: a structured account of why the investment is needed, what options were considered, what benefits, costs, risks and timescales attach to the preferred option, and why that option represents the best achievable value for the resources committed. PCI distinguishes it sharply from two documents with which it is routinely confused. The **project mandate** (or brief) expresses an intention; the **project charter** confers authority and resources on a named manager. The business case does neither — it justifies. It is the standing answer to the question every governance body must be able to ask at any point: is this investment still worth making?

From this follows the discipline's central principle, expressed in PRINCE2 as **continued business justification**: a project without a valid business case should not start, and a project whose case ceases to be valid should not continue. The business case is therefore not an initiation artefact to be filed once funding is secured; it is the reference point against which every stage gate, baseline change and termination decision is tested.

Ownership is unambiguous. The **sponsor** — in some frameworks the senior responsible owner — owns the business case and is accountable for the benefits it promises; the **project manager** manages its development and maintenance and is accountable for the outputs it requires. Assurance bodies and gate reviews — reflected in ISO 21502 and the APM and PMI bodies of knowledge — test, independently of the delivery team, whether the case remains desirable, viable and achievable.

4.2 The five dimensions of a complete case

PCI adopts the widely used **five-case model**, articulated most fully in HM Treasury's Green Book, as the structural standard a candidate must apply. A complete business case demonstrates that the investment is:

- **Strategic** — supported by a compelling case for change that fits the organisation's objectives, and traceable to the portfolio or programme strategy it serves;
- **Economic** — the best value option, evidenced by an appraisal of realistic alternatives against a do-nothing or do-minimum baseline;
- **Commercial** — viable as a procurement: the market can supply it, and a workable contracting and risk-allocation route exists;
- **Financial** — affordable: funded, within budgetary limits, with cash-flow consequences the organisation can absorb;

- **Management** — achievable: governed, resourced, planned with a credible delivery approach, and equipped with arrangements to realise and measure benefits.

The dimensions are interdependent, not sequential chapters. The strategic case anchors the whole; the economic case selects the option; the commercial case tests whether it can be bought; the financial case tests whether it can be paid for; the management case tests whether it can be delivered. A case that is strong on four dimensions and silent on the fifth is not four-fifths complete — it is unapproved.

4.3 Options appraisal and the investment logic

The economic heart of the case is **options appraisal**: the disciplined comparison of alternative ways of meeting the need, always including the counterfactual of doing nothing or the minimum. Foundational concepts include the **long list** filtered to a **short list** against critical success factors; **whole-life cost**, which counts operation, maintenance and disposal alongside capital expenditure; **opportunity cost**, the value of the best alternative forgone; **optimism bias**, the demonstrated tendency to understate costs and overstate benefits, corrected with explicit adjustments and reference-class comparisons; and the classification of **benefits** as cash-releasing, non-cash-releasing or qualitative, each demanding a different evidencing standard. Value for money is a judgement across all of these — never the cheapest option by default.

4.4 The financial appraisal toolkit

Because project costs and benefits fall in different years, they cannot be compared at face value: money has a **time value**. **Discounting** converts future cash flows to present values using a discount rate reflecting the organisation's cost of capital or a mandated public-sector rate. Three techniques form the examinable core. **Net present value (NPV)** is the sum of all discounted cash inflows and outflows; a positive NPV indicates the investment creates value at the chosen rate. **Internal rate of return (IRR)** is the discount rate at which NPV equals zero — the break-even return, compared against a hurdle rate. **Payback period** is the time taken for cumulative net inflows to recover the initial outlay; simple and intuitive, but blind to the time value of money and to everything after payback. The techniques are complements, not rivals: NPV ranks options, IRR communicates return, payback signals exposure; the worked example in section 4.6 applies all three.

4.5 Method: constructing and appraising the case step by step

The Institute expects candidates to follow a disciplined sequence rather than assemble a persuasive document after the decision has effectively been taken.

1. **Confirm strategic fit.** State the problem or opportunity, the contribution to organisational objectives and the consequence of doing nothing. This anchors the strategic dimension.

2. **Generate options.** Build a long list that always includes do nothing and do minimum as reference cases; short-list three to five credible options against agreed criteria.
3. **Estimate whole-life costs and benefits.** For each short-listed option, forecast incremental cash flows over a defined appraisal horizon, on a consistently real or nominal basis. Where AI-enabled tools generate benefit forecasts or probabilistic ranges, the appraiser validates the assumptions and owns the figures.
4. **Discount to present value.** Apply the organisation's cost of capital or mandated discount rate to every future cash flow.
5. **Apply the investment criteria.** Calculate NPV, IRR and payback period for each option.
6. **Test sensitivity and select.** Flex the critical assumptions, identify the switching values at which the recommendation changes, and select the preferred option with reasons.
7. **Complete the remaining dimensions.** Confirm the commercial route to supply, financial affordability (the funding and cash-flow profile, distinct from economic worth) and the management arrangements for delivery and benefits realisation.

4.6 Worked example: field-service scheduling investment

Meridian Utilities, a regulated network operator, is appraising two mutually exclusive options for improving field-force scheduling. Its discount rate is 9 per cent and the appraisal horizon is five years.

Option B — replace the scheduling system requires £1,200,000 at Year 0 and yields net cash inflows of £280,000, £360,000, £420,000, £420,000 and £380,000 in Years 1 to 5.

Year	Net cash flow (£)	Discount factor at 9%	Present value (£)
0	(1,200,000)	1.0000	(1,200,000)
1	280,000	0.9174	256,872
2	360,000	0.8417	303,012
3	420,000	0.7722	324,324
4	420,000	0.7084	297,528
5	380,000	0.6499	246,962

Present value of inflows = 256,872 + 303,012 + 324,324 + 297,528 + 246,962 = **£1,428,698**.

NPV = 1,428,698 – 1,200,000 = **+£228,698**. The option adds value at the 9 per cent hurdle rate.

IRR is the rate at which NPV falls to zero. Recalculating at 15 per cent gives NPV of +£20,926; at 16 per cent, –£9,082. Interpolating between the two:

$$\text{IRR} \approx 15\% + [20,926 \div (20,926 + 9,082)] \times 1\% = 15\% + 0.70\% \approx \mathbf{15.7\%}$$

Payback period uses undiscounted cumulative cash flows: £280,000; £640,000; £1,060,000; £1,480,000. The outlay is recovered during Year 4: $3 + (140,000 \div 420,000) = \mathbf{3.33 \text{ years}}$, about three years and four months. On a discounted basis, recovery takes approximately 4.1 years.

Option C — enhance the existing system costs £500,000 and produces NPV of +£96,000, IRR of 17.2 per cent and payback of 2.8 years.

Selection. Option C pays back faster and shows the higher IRR, but the options are mutually exclusive and funding is available, so the governing criterion is absolute value created: Option B's NPV of £228,698 exceeds Option C's £96,000. Sensitivity analysis shows Option B's benefits could fall by about 16 per cent ($228,698 \div 1,428,698$) before its NPV turned negative — an acceptable margin. Option B is recommended, and its assumptions and switching values are recorded in the economic dimension of the case.

4.7 Assuring continued viability at the stage gate

At each gate the sponsor, as owner of the case, re-tests it on a **forward-looking** basis: sunk costs are excluded, and the question is whether the present value of the remaining benefits still exceeds the remaining cost to complete. Suppose Meridian has spent £700,000 mid-delivery, the cost to complete has risen to £650,000 and re-forecast benefits have a present value of £1,100,000: the forward NPV is +£450,000, so continuation is justified even though the whole-life position has deteriorated. The gate recommendation — continue, change (descope, rephase, refinance) or terminate — must cite the re-forecast evidence, never the original promise.

4.8 Common pitfalls

- **Optimism bias** — systematically understated costs and overstated benefits; counter with explicit adjustments and reference-class comparisons.
- **Omitting the do-nothing baseline**, so every option looks attractive against an unstated alternative.
- **Confusing affordability with worth** — a positive NPV does not prove that the in-year funding profile can be met.
- **Honouring sunk costs at gates** — continuing because of money already spent rather than value still recoverable.
- **Ranking mutually exclusive options by IRR or payback** when NPV should govern the choice.
- **Double-counting benefits** claimed by more than one project from the same operational saving.

- **Treating the case as a bid document** — approved once, then never maintained as delivery assumptions change.
- **Adopting unvalidated AI-generated forecasts** — model outputs with no named, competent owner able to explain and defend them at the gate.

4.9 AI in business case practice

What changes. Predictive and generative tooling now reaches every part of business case work. Machine-learning models trained on completed-project portfolios generate reference-class forecasts of cost, duration and benefit that counter the optimism endemic to promoter-written cases. Monte Carlo engines run thousands of scenario sweeps across discount rates, benefit ramp-up profiles and contingencies. Generative assistants draft all five dimensions from structured inputs, check them for consistency and flag where the appraisal contradicts the strategic rationale, while gate analytics compare assumptions with actual performance and surface benefit erosion earlier than periodic manual review. None of this changes what a business case is — a justification owned by an accountable individual. It changes how quickly evidence can be assembled, and how easily a weak case can be made to look rigorous.

Automate the arithmetic; validate the assumptions. The disciplined division of labour is arithmetic versus judgement. A project delivery leader appropriately automates the computation of NPV, IRR and payback across an option set, sensitivity and scenario analysis, document assembly, and the tracking of benefit realisation against forecast. The leader must personally validate what the arithmetic rests on: the discount rate and its justification, the completeness and attribution of benefits, the realism of cost and schedule inputs, the framing of the options themselves (including the do-nothing baseline), and the final recommendation. An algorithm will calculate an internal rate of return flawlessly on flawed cash flows. The examination therefore tests interpretation and challenge, not keystrokes: given an AI-generated appraisal, the candidate must identify which assumptions drive the answer and whether they survive scrutiny.

Model risk and data caveats. Business case analytics carry specific model risks. Historical project data embeds past estimating bias, so a model trained on it can launder optimism into apparently objective forecasts. Comparable samples are often small and survivorship bias is acute: terminated projects are poorly documented, so training data over-represents success. Generative drafting adds a distinct hazard — fluent, plausible narrative that conceals weak investment logic, or benchmark figures fabricated rather than retrieved — and market-facing inputs to the commercial dimension go stale quickly. Every model, its version, its training-data cut-off and its assumptions should be recorded and controlled with the same rigour as the cost and schedule baseline, and recalibrated at each gate.

Accountability at the gate. Accountability for the business case is not transferable to a tool or its vendor. At each gate the accountable owner must be able to state which

figures were machine-generated, how each was validated, and why they are believed — every AI output in the case must be explainable, reproducible and traceable to its inputs. A recommendation to continue, change or terminate is a professional judgement, made and signed by a competent human on evidence that human has tested. Where the decision proves wrong, no governance body will accept "the model said so" as a defence.

AI proposes; the professional disposes. Applied to the business case: AI may draft the five dimensions, compute the appraisal and flag the failing assumption — but the option selected, the viability judgement at each gate, and the recommendation to continue, change or terminate belong to an accountable professional who can explain, evidence and own every figure in the case.

Key points

- A business case justifies an investment (a mandate expresses intention; a charter confers authority); under continued business justification, a project without a valid case should not start or continue.
- A complete case is demonstrated across five interdependent dimensions — strategic, economic, commercial, financial and management; silence on any one leaves the case unapproved.
- Options appraisal always includes a do-nothing or do-minimum baseline, uses whole-life costs, and corrects for optimism bias with explicit adjustments and reference-class comparisons.
- NPV, IRR and payback are complements: NPV governs choices between mutually exclusive options, IRR communicates return, payback signals exposure — and affordability (financial case) is distinct from worth (economic case).
- Stage-gate viability is tested forward-looking: sunk costs are excluded, and continuation is justified only if the remaining benefits exceed the remaining cost to complete.
- The sponsor (senior responsible owner) owns the case and its benefits; the project manager manages its development; independent assurance tests whether it remains desirable, viable and achievable.
- AI may automate the arithmetic — appraisal, sensitivity analysis, drafting, benefit tracking — but every AI output must be explainable, validated and owned by a competent human: AI proposes; the professional disposes.

Practice questions

Question 1. An organisation must choose between two mutually exclusive options and has funding available for either. Option X: NPV +£96,000, IRR 17.2 per cent, payback 2.8 years. Option Y: NPV +£228,698, IRR 15.7 per cent, payback 3.33 years. Which option should be recommended, and why?

A. Option X, because it has the higher IRR B. Option X, because it pays back faster C. Option Y, because it has the higher NPV D. Option Y, because a longer payback indicates a longer benefit stream

Answer: C. For mutually exclusive options with funding available, absolute value created — NPV — governs the choice; IRR and payback are secondary signals.

Question 2. At a stage gate, a project has spent £700,000, the cost to complete has risen to £650,000, and the present value of the remaining benefits is re-forecast at £1,100,000. The whole-life position is worse than the approved case. What should the gate recommend?

A. Terminate, because the whole-life NPV has deteriorated against the approved baseline B. Continue, because the forward-looking NPV of the remaining investment is +£450,000 C. Continue, because terminating would waste the £700,000 already invested D. Suspend the project until the original whole-life NPV can be restored

Answer: B. Gate decisions are forward-looking — sunk costs are excluded, and continuation is justified when remaining benefits exceed the remaining cost to complete; option C is the sunk-cost fallacy.

Question 3. An AI tool trained on the organisation's historical project data produces a benefit forecast for inclusion in a business case. What must be true before the forecast can enter the case?

A. The tool has been approved by the project management office B. The forecast is explainable, validated against independent evidence, and owned by a named accountable person C. The model's stated confidence level exceeds an agreed threshold D. The forecast is more conservative than the manual estimate it replaces

Answer: B. AI proposes; the professional disposes — every AI output entering a case must be explainable, validated and owned by a competent human.

5. Project Governance

Every project is a temporary organisation embedded in a permanent one. Project governance is the framework through which the permanent organisation retains ownership of that temporary organisation: it defines who directs the project, who is accountable for its outcomes, what each party may decide, and how the flow of information keeps direction and delivery honestly connected. This chapter establishes the concepts, sets out a design method, works a numerical example, and examines how AI changes the tempo — but never the accountability — of governance.

Governance and management: the essential distinction

Project governance is the system of authority, accountability and control within which a project is directed and overseen. It comprises four elements: the **structures** (sponsor, board, assurance and delivery roles, and their reporting lines); the **decision rights** allocated to each — the classes of decision it is empowered to take; the **delegated limits** (often called tolerances) that bound those rights, expressed against cost, time, scope, quality, risk and benefits; and the **assurance and reporting arrangements** through which those who direct obtain a true picture of what those who deliver are doing.

Governance must be distinguished from management. Governance directs and constrains: it sets objectives, approves the business case, allocates authority and verifies performance. Management plans and executes within the authority so granted. The delivery leader manages; the sponsor and board govern, and when the two blur — a board that manages, a delivery leader who self-governs — the control system loses the separation on which it depends. Distinguish, too, the **governance of an individual project** from the **governance of project management** — the organisation's standing framework of methods, gate regimes and delegation schemes within which every project is governed.

The framework and standards context

The candidate is expected to know the consensus landscape, not any single proprietary method. ISO 21505 addresses the governance of projects, programmes and portfolios, framing project governance as an extension of corporate governance: the governing body remains ultimately answerable for the projects it funds, and delegates — never abandons — that accountability. ISO 21502 situates governance within wider project management practice, and method families such as PRINCE2 contribute defined roles and responsibilities, management by exception, and stage-based authorisation. Professional bodies' governance principles converge on the same tenets: a single accountable sponsor, a clear link from project to strategy through the business case, decision-making at the lowest competent level, and independent assurance proportionate to risk. The assurance dimension draws additionally on the "three lines" model — delivery management (first line), oversight and standards functions (second

line) and independent review (third line) — while public-sector gateway regimes model staged independent review at key investment decisions. The Institute's examination tests understanding of these frameworks' principles; it neither presupposes nor certifies competence in any vendor's product, and PCI claims no accreditation or endorsement by their owners.

The core structures and how they relate

Four roles anchor every well-formed governance design. The **sponsor** is the single named individual accountable for the business case and the realisation of benefits; the sponsor owns the "why" and holds the delivery leader to account for the "how". The **project board** (or steering committee) is the collective directing body through which the sponsor exercises governance, bringing together the interests — typically user, supplier and business — that must consent to major decisions. The **project delivery leader** is accountable for delivering the agreed outputs within the limits the board delegates. The **assurance function** provides those who govern with confidence, independent of those who deliver, that the information they receive is accurate and the project is competently run; it reports to the governance tier, never to the delivery leader whose work it examines.

These pieces relate through a chain of delegation and a mirrored chain of escalation. Authority flows downward — corporate body to sponsor, sponsor and board to delivery leader, delivery leader to team — each step with explicit limits. **Escalation** is the return path: the obligation to refer a decision upward the moment a forecast breaches a delegated limit, so that no tier is surprised by an exception it was entitled to decide. **Stage gates** punctuate this chain in time: predefined points at which continued investment must be positively re-authorised on current evidence rather than allowed to continue by inertia.

Governance operates entirely on information, which is why the reporting regime is a governance artefact, not an administrative one. Its foundations are accuracy (a single, reconciled version of performance truth), timeliness (information arriving while decisions can still alter outcomes) and proportionate summarisation (each tier receiving the detail its decision rights require). AI-enabled reporting can strengthen all three — but no decision right is ever allocated to a system.

Designing a governance structure: a step-by-step method

Governance is designed, not inherited. The method below scales from a small internal change to a major programme; what varies is the weight of each step, never its omission.

1. **Tier the project by scale and risk.** Classify the project (for example low / medium / high) using objective criteria: budget, duration, novelty, regulatory exposure, degree of AI-enabled or safety-critical content. The tier drives everything that follows — board seniority, gate frequency, assurance depth.

2. **Name the roles — one accountable individual per role.** A single sponsor; a board small enough to decide (typically three to six voting members); a delivery leader accountable within delegated limits; an assurance function reporting to the sponsor or board, never to the delivery leader.
3. **Allocate decision rights in a decision-rights matrix.** For each recurring decision class — baseline change, contract award, risk acceptance, gate passage, benefit revision, deployment of an AI-generated forecast into the performance record — record who decides, who is consulted and who is informed. Every AI-assisted decision must show a named human owner.
4. **Set delegated tolerances.** Express limits for cost, schedule, scope and risk as both a percentage of baseline and an absolute cap, at each level: delivery leader, sponsor, board. Track consumption cumulatively, not change by change.
5. **Define escalation routes and clock speeds.** Escalation is triggered by a forecast breach of tolerance, not an actual one; each route states who is notified, within how many working days, and with what minimum information.
6. **Fix the gate and assurance schedule.** Position stage gates at commitment points (funding release, contract award, go-live) and commission independent review ahead of the highest-risk gates.
7. **Specify the reporting regime.** One agreed data set, a fixed cycle, exception-based highlighting, and a stated owner for every figure — including any drafted or flagged by an AI tool, validated before it enters the record.
8. **Document, approve and rehearse.** Write the structure into the initiation documentation, have the board approve it, and walk one hypothetical escalation through it before delivery begins.

Worked example: setting delegated limits and testing an escalation

Setup. A systems-replacement project has an approved budget at completion of \$12,000,000 and a 20-month schedule. It is tiered medium-high risk. The board sets: delivery-leader cost tolerance $\pm 2.5\%$, schedule tolerance ± 4 weeks; sponsor delegated limit $+7.5\%$ cumulative cost and $+10$ weeks; anything beyond, or any scope or benefits change, is reserved to the board.

Arithmetic — setting the limits.

- Delivery-leader cost tolerance: $2.5\% \times 12,000,000 = \text{\textbf{\$300,000}}$
- Sponsor cumulative delegation: $7.5\% \times 12,000,000 = \text{\textbf{\$900,000}}$

Arithmetic — testing an escalation. In month 9 a supplier fails; the recovery estimate is +\$340,000 and +6 weeks. Changes already approved to date total \$180,000.

- Single-event test: $\$340,000 > \$300,000 \rightarrow$ the event exceeds the delivery leader's tolerance and **must be escalated to the sponsor**. The 6-week slip also exceeds the 4-week schedule tolerance, confirming escalation on both dimensions.
- Cumulative test at sponsor level: $\$180,000 + 340,000 = \$520,000$; $\$520,000 < \$900,000 \rightarrow$ the decision sits **within the sponsor's delegation**. The sponsor may approve without convening the board, notifying it by exception report at the next cycle.
- Remaining sponsor headroom after approval: $\$900,000 - 520,000 = \$380,000$.

Interpretation. The regime placed the decision at the lowest competent level: the delivery leader escalated because the forecast breached tolerance; the sponsor decided because cumulative consumption remained inside delegation; the board was informed, not convened. Had the cumulative figure reached \$900,000 — or had the forecast trend made a breach probable — the decision would have risen to the board before the money was committed.

Common pitfalls

- **Sponsor as super-project-manager.** A sponsor directing daily work collapses the owner-deliverer separation, leaving no one to hold delivery to account.
- **Boards built for representation, not decision.** Twelve observers and no quorum rules produce briefings, not decisions.
- **Escalating on actuals.** Waiting until tolerance is actually breached escalates history; the trigger must be the forecast.
- **Salami-slicing delegation.** Ten changes each within individual limits can exhaust cumulative delegation unnoticed — track headroom cumulatively.
- **Uniform tolerances regardless of risk.** A $\pm 2.5\%$ tolerance fit for a routine project is negligent for a novel, safety-critical or heavily AI-dependent one.
- **Assurance reporting to the delivery leader.** Independence is lost the moment the reviewed party commissions, filters or edits the review.
- **"Watermelon" reporting.** Green on the cover, red inside — the symptom of a regime that punishes escalation rather than rewarding early warning.
- **Unowned AI outputs.** Machine-generated narratives, forecasts or risk scores entering the record unvalidated and with no named owner.
- **Governance by document, not behaviour.** A decision-rights matrix bypassed by informal approvals is not governance; test the routes, and use them.

How AI changes the governance competency

Predictive and generative tooling changes the tempo of governance more than its architecture. Where boards once governed on periodic, backward-looking reports, AI-enabled environments offer continuous signals: forecasts of cost and schedule outturn, anomaly detection, consistency checks between reporting narrative and underlying numbers, and portfolio-level pattern recognition that surfaces drifting projects before their own reports admit it. Well governed, this shifts oversight from retrospective inspection towards exception-based supervision; badly governed, it produces a torrent of unvalidated indicators, dashboards mistaken for assurance, and decisions quietly ceded to scores nobody present can explain. The principles of this chapter are unchanged: no model holds a decision right, sits on a board or exercises a tolerance. What changes is the information on which accountable people act — and the discipline required before machine-generated information may inform a governance decision.

The delivery leader may appropriately automate the mechanical layer of governance: assembling reporting packs from source systems; drafting status narratives, minutes and decision logs; reconciling RAG ratings against the thresholds that define them; checking gate documentation for completeness; and tracking actions, decisions and escalations against owners and due dates. These uses reduce latency and transcription error and free attention for judgement. Validation, by contrast, is not delegable. Before an AI output reaches a sponsor, board or gate review, a competent professional must have traced any forecast to its inputs and assumptions; confirmed that a generated summary is faithful to its sources, not merely fluent; tested exception flags before they trigger escalation; and verified that sensitive or personal information has not reached unapproved tools. An unvalidated output presented to a decision-maker is a reporting failure, whatever produced it.

Model risk, data caveats and human accountability

Predictive models learn from past projects and inherit their pathologies: optimism bias in estimates, sandbagged forecasts and thin samples of comparable work. Their outputs are probabilistic, degrade when context shifts — a new supplier, a novel technology — and can be systematically wrong while appearing precise. Boards face a specific hazard in automation bias: an unexplained model rating can anchor a gate decision more powerfully than the evidence justifies. Governance therefore extends to the tooling itself: models, prompts and thresholds belong under version control as governed artefacts; data lineage and access rights must be documented; and the assurance function should review the AI pipeline — data, calibration, failure history — with the same independence it applies to any other source of management information. Where an AI-derived indicator informs a gate recommendation, the gate record should say so.

Accountability does not transfer to a tool. The sponsor remains singly accountable for the investment, the board for decisions within its terms of reference, and the delivery

leader for the integrity of information presented to both. Every AI-derived item in a governance pack must carry a named human owner able to explain and stand behind it, and AI use in preparing governance information must itself be disclosed and auditable. An escalation raised by an algorithm is a prompt to judgement, never a substitute for it — the classic failures of confused roles and orphaned decisions are amplified rather than cured when "the system flagged it" replaces a named owner.

AI proposes; the professional disposes. AI may draft the pack, flag the exception and forecast the outturn; only accountable people hold decision rights, exercise tolerances and approve gates. Every machine-generated signal placed before a board must be explainable, validated and owned by a competent professional — a score that no one can explain has no place in a decision that someone must own.

Key points

- Governance directs and constrains; management plans and executes. The sponsor and board govern; the delivery leader manages within delegated limits.
- Four structures anchor every design: a single accountable sponsor, a board small enough to decide, a delivery leader, and an assurance function reporting to the governance tier — never to the delivery leader it examines.
- Authority flows down through explicit delegated tolerances; escalation is the mirrored return path, triggered by a forecast breach, not an actual one.
- Track tolerance consumption cumulatively as well as per event: single-event tests decide who is escalated to, cumulative tests who may approve.
- Stage gates force positive re-authorisation of continued investment at commitment points, supported by independent assurance proportionate to risk.
- AI changes the tempo of governance, not its architecture: continuous forecasts and exception signals, but never a decision right allocated to a system.
- Every AI-generated figure, forecast or flag before a governance body must be explainable, validated and owned by a named professional — AI proposes; the professional disposes.

Practice questions

1. A project has a budget at completion of \$8,000,000. The delivery leader holds a cost tolerance of $\pm 3\%$; the sponsor holds a cumulative delegation of $+7.5\%$. Previously approved changes total 150,000. A new event is forecast to cost $+260,000$. Who decides?

- A. The delivery leader, because cumulative changes remain under the sponsor's limit

- B. The sponsor, because the event exceeds the delivery leader's tolerance but cumulative consumption is within the sponsor's delegation
- C. The board, because cumulative changes now exceed \$400,000
- D. The delivery leader, with the board notified at the next cycle

Correct answer: B. The event exceeds the delivery leader's \$240,000 tolerance ($3\% \times 8,000,000$), while cumulative consumption (410,000) remains within the sponsor's \$600,000 delegation.

2. Under management by exception, when must the delivery leader escalate to the sponsor?

- A. When actual costs have breached the delegated tolerance
- B. When the board requests an updated forecast at a stage gate
- C. As soon as a forecast indicates that a delegated tolerance will be breached
- D. Only when both cost and schedule tolerances have been breached simultaneously

Correct answer: C. Escalation triggers on a forecast breach so the higher tier can decide before commitment; waiting for actuals escalates history, and any single tolerance dimension suffices.

3. An AI tool assigns a project a high-confidence "red" delivery-risk score ahead of a stage gate. What must be true before the score may inform the gate decision?

- A. The tool must be formally accredited by the Institute
- B. The score must be validated, explainable and owned by a named, competent professional
- C. The gate decision must be deferred until the score returns to amber or green
- D. The score may be used as-is, since gate authority remains with the board in any case

Correct answer: B. Under "AI proposes; the professional disposes", machine-generated signals inform decisions only once a named human has validated and can explain them; the Institute accredits no tools, and an unexplained score must neither delay nor silently anchor a decision.

Chapter 6: Scope and Requirements Management

Scope and requirements management converts stakeholder need into a controlled, deliverable commitment: needs are elicited as requirements; requirements are analysed, prioritised and structured; the structure is baselined or ordered; and all subsequent change, delivery and acceptance is judged against it. This chapter sets out the definitions, the operating procedure, a worked prioritisation example, and the boundary between what AI may draft and what only an accountable professional may decide.

6.1 Requirements and scope: precise definitions

A **requirement** is a documented, verifiable statement of a need, capability or condition that a solution must satisfy to deliver value to its stakeholders. Requirements are conventionally layered: **business requirements** state why the organisation is investing; **stakeholder requirements** state what particular groups need from the outcome; **solution requirements** state what the delivered product or service must do (**functional**) and how well it must do it (**non-functional** — performance, security, usability, compliance); and **transition requirements** state what is needed temporarily to move from the current to the future state.

Scope is the boundary drawn around what the project will and will not deliver. The discipline distinguishes **product scope** — the features and functions of the deliverable itself — from **project scope** — the work that must be performed to produce it. The two are inseparable: every element of project scope should exist to satisfy an element of product scope, and every requirement should trace to work that delivers it.

Elicitation is the structured activity of drawing requirements out of stakeholders, documents, systems and observation. The Institute deliberately prefers elicitation to the weaker gathering: requirements are rarely lying about waiting to be collected. They must be surfaced through recognised techniques, **analysed** for completeness, consistency and feasibility, and **prioritised**, because stakeholder demand almost always exceeds the capacity of any funded delivery.

6.2 The standards context

The candidate should understand, at working level, where these practices sit in the wider body of professional guidance. ISO 21502 treats scope definition and control as core practices within a governed life cycle. The PMI PMBOK Guide tradition frames scope management as a flow from planning through requirements collection, scope definition, work breakdown, validation and control. ISO/IEC/IEEE 29148 provides the engineering-grade treatment of requirements quality — each requirement singular, unambiguous, feasible and verifiable. Business analysis guidance such as the IIBA BABOK Guide elaborates elicitation and analysis technique, while the Scrum framework defines the **product backlog** as the single, ordered source of work in

adaptive delivery. The PML-AI examination does not test recall of clause numbers; it tests application of the shared discipline these sources describe in predictive, agile and hybrid settings alike. Reference to these publications is contextual only and implies no accreditation by, or affiliation with, their issuing bodies.

6.3 Running an elicitation and prioritisation cycle

Requirements work is a repeatable procedure, not an inspired conversation:

1. **Segment the stakeholders.** From the stakeholder register, group parties by interest and knowledge, then match technique to group: structured interviews for sponsors and senior users; facilitated workshops for cross-functional groups where conflict must surface; surveys for large user populations; observation for operational staff whose practice is tacit; document and system analysis for regulatory and legacy constraints; prototyping where needs are genuinely uncertain.
2. **Capture into a single register.** Record every candidate requirement with a unique identifier, source, statement of need (not solution), rationale, type and owner.
3. **Analyse.** Remove duplicates, test each statement for ambiguity and verifiability, and resolve conflicts by facilitated negotiation — escalating irreconcilable conflicts to the sponsor rather than quietly choosing a side.
4. **Prioritise.** Use MoSCoW (Must, Should, Could, Won't-this-time), a weighted scoring model, or cost-of-delay ranking, as fits the delivery approach.
5. **Confirm.** Walk the prioritised set back through the contributing stakeholders and record sign-off.

AI-mined candidate requirements enter this cycle only as proposals at step 2; the validation and ownership duties of section 6.7 apply in full.

6.4 From requirements to structure: breakdown or backlog

Once analysed and prioritised, requirements must be organised into a structure that can be planned, estimated and controlled. The choice is not stylistic: it follows from the delivery approach selected during planning, and the competent professional can justify which structure fits which component of the work.

Predictive route: the work breakdown structure (WBS). The WBS is a deliverable-oriented, hierarchical decomposition of the total scope. Decompose top-down: level 1 the project, level 2 the major deliverables, continuing until each **work package** can be reliably estimated and assigned to a single owner (the 8/80-hour heuristic). Enforce the 100 per cent rule at every level — children must sum to exactly the parent's scope, no more and no less — and write a WBS dictionary entry (boundaries, acceptance criteria, exclusions) for each work package before mapping packages to the organisation as control accounts.

Adaptive route: the prioritised backlog. The backlog is an ordered, emergent list in which position expresses priority and detail is added just in time. Decompose epics into features and features into user stories, each carrying acceptance criteria. Keep the backlog appropriately detailed (fine near the top, coarse below), estimated and prioritised, and admit stories to a sprint only once they meet the definition of ready.

Hybrid route. Hold stable, contracted scope in a WBS and evolving exploratory scope in a backlog, mapping each backlog area to the WBS element it rolls up to so that cost and progress still integrate.

6.5 Worked example: MoSCoW prioritisation with a capacity check

1. **Setup.** An e-permitting platform release has 10 two-week sprints available; the team's demonstrated velocity is 48 story points per sprint. After elicitation and analysis, the prioritised backlog totals: Must-have 274 points, Should-have 130 points, Could-have 96 points (Won't-this-time items are excluded from the release).
2. **Method.** Delivery capacity = velocity × sprints. Governance guideline (as in DSDM-style MoSCoW practice): Must-have effort should not exceed ~60 per cent of capacity, so the Should and Could items form the contingency pool.
3. **Substitution.** Capacity = $48 \times 10 = \mathbf{480 \text{ points}}$. Must-have ceiling = $0.60 \times 480 = \mathbf{288 \text{ points}}$; actual Must-have demand = $274 \leq 288$, a margin of $288 - 274 = \mathbf{14 \text{ points}}$. Committed scope = Must + Should = $274 + 130 = \mathbf{404 \text{ points}}$, leaving $480 - 404 = \mathbf{76 \text{ points}}$ for Could-have items. Total demand = $274 + 130 + 96 = \mathbf{500 \text{ points}}$, an oversubscription of $500 - 480 = 20 \text{ points}$ (104.2 per cent of capacity).
4. **Result.** The release is baselined at 480 points: all Must and Should items plus 76 points of Could items; the remaining $96 - 76 = \mathbf{20 \text{ points}}$ of Could items sit below the cut line.
5. **Interpretation.** The Must-have set passes the 60 per cent test, so even significant estimating error should not threaten the guaranteed minimum scope. The 20 deferred points remain on the backlog — deferral is a visible, sponsor-approved decision, not deletion.

6.6 Baselining, change control and the golden thread

In predictive and hybrid settings, the approved scope statement, WBS and WBS dictionary together form the **scope baseline** — the reference against which delivery performance and all proposed change are judged through integrated change control. Establish it through formal approval and version control. Thereafter, every change follows one path: log it; assess impact on scope, cost, schedule, risk and benefits, using the traceability matrix to find every affected element and test; decide at the delegated authority level; if approved, update the baseline, the matrix and the affected parties.

Adaptive delivery does not freeze scope, but it is not uncontrolled: the backlog changes only through the product owner's ordering decisions, and the iteration commitment and definition of done provide the control surface.

In both modes, the connective tissue is **traceability**: the maintained, bidirectional linkage from business need through requirement, scope element and deliverable to test and acceptance. Maintain the traceability matrix from day one — requirement ID → design element → build item → test case → acceptance result. Each requirement carries **acceptance criteria**: objective, verifiable conditions that must be satisfied before acceptance. Write them as testable thresholds ("permit application processed end-to-end in ≤ 3 working days at 1,000 applications per week"), never adjectives. This enables the crucial distinction between **verification** (the deliverable was built correctly, to specification) and **validation** (the correct deliverable was built, satisfying the need); validation then becomes an audit trail, each requirement confirmed against its recorded criterion. Traceability also makes scope control auditable: an unauthorised addition with no requirement behind it is **scope creep**; a requirement with no deliverable against it is a delivery gap.

6.7 AI in scope and requirements work

Scope and requirements management is among the most text-intensive competencies in project delivery, and therefore among the most exposed to AI assistance. Language-model and analytics tooling can mine contracts, transcripts, service tickets and legacy specifications for candidate requirements; cluster duplicates and flag conflicts; draft user stories with first-cut acceptance criteria; propose a WBS decomposition or an ordered backlog; and populate a draft traceability matrix. Predictive models can flag requirements likely to prove volatile, estimate the downstream effect of a proposed change, and detect drift between delivered artefacts and baselined scope. Used well, this shifts the leader's effort from clerical capture towards analysis, negotiation and decision — precisely the activities only a professional can perform.

The automation boundary. The candidate is expected to draw this boundary deliberately. Automate the mechanical layers: extraction of candidate requirements, de-duplication and clustering, first-draft story and criteria wording, traceability-matrix population, and continuous consistency checks — orphaned requirements, untraced deliverables, criteria with no planned verification, breaches of the 100 per cent rule. What the leader must validate, and never delegate, are the judgements those drafts feed: whether the elicited set is genuinely complete for this project's stakeholders, whether wording is testable and unambiguous, whether prioritisation reflects real business value rather than historical pattern, whether a decomposition suits the chosen delivery approach, and — above all — whether a baseline change is warranted and whether a deliverable is accepted. An AI-generated backlog can look complete while omitting the one requirement that matters most; only professional review against the stakeholders and the contract exposes the gap.

Model risk and data caveats. Three caveats govern this competency. First, extraction and generation models produce plausible text: a hallucinated or subtly misworded requirement is visually indistinguishable from a sound one, and a fabricated acceptance criterion is more dangerous than a missing one because it invites false confirmation of delivery. Second, decomposition and prioritisation suggestions inherit the biases of their training corpus — a model trained on unlike projects will confidently propose an unsuitable structure, and unstated stakeholder needs, by definition, never appear in the documents a model reads. Third, requirements material is frequently commercially sensitive or personal; feeding it to external tooling without confidentiality, retention and data-protection safeguards is a governance failure regardless of the analytical benefit. AI-assisted outputs used in scope work must therefore be version-controlled alongside the baseline they inform.

Human accountability. Accountability is not transferable to tooling. The scope baseline is approved by named governance authorities; change is disposed by the accountable board on a professional's recommendation; acceptance is confirmed by the designated acceptor against verifiable criteria. Where AI contributed to any of these, the delivery leader owns the result as if authored personally, records that AI assistance was used, and must be able to explain and defend the decision without reference to the tool. An unvalidated AI output in a scope baseline is not efficiency; it is unexamined scope.

AI proposes; the professional disposes. AI may draft the requirement, propose the decomposition, populate the traceability matrix and flag the creeping scope — but the requirement set is confirmed with stakeholders, the baseline is approved, change is authorised and delivery is accepted only by an accountable professional who can explain, evidence and own every disposition.

6.8 Common pitfalls

- Eliciting only from the loudest or most senior stakeholders, then discovering operational requirements during acceptance.
- Recording solutions ("use product X") instead of needs, foreclosing better options.
- Breaking the 100 per cent rule, or structuring the WBS by department rather than deliverable, so gaps and overlaps hide.
- Prioritisation collapse: allowing every requirement to be classed Must-have, which destroys the contingency pool.
- Scope creep through undocumented "small favours" and gold-plating that never pass through change control.
- Baselining before known conflicts are resolved, guaranteeing early change traffic.

- Unverifiable acceptance criteria ("fast", "user-friendly") that make validation a negotiation instead of a test.
- Building the traceability matrix retrospectively for the audit, too late to control anything.
- Accepting AI-generated requirements, classifications or impact assessments without human validation and named ownership.

Key points

- Requirements are layered — business, stakeholder, solution (functional and non-functional) and transition — and product scope is distinct from project scope; each must trace to the other.
- Requirements are elicited through techniques matched to stakeholder groups, captured as needs rather than solutions, analysed for ambiguity and conflict, prioritised and confirmed with stakeholders.
- Predictive delivery structures scope as a WBS governed by the 100 per cent rule and a WBS dictionary; adaptive delivery uses an ordered, emergent backlog; hybrid delivery uses both, mapped together so cost and progress integrate.
- In MoSCoW practice, Must-have effort should not exceed roughly 60 per cent of demonstrated capacity, so the Should- and Could-have items form the contingency pool; items below the cut line are visibly deferred, not deleted.
- The scope baseline (or the product-owner-ordered backlog) is the sole reference for change: every proposed change is logged, impact-assessed via the traceability matrix, decided at the delegated authority level and reflected in an updated baseline.
- Acceptance criteria must be verifiable thresholds, enabling the distinction between verification (built to specification) and validation (satisfies the need); orphaned work signals scope creep, and untraced requirements signal delivery gaps.
- AI may automate the mechanical layers of requirements work, but completeness, testability, prioritisation, change and acceptance judgements remain with a named, accountable professional: AI proposes; the professional disposes.

Practice questions

Question 1. A release has 8 sprints available and a demonstrated velocity of 50 story points per sprint. The prioritised backlog shows Must-have demand of 260 points. Applying the MoSCoW governance guideline in this chapter, which statement is correct?

- A. The Must-have set is acceptable, because 260 points is below the 400-point capacity.
 B. The Must-have set is acceptable, because Must-haves may consume up to 80 per cent of capacity.
 C. The Must-have set breaches the guideline, because 260 points

exceeds the 240-point ceiling (60 per cent of 400). D. The guideline cannot be applied, because velocity is relevant only to Could-have items.

Answer: C. Capacity is $50 \times 8 = 400$ points and the Must-have ceiling is $0.60 \times 400 = 240$ points, so a 260-point Must-have set erodes the contingency pool and must be renegotiated.

Question 2. At handover a deliverable meets every documented specification, yet the operational users demonstrate that it does not resolve the business need that justified the investment. Which statement best describes the situation?

A. Verification has failed and validation has passed. B. Verification has passed but validation has failed. C. Both verification and validation have passed, since the specification was met. D. This is scope creep, because the deliverable exceeded its requirements.

Answer: B. Verification confirms the deliverable was built correctly to specification, which occurred; validation confirms the correct deliverable was built to satisfy the need, which it was not.

Question 3. An AI tool drafts requirements from workshop transcripts; one drafted requirement is later found to be hallucinated after being admitted to the baseline. Under the Institute's doctrine, who is accountable?

A. The tool vendor, since the model generated the erroneous text. B. No one, since hallucination is a known and accepted model limitation. C. The stakeholders who attended the workshop recorded in the transcript. D. The professional who admitted it to the baseline without validation and named ownership.

Answer: D. Accountability is not transferable to tooling: every AI output must be explainable, validated and owned by a competent human before it enters a baseline — AI proposes; the professional disposes.

Chapter 7 — Planning and Execution

This chapter examines the competency at the centre of delivery leadership: constructing an integrated plan, estimating with declared confidence, executing to a controlled cadence, and taking proportionate corrective action when performance diverges from commitment. It also establishes how AI-enabled planning and forecasting tools are used within the Institute's doctrine of professional accountability.

7.1 The integrated plan: a single system of commitments

Planning, in the sense examined by this credential, is not the production of documents but the construction of a single, internally consistent model of the work. An **integrated project plan** is one in which four elements — scope, schedule, resource and cost — are derived from one another and reconcile without contradiction. Scope defines what will be delivered; the schedule defines when; the resource plan defines who and with what; the cost plan defines at what expenditure. The test of integration is traceability: every scheduled activity traces to a scope element, every scope element carries a resource assignment, and every resource assignment generates a cost. A plan that fails this test — a schedule containing work absent from the scope statement, or a budget that does not reflect the resource histogram — is not a plan but a collection of unconnected assertions.

The structural spine of integration is the **work breakdown structure (WBS)**: a hierarchical, deliverable-oriented decomposition of the total scope. The WBS provides the common coding structure through which schedule activities, cost accounts and responsibility assignments are cross-referenced. Once the four elements are mutually consistent, they are **baselined**: formally approved, placed under change control, and thereafter changed only through the governance mechanisms established at initiation (Chapters 3 and 4). The baseline is the reference against which all subsequent performance measurement is made; without it, the terms variance, progress and forecast have no defined meaning.

7.2 Standards context

The concepts in this chapter align with the international consensus expressed in ISO 21502 (project management guidance) and ISO 21508 (earned value management), and with the bodies of practice maintained by the major professional associations, which distinguish predictive (plan-driven), adaptive (agile) and hybrid life cycles. PRINCE2-style methods contribute the discipline of stage-based authorisation and management by exception; earned value standards contribute the performance measurement baseline and its derived indices. PCI's syllabus draws on these sources generically: candidates are examined on the underlying logic, not on any single proprietary method, and PCI makes no claim of accreditation by, or affiliation with, the bodies that publish them.

7.3 Producing the integrated plan: the method

An integrated plan is one dataset expressed several ways, not separate documents reconciled after the fact. The Institute expects candidates to follow six steps in order.

1. **Fix the scope baseline.** Decompose the approved scope into a WBS to work-package level; give every package an owner and acceptance criteria. Anything not in the WBS is not in the plan.
2. **Sequence the work.** Build the activity network from the work packages, apply dependencies, and identify the critical path and total float.
3. **Resource-load the schedule.** Assign roles with real availability and level the histogram before baselining. **Levelling** extends dates to respect capacity limits; **smoothing** redistributes effort within float. A schedule that assumes 120% utilisation is not a plan; it is a wish.
4. **Cost the resourced schedule.** Derive the time-phased budget from the loaded schedule so that scope, dates and money share one source. This yields the **performance measurement baseline**.
5. **Add contingency explicitly.** Hold contingency against identified risks at a named level of the budget, never smeared invisibly through the estimates.
6. **Baseline under change control**, recording the maturity class and confidence attached to every estimate.

The consistency test is simple: change any one element and the others must be repriced. If scope can grow while the budget stands still, the plan was never integrated.

7.4 Estimating and estimate maturity

An **estimate** is a quantified prediction — of duration, effort or cost — made under uncertainty and accompanied by a stated basis and confidence. Three technique families recur throughout the syllabus, with a fourth, machine-assisted, mode of application.

- **Analogous estimating** scales from the actual outcomes of comparable past work. It is rapid, inexpensive and defensible early, but only as reliable as the analogy; a wide band should be declared (for example, of the order of –30% to +50% at concept stage).
- **Parametric estimating** applies a validated statistical relationship between historical data and a driving variable (cost per unit, hours per interface). Its precision depends entirely on the quality of the calibration data and the stability of the underlying relationship.
- **Three-point (PERT) estimating** captures uncertainty explicitly through optimistic (O), most likely (M) and pessimistic (P) values, typically combined

through a weighted beta distribution: expected value $E = (O + 4M + P) \div 6$, with standard deviation $\sigma \approx (P - O) \div 6$, giving an explicit confidence range.

- **AI-assisted estimating** uses machine-learning tools to propose reference-class ranges and flag optimism in human inputs. The estimator must be able to explain the drivers, validate them against project data, and own the published figure.

Estimates mature. Early, top-down figures produced for feasibility purposes carry wide ranges — often expressed as estimate classes with declared accuracy bands — and are progressively refined into bottom-up, definitive estimates as scope definition improves. The professional obligation is candour: every estimate presented to a sponsor must state its basis, its assumptions, its class and its confidence interval. A single-point figure offered without a range misrepresents knowledge the estimator does not possess; a number without a stated confidence is an opinion.

7.5 Worked example: from estimate to baseline to corrective action

A delivery leader plans a work package covering the build and test of **60 system interfaces**.

Estimate (parametric). Validated history shows 14 hours per interface at a blended rate of USD 95 per hour.

- Effort = $60 \times 14 = \mathbf{840 \text{ hours}}$
- Budget at completion (BAC) = $840 \times 95 = \mathbf{USD 79,800}$

Duration (three-point). O = 9 weeks, M = 12 weeks, P = 21 weeks.

- $E = (9 + 4 \times 12 + 21) \div 6 = 78 \div 6 = \mathbf{13 \text{ weeks}}$
- $\sigma \approx (21 - 9) \div 6 = \mathbf{2 \text{ weeks}}$

The published estimate is therefore 13 weeks $\pm$ 2 weeks (approximately 68% confidence of completion between weeks 11 and 15), and the plan is baselined on this basis.

Status at week 6. The baseline planned 30 interfaces complete; 24 are actually complete and 384 hours have been booked.

- Planned Value: $PV = (30 \div 60) \times 79,800 = \mathbf{USD 39,900}$
- Earned Value: $EV = (24 \div 60) \times 79,800 = \mathbf{USD 31,920}$
- Actual Cost: $AC = 384 \times 95 = \mathbf{USD 36,480}$
- Cost Variance: $CV = 31,920 - 36,480 = \mathbf{-USD 4,560}$; $CPI = 31,920 \div 36,480 = \mathbf{0.875}$
- Schedule Variance: $SV = 31,920 - 39,900 = \mathbf{-USD 7,980}$; $SPI = 31,920 \div 39,900 = \mathbf{0.80}$

- Estimate at Completion: $EAC = BAC \div CPI = 79,800 \div 0.875 = \text{USD } 91,200$;
 $VAC = 79,800 - 91,200 = -\text{USD } 11,400$

Interpretation and corrective action. Productivity is the root cause: $384 \div 24 = 16$ hours per interface against 14 planned. The proportionate response is a corrective-action ladder, escalated only as far as necessary: first, diagnose why productivity is below plan (here, test-environment availability); second, re-sequence remaining interfaces to protect the critical path; third, adjust resourcing; only if these fail, seek de-scoping or a re-baseline through formal change control. The revised EAC and completion forecast are reported against tolerance immediately, not concealed until they harden into fact.

7.6 Execution, monitoring and control

Execution is the conversion of the baselined plan into delivered outputs; **monitoring and control** is the feedback loop that keeps the two aligned. The loop has four steps: measure actual performance; compare against the baseline to identify **variances**; analyse cause and forecast consequence; and select a proportionate response — corrective action within tolerance, escalation when tolerance is threatened, or a formal change to the baseline when the plan itself is no longer viable. Earned value management supplies the standard vocabulary — planned value, earned value and actual cost, with the variances and performance indices derived from them, as applied in the worked example and treated further in Chapter 8. Control is proportionate when the cost and intrusiveness of the response match the significance of the variance; management by exception exists precisely to prevent governance attention being consumed by noise.

In practice, control is a rhythm, not an event: a weekly team-level review of progress and impediments; a monthly baseline-variance review with thresholds agreed in advance (for example, variance beyond $\pm 5\%$ triggers analysis, beyond $\pm 10\%$ triggers escalation); and a standing rule that every red indicator carries a named owner and a dated action. In agile and hybrid delivery, cadence — fixed-length iterations, regular reviews and stable team capacity — performs the same integrating function that the baseline performs in predictive work: it makes throughput measurable and forecasting empirical.

7.7 AI in planning and execution

AI changes the economics of this competency; it does not change its logic. Machine-assisted planning tools can draft schedule networks from scope statements, propose resource-loaded sequences and run automated quality checks on plan integrity — dangling logic, missing dependencies, implausible durations, over-allocated resources and misaligned cost and schedule phasing. Predictive models trained on historical project data can generate parametric estimates with quantified ranges and, during execution, produce probabilistic forecasts of completion dates and costs, detect

anomalies in progress data and surface emerging variances before they appear in period-end reporting. The integrated plan thus becomes a living dataset rather than a static document. The learning objectives of this chapter stand unchanged: the plan must remain mutually consistent and formally baselined, every estimate must carry a stated confidence, and every corrective action must be proportionate and defensible — whoever, or whatever, first proposed it.

What the leader automates, and what the leader validates. The delivery leader should automate the mechanical layer: assembly of progress data from source systems, schedule-quality and resource-loading checks, first-draft parametric and analogous estimates, variance detection within the delivery cadence, and the drafting of routine progress narratives. The leader must validate, before use, the layer on which judgement depends: the assumptions, exclusions and reference class behind any machine-generated estimate; the plausibility of a proposed critical path; the causes — not merely the existence — of a flagged variance; and any recommended corrective action, tested against its cost, risk and stakeholder consequences. Certain decisions are never delegated to tooling: approval of the baseline, the confidence statement attached to an estimate, the selection of corrective action, and any re-baselining, which passes through the change-control route established in the governance chapters.

Model risk and data caveats. Predictive outputs are only as representative as the data beneath them. An estimating model trained on projects of a different sector, scale, delivery method or economic era will produce confident but biased figures; the leader must ask what the model learned from and whether that population resembles the current project. Forecasting models degrade silently when progress data is poor — optimistic percent-complete reporting corrupts every downstream projection. Two behavioural risks deserve particular attention: **automation bias**, the tendency to accept a machine-generated number because it is precise, and **anchoring**, where an early AI estimate frames all subsequent negotiation. A spuriously exact figure is still an estimate and must be presented with its range and confidence. Generated narrative is a further hazard: drafted commentary can assert what the data does not support, so every generated statement must be traceable to source records. An output that cannot be explained in plain terms to a sponsor or an assurance reviewer cannot support a baseline decision.

Human accountability. Under the Institute's doctrine, accountability for the plan is non-transferable. The leader who baselines an integrated plan owns every number in it, however it was produced, and the leader who acts on a predictive warning — or declines to — owns that judgement equally. Good practice is to record which planning artefacts were AI-assisted, retain the validation evidence alongside the basis of estimate, and present forecasts with their provenance and confidence visible. The examination tests this directly: candidates are expected to determine when an AI-generated schedule, estimate or forecast should be accepted, adjusted or rejected, and to justify the decision.

AI proposes; the professional disposes. A model may draft the schedule, price the estimate and flag the variance first. The delivery leader interrogates the data and assumptions, sets the confidence, approves the baseline and selects the corrective action — and remains personally accountable for every commitment the plan makes.

7.8 Common pitfalls

- **Baselining before resource levelling**, producing dates no available team can meet.
- **Hidden padding at every level** instead of explicit, risk-based contingency owned at a named tier.
- **Single-point estimates** published without class or confidence band.
- **Anchoring the three-point on M**, with a token pessimistic value that understates σ .
- **Reading SV as time**: schedule variance in currency trends to zero at completion even on a late project; always check the critical path as well.
- **The rubber baseline** — re-baselining to erase variance rather than to reflect approved change.
- **Self-reported percent complete** without objective earning rules, which inflates EV.
- **Accepting AI-generated estimates unvalidated** — an unexplained figure fails the doctrine that AI proposes and the professional disposes.

Key points

- An integrated plan is a single, mutually consistent dataset in which scope, schedule, resource and cost derive from one another; the WBS is its structural spine and traceability is its test.
- The baseline is formally approved and held under change control; without it, variance, progress and forecast have no defined meaning.
- Estimates are published with basis, class and confidence: analogous for speed, parametric for calibrated precision, three-point (PERT: $E = (O + 4M + P) \div 6$) for explicit uncertainty.
- Monitoring and control is a four-step feedback loop — measure, compare, analyse, respond — governed by management by exception and executed to a fixed cadence.
- Corrective action follows a ladder of proportionality: diagnose, re-sequence, re-resource, and only then de-scope or re-baseline through formal change control.

- AI tools can draft plans, price estimates and flag variances early, but model risk, automation bias and anchoring demand validation of every output before use.
- Accountability is non-transferable: AI proposes; the professional disposes.

Practice questions

1. A work package has an optimistic duration of 4 weeks, a most likely duration of 6 weeks and a pessimistic duration of 14 weeks. Using PERT, the expected duration is:

A. 6 weeks B. 7 weeks C. 8 weeks D. 9 weeks

Answer: B. $E = (4 + 4 \times 6 + 14) \div 6 = 42 \div 6 = 7$ weeks; the beta weighting counts the most likely value four times.

2. A project has BAC = USD 200,000, EV = USD 80,000 and AC = USD 100,000. Assuming current cost performance continues, the estimate at completion (EAC) is:

A. USD 200,000 B. USD 220,000 C. USD 250,000 D. USD 280,000

Answer: C. $CPI = 80,000 \div 100,000 = 0.80$, and $EAC = BAC \div CPI = 200,000 \div 0.80 = USD 250,000$.

3. An AI planning tool generates a parametric cost estimate for a work package. Under the Institute's doctrine, the estimate may enter the baseline only when:

A. The tool's vendor certifies the model's accuracy for this class of project. B. The figure lies within the sponsor's approved budget tolerance. C. A competent professional can explain its derivation, has validated its inputs and calibration, and accepts ownership of it. D. The tool has been trained on at least one comparable historical project.

Answer: C. AI proposes; the professional disposes — every AI output must be explainable, validated and owned by an accountable human before it becomes a commitment.

8. Stakeholder Leadership and Engagement

8.1 Stakeholders, engagement and leadership: precise definitions

A **stakeholder** is any individual, group or organisation that can affect, is affected by, or perceives itself to be affected by a decision, activity or outcome of the project. This definition, common in substance to the principal international standards, is deliberately broad in three respects. First, it includes those who merely perceive an impact: perception drives behaviour whether or not the underlying impact is real, so a group that wrongly believes it will lose out must still be engaged. Second, it embraces those who can affect the project as well as those affected by it — regulators, funders, interface owners and suppliers alongside end users and operational teams. Third, it is dynamic: the stakeholder population changes as the project moves through its life cycle, and an analysis that was accurate at initiation may be dangerously stale by execution.

Stakeholder engagement is the deliberate, two-way process of involving stakeholders so that their needs, expectations and concerns are understood and addressed, and their support for project outcomes is won and retained. The Institute follows contemporary usage in preferring engagement to management: senior stakeholders, regulators and communities cannot be "managed" in any directive sense, but they can be understood, informed, consulted and influenced. **Stakeholder leadership** goes further still. It denotes the active shaping of attitudes and alliances — mobilising sponsors, building coalitions for change and confronting resistance constructively — and it is this leadership dimension that distinguishes the delivery leader from the delivery administrator.

Communication is the principal instrument of engagement but is not synonymous with it. Engagement also encompasses consultation, negotiation, joint problem-solving and shared decision-making. The **communication plan** is the documented expression of the engagement strategy: it matches message, channel, frequency and sender to the needs of each stakeholder group, and it is derived from analysis rather than composed from habit.

8.2 The analytical dimensions: interest, influence and attitude

Rigorous engagement rests on three analytical dimensions:

- **Interest** — the degree to which the stakeholder's objectives, work or wellbeing are affected by the project: the size and nature of the stake.
- **Influence** (or power) — the stakeholder's capacity to affect decisions, resources, approvals or opinion, whether through formal authority or informal networks.

- **Attitude** — the stakeholder's current disposition towards the project, conventionally graded from unaware through resistant and neutral to supportive and leading.

The first two dimensions support classification, most familiarly through the power-interest grid associated with Mendelow. The third converts classification into strategy: a high-influence supporter and a high-influence resistor occupy the same cell of the grid yet demand entirely different treatment. Complementary lenses include the salience model of Mitchell, Agle and Wood, which ranks stakeholders by power, legitimacy and urgency, and the engagement assessment matrix, which records each stakeholder's current against desired engagement level so that the gap between the two defines where engagement effort must concentrate.

8.3 Governing frameworks and standards

Stakeholder engagement is an established practice area across the recognised bodies of knowledge. ISO 21502 treats it as an integral project management practice spanning identification, analysis and ongoing engagement; the PMBOK Guide addresses it as a dedicated performance domain interwoven with planning, delivery and measurement; PRINCE2 requires a documented communication management approach reviewed at each stage boundary; and the APM Body of Knowledge frames stakeholder engagement as a core interpersonal and professional discipline. Freeman's stakeholder theory supplies the underlying management thought: organisations create value for, and are accountable to, a constituency wider than their shareholders. The candidate is not examined on the clause structure of any one standard, but on the shared logic these frameworks encode — identify, analyse, plan, engage, monitor and adapt.

Engagement also operates within governance, not alongside it. The sponsor owns the business case and is simultaneously the project's most consequential stakeholder; boards and steering committees are standing engagement forums; and escalation routes defined in the governance framework are themselves engagement channels of last resort.

8.4 The engagement cycle

The elements of this chapter form a cycle, not a checklist. Identification and analysis produce a defensible picture of who matters, why, and with what disposition. That analysis drives a differentiated engagement strategy, which the communication plan operationalises. Delivery then generates evidence — attendance, responsiveness, sentiment, decision latency — against which effectiveness is evaluated and the approach adapted as attitudes and coalitions shift. Stakeholder analysis is perishable; the cycle is what keeps it current.

The discipline also integrates horizontally. Stakeholder attitudes are a recurrent source of both threat and opportunity in the risk register; benefits are realised only when

stakeholders change how they work; and requirements, acceptance and adoption all depend on the quality of engagement.

8.5 Method: scoring stakeholders by interest, influence and attitude

The analysis that underpins every engagement decision can be performed as a repeatable, auditable procedure. The Institute expects candidates to be able to reproduce each step, including the arithmetic, under examination conditions.

1. **Identify.** Assemble the stakeholder register from the organisation chart, the contract and supplier structure, the benefits map and a facilitated workshop. Record individuals where influence is personal; group where it is collective.
2. **Score influence and interest.** Rate influence I (ability to affect the project) and interest S (degree to which the project affects them) on a 1–5 scale, then compute a priority score $P = I \times S$.
3. **Assess attitude.** Rate current attitude A on a -2 (active resistance) to $+2$ (champion) scale, and set a realistic target attitude T . The engagement gap is $G = \max(0, T - A)$: a stakeholder already at or above target has a zero gap.
4. **Weight the effort.** Compute an effort weight $W = P \times G$ and allocate the team's finite engagement capacity in proportion to W .
5. **Derive the strategy.** High P with a positive gap indicates manage closely; high P with no gap, keep satisfied and protect; lower P , keep informed where interest dominates or monitor where both dimensions are low — refined, always, by attitude.
6. **Re-score on a fixed cadence** and re-allocate as attitudes and coalitions move.

8.6 Worked example: allocating engagement effort on a warehouse-system replacement

Setup. A logistics company is replacing its warehouse management system over 14 months, affecting roughly 600 staff. The delivery leader, change manager and sponsor can jointly commit **190 hours per quarter** to deliberate engagement. Six stakeholder groups are scored.

Stakeholder group	I	S	P = I × S	A	T	G	W = P × G
Executive sponsor board	5	3	15	+1	+2	1	15
Warehouse operations managers	4	5	20	−1	+1	2	40
Warehouse shift staff (via representatives)	2	5	10	−2	0	2	20
IT operations	3	4	12	0	+1	1	12
Finance and controlling	3	3	9	+1	+1	0	0
Key logistics customers	4	2	8	0	+1	1	8

Substitution. $\Sigma W = 15 + 40 + 20 + 12 + 0 + 8 = 95$. Hours per group = $190 \times W \div 95 = 2 \times W$.

Result. Operations managers receive $2 \times 40 = 80$ hours; shift staff 40; sponsor board 30; IT operations 24; customers 16; finance 0 from this formula — but professional judgement overrides the arithmetic: a zero-gap group still receives baseline reporting so a supportive attitude does not decay unnoticed.

Interpretation. The largest single investment goes not to the most senior group but to the operations managers, whose combination of high priority (20) and a two-point attitude gap makes them the pivotal coalition: if they move from −1 to +1, they carry the shift staff with them. At the quarterly re-score, suppose they have moved to 0: *c* falls to 1, *w* falls to 20, and roughly 40 hours are released to redeploy — evidence, not intuition, drives the adaptation.

8.7 From analysis to the communication plan

Convert each register row into a plan row that fixes **message, channel, frequency and sender**. For the two highest-weight groups above: operations managers receive an operational-impact briefing (what changes on their shift floor, and what they control), face-to-face, fortnightly, delivered by the operations director — a trusted peer authority — not by the project office; shift staff receive short visual updates through their representatives, monthly, co-presented with a respected supervisor. Sender selection is a design decision of equal weight to message content: the same message from the wrong sender routinely hardens resistance.

8.8 Sustaining and adapting engagement

Evaluate effectiveness with observable evidence — attendance and contribution rates, survey attitude scores, issue-raising behaviour (silence from a high-interest group is a warning, not a success) — and re-run the scoring at least quarterly or after any major event such as re-baselining. Where AI-enabled tools contribute attitude ratings or drafts to this cycle, the disciplines of the following sections apply in full. Stakeholder attitude data is sensitive personal information and must be handled under the confidentiality standards of Chapter 2.

8.9 AI-enabled stakeholder intelligence

Artificial intelligence is changing stakeholder leadership from a periodically refreshed mapping exercise into a continuously informed discipline. AI-enabled tools can propose draft stakeholder maps from organisational data, and analyse large volumes of correspondence, meeting minutes, consultation responses and collaboration-platform activity to surface sentiment trends, emerging concerns and shifts in tone among key groups. Predictive models can flag early indicators of disengagement — declining meeting attendance, slower response times, reduced participation in reviews — before they harden into resistance. Generative tools can draft tailored communications for different audiences, propose engagement approaches for a given stakeholder profile, and summarise long consultation threads into decision-ready briefings. Used well, this augments the leader's situational awareness; it does not replace the judgement, empathy and political acumen that stakeholder leadership demands: trust cannot be delegated to a model.

8.10 What the leader automates — and what the leader validates

The certified Project Management Leader – AI draws a deliberate boundary between automation and validation. Suitable for automation are the mechanical layers of the competency: collation of stakeholder data, first-draft communication tailoring, sentiment and engagement analytics, distribution-list hygiene, and the routine tracking of actions arising from engagement events. Reserved for human validation are the layers where meaning and relationships are at stake: the interpretation of sentiment signals, the classification of any stakeholder's attitude or influence, the wording of sensitive messages, and every decision about who is engaged, when, and by whom. A model may suggest that a sponsor's confidence is declining; only the leader can test that inference in conversation, understand its causes and decide how to respond. No AI-generated communication should reach a stakeholder without review by an accountable professional, and no stakeholder should ever discover that a message concerning them was machine-authored and unowned.

8.11 Model risk, data protection and human accountability

Stakeholder analytics carries distinctive model risk. Sentiment models are trained on general language corpora and routinely misread irony, cultural register, professional politeness and multilingual nuance; a courteous refusal may score as positive while candid, committed challenge scores as negative. Engagement data is partial by construction — it captures written, digital interaction and is silent on corridor conversations, informal coalitions and the views of stakeholders who simply do not write things down. Treating such outputs as objective truth risks systematically misjudging precisely those stakeholders who matter most. There are also serious data-protection and ethical constraints: analysing individuals' communications and behaviour may engage privacy law, employment regulation and workforce-representation requirements, and covert sentiment monitoring can destroy the very

trust engagement is meant to build. The leader ensures that any stakeholder analytics is transparent, proportionate, lawfully based and consistent with organisational policy — and records the provenance and limitations of the data on which conclusions rest.

Accountability for stakeholder relationships cannot be transferred to a tool. The professional who relies on AI-derived stakeholder insight must be able to explain, in plain language, what the tool analysed, what it inferred, how confident that inference is and why the resulting engagement decision was justified. Where an AI-informed judgement affects an individual's standing — for example, labelling a stakeholder a "blocker" — the leader treats the label as a hypothesis to be tested respectfully, never a verdict to be filed. Governance forums should be told when material stakeholder assessments are AI-assisted, and the engagement strategy remains a document owned, signed and defended by the project leader. In the examination, candidates are expected to demonstrate this boundary: knowing what to delegate to the machine, and what may never be.

AI proposes; the professional disposes. AI may map the stakeholder landscape, score sentiment, and draft the message. The professional validates the inference, owns the relationship, signs the communication and answers for the trust the project builds or loses. Insight can be automated; leadership cannot.

8.12 Common pitfalls

- **Static analysis.** Scoring once at initiation and never re-scoring; attitudes and coalitions move, and last quarter's plan quietly becomes wrong.
- **Confusing volume with influence.** The loudest stakeholder is scored highest while quiet, decisive influencers are underweighted.
- **Broadcast mistaken for engagement.** Communications sent is measured; attitude moved is not.
- **Neglecting supporters.** All effort flows to resisters while champions, receiving nothing, drift to neutral.
- **Unrealistic targets.** Setting $\tau = +2$ for every group inflates gaps and spreads effort too thinly to move anyone.
- **Sender mismatch.** Peer-credibility messages issued from the project office rather than a trusted equivalent.
- **Careless register handling.** Candid attitude assessments left accessible to their subjects — or sanitised until analytically useless.
- **Letting the arithmetic decide.** Treating w as an instruction rather than a prompt for judgement, including accepting AI-derived sentiment scores without human validation.

Key points

- A stakeholder is anyone who can affect, is affected by, or perceives themselves to be affected by the project; perception drives behaviour, and the population changes across the life cycle, so analysis is perishable.
- Engagement is two-way and broader than communication; stakeholder leadership — shaping attitudes, coalitions and sponsorship — distinguishes the delivery leader from the administrator.
- Analysis rests on interest, influence and attitude: the power-interest grid classifies, while attitude and the current-to-target gap convert classification into strategy.
- The scoring method $P = I \times S$, $G = \max(0, T - A)$, $W = P \times G$ allocates finite effort in proportion to weight, re-scored on a fixed cadence; the arithmetic prompts judgement, never replaces it.
- The communication plan fixes message, channel, frequency and sender for each group; sender credibility is as consequential as message content.
- AI can collate data, score sentiment, predict disengagement and draft communications, but carries model risk, partial data and privacy obligations; interpretation, attitude classification, sensitive wording and engagement decisions remain human.
- AI proposes; the professional disposes: every AI-derived stakeholder insight must be explainable, validated and owned by an accountable professional.

Practice questions

1. A residents' association believes, incorrectly, that a project will close its access road, and analysis confirms no such impact will occur. According to the definition of a stakeholder, the delivery leader should:

A. Exclude the association from the register, since no real impact exists
B. Record the association but take no action until it escalates formally
C. Engage the association, because perceived impact drives behaviour whether or not it is real
D. Refer the association to the legal team to correct the misunderstanding

Answer: C. The definition explicitly includes those who perceive themselves affected, because perception drives behaviour regardless of the underlying facts.

2. Using the Institute's scoring method, a stakeholder group is rated influence 4, interest 3, current attitude -1 and target attitude $+1$. Its effort weight w is:

A. 12 B. 14 C. 24 D. 48

Answer: C. $P = 4 \times 3 = 12$, $G = (+1) - (-1) = 2$, so $W = 12 \times 2 = 24$.

3. Under the doctrine "AI proposes; the professional disposes", which activity is appropriate to delegate to an AI-enabled tool?

A. Final classification of a sponsor's attitude in the stakeholder register B. Releasing a tailored communication to a resistant group without human review C. First-draft collation of engagement data and sentiment trends for human validation D. Deciding which stakeholders are engaged, when, and by whom

Answer: C. Mechanical collation and first-draft analytics are suitable for automation; attitude classification, message release and engagement decisions require a validating, accountable human.

9. Procurement

Procurement is the discipline through which a project acquires the goods, services and works it cannot, or should not, produce itself. On most projects of any scale the majority of the budget is spent through suppliers, which makes procurement one of the principal channels through which cost, schedule and risk actually flow. A project delivery leader who cannot shape a procurement strategy, run a defensible tender and read a contract's risk allocation is managing only a fraction of the delivery system. This chapter establishes the concepts the examination expects a candidate to command, applies them through step-by-step methods and a worked example, and closes with the governance of AI-assisted sourcing.

Core concepts defined

Procurement is the end-to-end process of obtaining goods, services or works from parties external to the delivering organisation, from identification of need through contract award to contract closure. It is broader than purchasing (the transactional act of ordering and paying) and narrower than commercial management (which also embraces revenue, claims and the wider business relationship).

A **procurement strategy** is the documented, approved approach that answers the structural questions before any supplier is engaged: what will be bought and what retained in-house; how requirements will be packaged into contracts; which sourcing route each package will use; what pricing and risk-allocation model each contract will carry; and how the supply market's capacity, competitiveness and appetite condition these choices. The strategy must be traceable to the project's objectives and risk appetite — maximising price competition on a schedule-critical package may be internally coherent yet wrong for the project.

The **make-or-buy decision** is the structured analysis that precedes strategy: a comparison of performing work internally against acquiring it externally, weighing relative cost, capacity and capability, control over intellectual property and sensitive data, the strategic importance of retaining the competence, and the transfer or retention of risk. Its output — the buy list — defines the scope over which the rest of the procurement process operates.

A **sourcing route** is the mechanism by which suppliers are approached: open competition, restricted (pre-qualified) competition, negotiated procedures, framework agreements with call-offs, or single- and sole-source awards. **Tendering** is the formal cycle of soliciting and receiving offers — typically through a pre-qualification questionnaire, an invitation to tender or request for proposal, and a request for quotation — conducted under published rules. **Supplier evaluation and selection** is the disciplined comparison of offers against pre-declared, weighted criteria, balancing price against quality, capability, financial standing and whole-life cost rather than lowest price alone. The resulting **contract** is the legal instrument that fixes scope,

price mechanism and risk allocation; the choice among fixed-price, cost-reimbursable, time-and-materials and target-cost arrangements is, at root, a decision about which party carries which risk and what behaviour the incentive structure will produce.

Ethical procurement denotes the integrity conditions under which all of the above must be conducted: fairness, transparency, impartiality, equal treatment of bidders, managed conflicts of interest, and the absolute exclusion of bribery, collusion and bid manipulation. **Sustainable procurement** extends the evaluation lens to environmental, social and economic impacts across the supply chain and the asset's life.

The governing frameworks

Procurement sits within a recognised professional and legal architecture. PMI's PMBOK Guide addresses the planning, conduct and control of procurements; the APM Body of Knowledge situates procurement within integrated commercial and contract management. ISO 20400 gives guidance on embedding sustainability in procurement, and ISO 37001 specifies anti-bribery management systems; legislation such as the UK Bribery Act 2010, the OECD Anti-Bribery Convention and modern-slavery disclosure regimes gives ethical obligations statutory force. Public-sector procurement is additionally governed by jurisdictional regimes — the UK's procurement legislation, the EU directives, the UNCITRAL Model Law — whose common principles of transparency, proportionality and non-discrimination this Body of Knowledge treats as good practice for any sector. Standard contract families such as NEC and FIDIC supply tested risk-allocation templates, and the codes of the procurement profession's chartered bodies define the conduct standards the examination assumes. Candidates are examined on principles, not on the clause-level content of any statute or standard form.

How the elements relate

The elements form a deliberate sequence in which each decision constrains the next. Make-or-buy defines what is bought; the strategy decides how the buying is structured; the sourcing route and tender generate offers under conditions that keep competition genuine; evaluation converts offers into a defensible selection; and the contract converts selection into an enforceable risk allocation that the delivery team must then manage. Ethics and sustainability are not a final step but a property of the whole chain: a technically sound evaluation built on a conflicted panel is void of integrity, and integrity failures are rarely recoverable.

Step by step: conducting a make-or-buy analysis

A defensible make-or-buy analysis proceeds in six steps.

1. **Specify the requirement in output terms.** Define what the project must obtain — capability, capacity, quality, timescale — without presuming how it will be produced.
2. **Establish whole-life cost for each option over a common period.** For the make option, include development, ongoing support, infrastructure and management overhead; for the buy option, include acquisition, implementation, recurring charges, contract management and exit costs.
3. **Risk-adjust both options.** Identify the principal risks of each route, estimate expected values (probability multiplied by impact) and add them to the base cost, so the comparison reflects exposure and not merely estimates.
4. **Assess non-financial factors.** Consider strategic fit and core competence, intellectual property, internal capacity and its opportunity cost, speed to delivery, supply-market maturity, and dependency or lock-in.
5. **Test sensitivity.** Vary the key assumptions across plausible ranges. If the ranking reverses within normal estimating uncertainty, the financial case alone is not decisive and the recommendation must rest on the qualitative factors.
6. **Recommend, document and seek approval.** Record the recommended strategy — route to market, contract type and risk allocation — in the procurement management plan and obtain governance sign-off before approaching the market.

Step by step: designing the tender and evaluation

1. Select a sourcing route (open, restricted, framework call-off or negotiated) proportionate to value, risk and market conditions.
2. Fix the evaluation criteria, weightings, scoring scale and price formula **before** the invitation to tender is issued, and publish them to all bidders.
3. Separate mandatory pass/fail gates (financial standing, legal compliance, minimum technical thresholds) from scored criteria.
4. Provide narrative descriptors for each point on the scoring scale so that scores can be explained and audited.
5. Have panel members score independently, then moderate to consensus, recording the rationale for every score.
6. Manage integrity throughout: declared conflicts of interest, a single communication channel, equal information to all bidders, and a complete audit trail.
7. Where AI tools are used to summarise bids or flag anomalies, evaluators must still read, validate and own every score.

Worked example: from make-or-buy to contract award

A programme requires a document management capability for five years.

Make option. Internal build £480,000; support £60,000 per year $\times 5 = £300,000$; base cost £780,000. Risk adjustment: a 30% probability of a 25% development overrun (£120,000) gives an expected value of $0.30 \times £120,000 = \text{£}36,000$. Risk-adjusted total: $£780,000 + £36,000 = \text{£}816,000$.

Buy option. Implementation £350,000; subscription £95,000 per year $\times 5 = £475,000$; base cost £825,000. Risk adjustment: a 10% probability of a £50,000 integration overrun gives $0.10 \times £50,000 = \text{£}5,000$. Risk-adjusted total: $£825,000 + £5,000 = \text{£}830,000$.

The difference of £14,000 is about 1.7% of the make option — well inside estimating tolerance, so the financial case is not decisive. Because the internal team is fully committed to core delivery and the supply market is mature, the recommended strategy is **buy**, and the analysis, including this reasoning, is recorded for governance approval.

The tender attracts two compliant bids with five-year whole-life prices of £825,000 (Supplier A) and £760,000 (Supplier B). Published weightings: technical 40%, whole-life cost 30%, delivery capability 20%, sustainability and social value 10%. The published price formula awards 10 to the lowest price, others pro rata: Supplier A scores $760,000 \div 825,000 \times 10 = 9.21$.

Criterion (weight)	A raw	A weighted	B raw	B weighted
Technical (40%)	8.00	$8.00 \times 0.40 = 3.20$	7.00	2.80
Whole-life cost (30%)	9.21	$9.21 \times 0.30 = 2.76$	10.00	3.00
Delivery capability (20%)	7.00	1.40	8.00	1.60
Sustainability (10%)	6.00	0.60	8.00	0.80
Total		7.96		8.20

Supplier B is recommended for award at 8.20 against 7.96. The moderation record explains each score against the published descriptors, and the unsuccessful bidder receives a structured debrief — the process can withstand challenge because every element was fixed in advance and evidenced throughout.

Common pitfalls

- **Comparing acquisition price rather than whole-life cost**, omitting support, contract management and exit costs from the buy option.
- **Ignoring risk adjustment and optimism bias**, which systematically flatters the make option.

- **Treating small financial margins as decisive** when the difference lies within estimating uncertainty.
- **Setting or altering weightings after bids are opened**, which destroys the defensibility of the award and may be unlawful in regulated regimes.
- **Vague scoring descriptors**, producing scores that evaluators cannot explain under challenge.
- **Untested price formulas**, where an outlier bid distorts relative scores; model the formula before publication.
- **Integrity breaches during a live procurement** — undeclared conflicts, hospitality from bidders, or side-channel communications.
- **Delegating evaluation to AI outputs** — summaries and anomaly flags may inform the panel, but unvalidated machine scoring is indefensible.
- **Overlooking sustainability and lock-in** — awarding on price while ignoring exit barriers, supply-chain conduct and environmental impact.

How AI changes the procurement competency

AI and predictive tooling now touch every stage of the procurement lifecycle. Market-intelligence engines scan supplier landscapes, price indices and capacity signals faster than any category team; generative tools draft specifications, tender schedules and contract clauses in minutes; analytical models flag anomalous pricing and predict supplier delivery risk or financial distress from historical performance data. The competency shifts accordingly, from assembling procurement information to interrogating it: framing the make-or-buy question correctly, testing what the models have assumed, and defending the recommendation before governance boards, auditors and, in regulated contexts, formal challenge.

Sensible automation targets high-volume, rule-bound work in which errors are detectable and reversible: spend classification, supplier data cleansing, compliance checks against mandatory criteria, first drafts of standard tender documentation and first-pass screening of pre-qualification responses. Judgement-bound decisions must remain human: setting the strategy and risk allocation, fixing criteria and weightings before tenders are invited, scoring qualitative bid content, deciding award, and any decision to exclude a supplier. A leader who allows an algorithm to rank bids without independent scrutiny has not accelerated the evaluation; they have abdicated it. The validation disciplines are standard: sample-check automated classifications, require the rationale behind every AI-generated score or red flag, and record in the procurement file which outputs were machine-assisted and how each was verified.

Model risk, data caveats and human accountability

Procurement data is a hostile environment for naive models. Supplier performance histories encode incumbency bias: the suppliers an organisation has used most generate the most data and therefore the most confident predictions, quietly disadvantaging new entrants, smaller firms and diverse suppliers — an outcome that may breach equal-treatment obligations in public procurement. Price benchmarks decay rapidly in volatile markets, so a model trained on pre-shock data will mis-advise on both budget and negotiating posture. Generative drafting tools can produce plausible but legally defective clauses, or import terms from jurisdictions that do not apply. Confidentiality risk is acute: bid material, supplier pricing and evaluation deliberations must never be entered into tools that retain or train on user prompts. The leader should demand the provenance, coverage and refresh cycle of the data behind any predictive score, and treat an unexplainable ranking as inadmissible in an evaluation that must withstand scrutiny.

Fairness and transparency obligations do not relax because software performed the analysis. If an award is challenged, "the model scored it" is not a defence; the named evaluators and the accountable procurement authority own the outcome and must be able to explain it. Governance therefore requires disclosed use of AI where policy or regulation demands it; audit trails linking every automated output to a human reviewer; bias testing before screening tools touch live supplier data; and clear escalation routes when a professional disagrees with a machine recommendation. Candidates should expect examination scenarios that probe exactly this boundary — situations in which the AI output is convenient, confident and wrong, and the professionally defensible action is to overrule it and record why.

AI proposes; the professional disposes. In procurement, AI may scan markets, draft documents, screen submissions and score risk — but the strategy, the criteria, the award and the ethics of the deal belong to an accountable human being. Every machine-assisted judgement in the sourcing file must be explainable, validated and owned by a competent professional, because it is the professional, not the model, who answers to the supplier, the auditor and the tribunal.

Key points

- Procurement is the principal channel through which cost, schedule and risk flow on most projects; the strategy must trace to the project's objectives and risk appetite, not merely to price competition.
- The elements form a constraining sequence — make-or-buy, strategy, sourcing route and tender, evaluation, contract — and ethics and sustainability are properties of the whole chain, not a final step.

- Make-or-buy comparisons use risk-adjusted whole-life cost over a common period; a margin within estimating uncertainty is not decisive, and the recommendation then rests on qualitative factors.
- Evaluation criteria, weightings, scoring descriptors and the price formula are fixed and published before the invitation to tender and never altered after bids are opened.
- Contract type — fixed-price, cost-reimbursable, time-and-materials, target-cost — is fundamentally a decision about risk allocation and the behaviour the incentives will produce.
- AI legitimately accelerates rule-bound work (market analysis, drafting, first-pass screening), but judgement-bound decisions — strategy, criteria, scoring, award, exclusion — remain human and must be explainable, validated and owned.
- Beware incumbency bias in supplier data, decayed price benchmarks, defective generated clauses and confidentiality leakage; an unexplainable ranking is inadmissible.

Practice questions

1. A published price formula awards 10 points to the lowest compliant whole-life price, with other bids scored pro rata. The lowest bid is £760,000 and Supplier A bids £825,000. What is Supplier A's price score (to two decimal places)?

- A. 9.21
- B. 10.86
- C. 9.14
- D. 8.00

Correct answer: A. The pro-rata formula is $\text{lowest price} \div \text{bid price} \times 10 = 760,000 \div 825,000 \times 10 = 9.21$; option B inverts the ratio and the others do not follow the published formula.

2. Midway through scoring, the evaluation panel concludes that the published 30% weighting for whole-life cost favours one bidder and proposes increasing the technical weighting. What is the correct course of action?

- A. Amend the weightings and notify all bidders of the change
- B. Complete the evaluation under the published weightings and record the concern for future procurement strategy
- C. Amend the weightings only if doing so does not change the ranking
- D. Ask the AI screening tool to re-score the bids under the proposed weightings

Correct answer: B. Altering published criteria or weightings after bids are opened destroys the defensibility of the award and may be unlawful in regulated regimes, regardless of notification, effect on ranking, or the tool used.

3. An award decision is formally challenged, and it emerges that a predictive tool ranked the bids and the panel adopted the ranking without independent scoring. Which statement best reflects the Institute's doctrine?

- A. The award stands because the tool applied the published weightings consistently
- B. The award is defensible if the tool's developer certifies the model's accuracy
- C. The evaluation is indefensible because no accountable human validated, explained and owned the scores
- D. The evaluation is defensible provided the use of AI was disclosed to bidders

Correct answer: C. Under "AI proposes; the professional disposes", disclosure, consistency and third-party assurance cannot substitute for named evaluators who validate and own every score — "the model scored it" is not a defence.

10. Contracts and Commercial Delivery

This chapter follows the commercial arc of a project from contract selection through administration, variation, claim and dispute to close-out. Its four disciplines form one causal chain: the contract type chosen at award determines what must be administered; the quality of administration and records determines whether variations and claims can be managed on evidence rather than assertion; and the discipline of all three determines whether commercial close-out is a confirmation or a negotiation.

The contract as the instrument of delivery

A **contract** is a legally binding agreement under which one party (the **supplier**, contractor or service provider) undertakes to deliver defined scope, goods or services to another (the **client**, employer or buyer) in exchange for consideration, normally payment. In project delivery the contract is far more than a legal formality: it is the instrument that converts intent into enforceable obligation. It fixes what will be delivered, by when, to what standard, for how much and under which **payment mechanism**, and — critically — it allocates commercial **risk** between the parties and prescribes the remedies available when performance departs from promise. **Commercial delivery**, as this chapter uses the term, is the discipline of selecting, administering and closing contracts so that the commercial arrangement reinforces the project's delivery objectives rather than working against them. The delivery leader is rarely the drafting lawyer, but is accountable for ensuring that the contract can actually be delivered and that it is managed with discipline once signed.

Contract types, payment mechanisms and risk allocation

Every contract type sits on a spectrum defined by its payment mechanism — the rule that converts work performed into entitlement to payment. At one pole sit **price-based** arrangements (lump-sum or fixed-price contracts, including turnkey and EPC forms), in which the supplier is paid a predetermined price and therefore carries cost, quantity and productivity risk. At the other sit **cost-based** arrangements (cost-reimbursable and time-and-materials contracts), in which the client pays actual cost plus a fee and therefore retains most delivery risk. Between them lie hybrids: **unit-rate and remeasurement** contracts, where rates are fixed but quantities are measured as built; **target-cost** contracts with pain/gain share, which split overruns and underruns to a formula; and **framework agreements** with call-off contracts for repeated or evolving demand. Supporting mechanisms — milestone and stage payments, retention, performance bonds, liquidated damages, key performance indicators and service credits — tune cash flow and incentive within any of these forms.

The governing principle of risk allocation is that risk should sit with the party best able to foresee, control and price it. The single most important variable in applying that principle is **scope maturity**: a stable, well-specified scope can responsibly be priced fixed; an immature or evolving scope priced fixed merely purchases an inflated risk

premium and a pipeline of claims. This is why adaptive and agile delivery tends towards capacity-based or target-cost commercial models, while well-defined construction or supply scope tends towards fixed-price forms. Selecting the mechanism that matches scope maturity is the first examinable competence of this chapter.

Selecting the commercial model: a five-step method

Selection proceeds in five steps. First, assess scope maturity honestly: determine how completely the requirement can be specified today and how stable it will remain through delivery. Second, identify the principal delivery risks and allocate each to the party best placed to manage it — risk transferred to a supplier who cannot control it returns as price premium, claims or failure. Third, shortlist payment mechanisms against that allocation: firm fixed price where scope is mature and stable; target cost with a pain/gain share where scope is broadly defined but likely to evolve; cost-reimbursable or time-and-materials where scope is immature, always paired with strong governance, capped commitments and break points; and unit or output rates where volumes rather than definition are uncertain. Fourth, stress-test the shortlist against scenarios — overrun, major change, supplier distress — and confirm that the supplier's commercial incentive points in the same direction as the client's objectives. Fifth, record the selection and rationale in the commercial strategy and obtain governance approval before tendering.

Governing frameworks and standard forms

The chapter draws on established professional and contractual reference points, described at awareness level and always in the Institute's own words. Recognised project management bodies of knowledge treat procurement and contract management as an integral knowledge area, and ISO 21502 includes the management of suppliers within its project management practices. In engineering and construction, **standard forms of contract** — notably the FIDIC suite and the NEC family, alongside national forms — codify tested allocations of risk; the NEC forms are notable for embedding proactive administration through early-warning and compensation-event mechanisms. ISO 44001 addresses collaborative business relationships, relevant to alliancing and partnering models. Public-sector procurement regimes add rules on competition, transparency and remedies. Candidates are examined on principles and their application, not on the recall of clause numbers.

Administering the contract during delivery

Contract administration is the systematic post-award management of both parties' obligations. It begins with contract start-up: build an obligations and deliverables register from the executed contract, baseline milestones and service levels, and agree communication protocols, notice periods and delegated authorities. Thereafter run a disciplined monthly cycle: verify deliverables against acceptance criteria before

certifying milestones; apply the payment mechanism exactly as drafted, including retention and service-credit provisions; monitor service levels; and maintain contemporaneous records of instructions, notices and early warnings.

Supplier performance management builds on this foundation, measuring delivery against agreed obligations, milestones and service levels, and acting on shortfall through the contract's remedies and through structured performance reviews. The quality of records kept here is not clerical housekeeping: it decides whether later variations, claims and the final account are argued on evidence or on assertion. AI-enabled tools can propose obligation extractions, flag anomalies in cost submissions and draft performance summaries, but every such output must be validated and owned by the contract administrator.

Worked example: settling a target-cost contract

A data-platform build is let as a target-cost contract with an original target cost of £4,000,000, a fixed fee of £400,000 (10 %), a 50/50 pain/gain share on any difference between adjusted target and actual allowable cost, and supplier pain capped at the fee.

During delivery, two compensation events are approved through the contractual change mechanism: £250,000 for a regulatory data-residency change and £150,000 for client-instructed additional interfaces.

Adjusted target cost = £4,000,000 + £250,000 + £150,000 = **£4,400,000**

At completion, audited allowable actual cost is £4,720,000.

Overrun = £4,720,000 – £4,400,000 = **£320,000**

Supplier painshare = 50 % × £320,000 = **£160,000** (well within the £400,000 cap, so no adjustment).

Final price payable = actual cost + fee – painshare = £4,720,000 + £400,000 – £160,000 = **£4,960,000**

The supplier's realised margin is £400,000 – £160,000 = £240,000, or £240,000 ÷ £4,720,000 = **5.1 %** of actual cost against the 10 % planned — a genuine but survivable commercial consequence. Contrast the alternatives: under a firm price of £4,400,000 plus fee, the supplier would have absorbed the full £320,000; under pure cost-reimbursement the client would have paid £5,120,000 and the supplier would have felt no consequence at all. The pain/gain mechanism shares the overrun £160,000/£160,000 — precisely the allocation intended for partially mature scope.

Managing variations, claims and disputes

Because no contract survives delivery unchanged, every well-drafted agreement contains a **variation** (change) mechanism — the authorised route for instructing, pricing and incorporating change to scope, and the commercial counterpart of the

change control discipline established in this Body of Knowledge's governance and planning chapters. Route every change through that single mechanism: capture the proposed variation in writing; confirm entitlement against the relevant clause; quantify cost and time effects before instruction wherever practicable; approve at the correct delegated authority; and update the contract baseline and registers immediately.

A **claim** is an assertion by one party of entitlement to additional time or money arising from events for which the other party bears contractual risk; a **dispute** is a claim or difference that the parties' ordinary processes have failed to resolve. For claims, observe notice periods strictly, insist on contemporaneous records, and evaluate merit dispassionately — entitlement first, quantum second. Where agreement fails, resolution routes form an escalating ladder — commercial negotiation, senior-executive escalation, mediation, adjudication or dispute boards, and only then arbitration or litigation — rising in cost, formality and relational damage. Select the route by weighing value at stake, speed, cost, confidentiality and the importance of preserving the relationship; most project disputes settle fastest and cheapest at the lowest rung both parties take seriously.

Commercial close-out

Commercial close-out completes the cycle. Verify that every obligation in the register is discharged or formally waived; agree the final account, including all variations, claims and fluctuations; release retentions, bonds and guarantees on the contractual dates; document surviving obligations such as warranties, latent-defect liabilities and data-handling duties; record supplier performance for future sourcing decisions; and capture commercial lessons into organisational knowledge before the team disperses. Done on disciplined records, close-out confirms positions already evidenced; done late or on weak records, it becomes a negotiation.

AI in contracts and commercial delivery

AI and predictive tooling now reach every stage of this chapter's arc. At formation, contract-analytics tools can compare a draft against a standard-form library, flag unusual clauses and departures from precedent, and simulate how alternative payment mechanisms would distribute cost outcomes under different scope-maturity scenarios — sharpening, though never making, the risk-allocation decision. During administration, analytics platforms monitor supplier performance continuously — milestone attainment, service-level trends, defect and invoice patterns — and surface early-warning signals of deterioration well before a periodic review would. In variations and claims, document-intelligence tools assemble chronologies from contemporaneous records, cross-reference correspondence against notice provisions, and estimate the likely cost and schedule effect of a proposed change; some offer predictive ranges for claim outcomes based on historical settlements. At close-out, extraction tools accelerate reconciliation of the final account against instructed

variations. What changes is the speed and reach of commercial intelligence; what does not change is the locus of judgement, because the contract binds named parties and named certifiers, not models.

The delivery leader draws the automation boundary deliberately. Suitable for automation, with sampling checks: first-pass clause comparison and deviation flagging; obligation, milestone and service-level tracking; arithmetic verification of payment applications; assembly of variation and claims chronologies; close-out checklist reconciliation. Never delegable: selection of contract type and payment mechanism against actual scope maturity; whether a flagged clause matters commercially in this deal; whether a deteriorating supplier trend warrants relational intervention or contractual remedy; entitlement and quantum on any claim; the choice of dispute route; and certification of payment or of the final account. The leader validates an AI output as they would a junior analyst's work: what evidence supports this, what was assumed, and what would make it wrong?

Three caveats govern use of these tools. First, language models can hallucinate — inventing clause numbers, misquoting terms, or offering plausible but incorrect readings of entitlement — so every citation must be verified against the executed document itself. Second, predictive models trained on historical settlements embed the biases and jurisdictions of that history, and supplier-performance analytics inherit the quality of the site records feeding them: incomplete diaries produce confident but misleading signals. Third, contract terms, claims positions and settlement strategy are commercially sensitive and may be legally privileged — confirm a tool's data-handling and retention terms before any contract document is processed, and recognise that AI-generated analysis relied upon in a dispute may become disclosable and must therefore be explainable.

Contractual powers — instructing a variation, serving or responding to a notice, certifying payment, agreeing a final account — vest in named individuals, and no AI output transfers or dilutes them. An AI-drafted notice served out of time is still out of time; a certificate issued on unvalidated analytics is still the certifier's certificate. The examination expects candidates to demonstrate not only commercial competence but the governance of these tools: every AI output that informs a commercial position must be explainable, validated against the contract and the records, and owned by a competent, accountable professional.

AI proposes; the professional disposes. A model may draft the comparison, flag the slipping service level, assemble the claims file and predict the settlement range — but the contract knows nothing of models. Only a competent, named professional allocates risk, certifies payment, serves notice, agrees the variation and settles the account — and that professional answers for every AI-informed decision as fully as for their own analysis.

Common pitfalls

- Transferring risk the supplier cannot price or control, then paying for it anyway through contingency, claims or insolvency.
- Selecting a fixed price for immature scope and converting delivery into a variations negotiation from month one.
- Certifying milestones on effort or optimism rather than evidenced acceptance criteria.
- Issuing instructions informally, then discovering they constitute uncontrolled variations with cost and time entitlement.
- Missing contractual notice periods and forfeiting otherwise valid entitlements.
- Allowing the obligations register and contract baseline to drift from the change log.
- Accepting AI-generated contract summaries or claim valuations without human validation of clause references and arithmetic.
- Escalating prematurely to formal proceedings and destroying a relationship the programme still depends upon.
- Treating close-out as administrative housekeeping and releasing securities before obligations are verifiably discharged.

Key points

- The contract converts intent into enforceable obligation: it fixes scope, time, standard, price and payment mechanism, and allocates commercial risk between the parties.
- Payment mechanisms sit on a spectrum from price-based (supplier carries cost risk) through unit-rate, target-cost and framework hybrids to cost-based (client retains risk); risk belongs with the party best able to foresee, control and price it.
- Scope maturity is the decisive selection variable: mature, stable scope suits fixed price; evolving scope suits target-cost or capacity-based models; fixed-pricing immature scope buys a risk premium and a pipeline of claims.
- Disciplined administration — an obligations register, evidenced certification and contemporaneous records — is what allows variations and claims to be resolved on evidence rather than assertion.
- Route all change through the single contractual variation mechanism; on claims, observe notice periods strictly and establish entitlement before quantum; escalate disputes deliberately up the ladder from negotiation to litigation.
- Target-cost pain/gain mechanisms share overrun and underrun to a formula, keeping both parties' incentives aligned where scope is partially mature.
- AI proposes; the professional disposes: first-pass extraction, tracking and chronology assembly may be automated with sampling checks, but risk

allocation, entitlement, dispute-route choice and every act of certification remain with a named, accountable human.

Practice questions

1. A client must contract for a system integration whose requirement is broadly defined but certain to evolve as regulation develops. Which commercial model best matches the principle that risk should sit with the party best able to control it?

- A. Firm fixed price with liquidated damages
- B. Target cost with a 50/50 pain/gain share
- C. Uncapped cost-reimbursable with no break points
- D. Remeasurement against a fixed bill of quantities

Correct answer: B. Target cost with pain/gain shares the cost consequences of partially mature, evolving scope, whereas A transfers unpriceable risk, C leaves the supplier with no consequence, and D addresses quantity uncertainty rather than definitional uncertainty.

2. A target-cost contract has an original target of £2,000,000, a fixed fee of £200,000, a 50/50 pain/gain share and one approved compensation event of £100,000. Audited allowable actual cost is £2,300,000. What is the final price payable?

- A. £2,300,000
- B. £2,400,000
- C. £2,450,000
- D. £2,500,000

Correct answer: B. Adjusted target is £2,100,000, so the £200,000 overrun yields a £100,000 painshare, and the final price is $£2,300,000 + £200,000 - £100,000 = £2,400,000$; the distractors omit the painshare, the fee, or misapply the share percentage.

3. Under the Institute's AI doctrine, which of the following may never be delegated to an AI tool, however capable?

- A. First-pass comparison of a draft against a standard-form library
- B. Assembly of a claims chronology from contemporaneous records
- C. Certification of a payment application
- D. Arithmetic verification of a cost submission

Correct answer: C. Certification is a contractual power vested in a named individual and cannot be transferred to a model, whereas A, B and D are first-pass or mechanical tasks suitable for automation with human sampling checks and ownership of the output.

11. Integrated Cost, Schedule and Risk Management

Cost, schedule and risk are commonly taught as three disciplines; on a real project they are one system. The baseline is a prediction of cost and time; risk is the reason that prediction may prove wrong; earned value management measures how the prediction is faring; and integrated forecasting reconciles all three into a defensible estimate of where the project will finish. This chapter treats them as that single system. It fixes the vocabulary, sets out the integrated method step by step, works a full example, and examines how AI-enabled tooling changes — and does not change — the delivery leader's accountabilities.

Performance measurement baselines and the earned value vocabulary

A **baseline** is the approved, version-controlled plan against which performance is measured; it changes only through formal change control. The **performance measurement baseline (PMB)** is the time-phased budget for the authorised scope, built up through the work breakdown structure into **control accounts** — the management points at which scope, budget, actual cost and schedule integrate. The total of the PMB is the **Budget at Completion (BAC)**.

Earned value management (EVM) rests on three measures, all expressed in currency at a common data date: **Planned Value (PV)**, the budget for the work scheduled to date; **Earned Value (EV)**, the budget for the work actually performed; and **Actual Cost (AC)**, what that performed work actually cost. From these follow the variances ($CV = EV - AC$; $SV = EV - PV$), the performance indices ($CPI = EV \div AC$; $SPI = EV \div PV$), and the forecasting measures — **Estimate at Completion (EAC)**, **Estimate to Complete (ETC = EAC - AC)**, **Variance at Completion (VAC = BAC - EAC)** and the **To-Complete Performance Index (TCPI)**, the cost efficiency required on remaining work to meet a stated target. The foundational point is that EVM converts physical progress into money, so that scope, schedule and cost become directly comparable.

The vocabulary of risk

A **risk** is an uncertain event or condition that, if it occurs, has a positive or negative effect on objectives; ISO 31000 frames this generally as the effect of uncertainty on objectives. Negative risks are **threats**; positive risks are **opportunities**, and a competent delivery leader manages both. Individual risks are distinct from **overall project risk**, the combined uncertainty of the whole endeavour. The **risk register** records each identified risk with its cause, probability, impact, priority, a single named **risk owner** and an agreed response — avoid, transfer, mitigate (reduce) or accept for

threats; exploit, share, enhance or accept for opportunities. Risk remaining after response is **residual risk**; a response that creates a new risk creates a **secondary risk**.

Two reserves fund uncertainty, and the distinction is examinable. **Contingency reserve** funds identified residual risk: it sits within the project budget, is drawn down under the project's own authority strictly against realised risks, and every drawdown is logged. **Management reserve** covers genuinely unidentified work: it sits outside the PMB, is controlled by the sponsor or governance tier, and is never used to disguise overspend. Frameworks differ on whether contingency sits inside or alongside the PMB; the examinable principle is that reserves are held distinct from work-package budgets, controlled at the right level of authority, and traceable in drawdown.

From deterministic estimates to stated confidence

A single-point estimate of cost or duration is **deterministic**: it carries no statement of how likely it is to be achieved. Quantitative cost risk analysis (QCRA) and quantitative schedule risk analysis (QSRA) replace single points with distributions — typically three-point estimates on activities and cost elements, plus discrete risk events from the register — and combine them, most commonly by Monte Carlo simulation, into a probability distribution of project outcomes. The outputs are cumulative probability curves (S-curves), confidence levels such as **P50** and **P80** (the value with a 50% or 80% probability of not being exceeded), and sensitivity or tornado analyses showing which risks drive the spread. Contingency is then set as the difference between the funded confidence level and the base estimate — a governance decision expressed in probabilistic language, not a percentage plucked from habit.

Governing frameworks and standards

This chapter aligns with the internationally recognised bodies of guidance: the PMBOK Guide and PMI's practice standards for earned value management and risk management; ISO 21502 on project management, ISO 21508 on earned value management and ISO 31000 on risk management; the EIA-748 EVM system guidelines used in government-regulated environments; AACE International's Total Cost Management Framework and recommended practices on risk analysis and contingency determination; and the APM Body of Knowledge. Candidates are examined on the concepts and calculations as defined here, not on the clause structure of any single standard.

The integrated method, step by step

1. **Establish the PMB.** Decompose the scope into a work breakdown structure, assign each work package a budget and an earned value measurement technique, and time-phase the budget to form the planned value curve. Work-package budgets exclude reserves.

2. **Record performance data each period.** At control-account level, capture PV, EV and AC as at a single data date.
3. **Calculate and interpret the metrics.** Compute CV, SV, CPI and SPI, and investigate root causes before reporting: a number without an explanation is data, not analysis.
4. **Maintain the risk register.** Identify risks systematically, assess probability and impact, prioritise, assign a single named owner to each risk, and agree responses. Fund agreed responses within the baseline through change control, and quantify the residual exposure that remains.
5. **Run quantitative risk analysis on the remaining work.** Replace single-point estimates of remaining cost and duration with three-point ranges, add risk events from the register, model correlation between related items, and simulate to produce probability distributions for cost and completion date.
6. **Set contingency at a stated confidence level.** Contingency equals the value at the percentile mandated by governance minus the deterministic estimate. Draw it down only against realised risks, with an auditable log.
7. **Forecast and report on an integrated basis.** Reconcile the earned-value EAC with the QRA percentiles, and present variances, forecast, contingency status and top risks as one coherent picture — because a healthy CPI alongside a deteriorating risk register is not a healthy project.

The loop then closes: execution generates the next period's EV, AC and risk data, and the analysis repeats against a baseline that changes only through change control.

Worked example: a data platform programme at month six

A twelve-month data platform programme has a budget at completion (BAC) of £4,800,000, time-phased evenly for simplicity. At the month-six data date:

- PV = £2,400,000 (work scheduled to date)
- EV = £2,100,000 (budgeted value of work actually performed)
- AC = £2,300,000 (actual cost incurred)

Variances and indices.

- $CV = EV - AC = £2,100,000 - £2,300,000 = -£200,000$ (over cost)
- $SV = EV - PV = £2,100,000 - £2,400,000 = -£300,000$ (behind schedule)
- $CPI = 2,100,000 \div 2,300,000 = 0.913$ (£0.91 earned per £1 spent)
- $SPI = 2,100,000 \div 2,400,000 = 0.875$ (progressing at 87.5% of the planned rate)

Forecasts. Assuming current cost efficiency persists:

- $EAC = BAC \div CPI = 4,800,000 \div 0.913 = £5,257,000$ (rounded)
- $ETC = EAC - AC = 5,257,000 - 2,300,000 = £2,957,000$

- $VAC = BAC - EAC = -\text{£}457,000$

If schedule pressure is also expected to drive cost (for example, through extended time-related overheads), the combined-index formula gives $EAC = AC + (BAC - EV) \div (CPI \times SPI) = 2,300,000 + 2,700,000 \div 0.799 = \text{£}5,679,000$. The credible forecast range is therefore roughly £5.26m to £5.68m.

Risk and contingency. The register's top residual risk — migration rework, probability 40%, cost impact £350,000 — carries an expected monetary value of $0.40 \times \text{£}350,000 = \text{£}140,000$. A Monte Carlo simulation of the remaining work, combining three-point estimates with the register's risk events, yields a cost-to-complete distribution with $P50 = \text{£}3,120,000$ and $P80 = \text{£}3,410,000$ against the deterministic ETC of £2,957,000. With governance requiring 80% confidence, cost contingency is set at $3,410,000 - 2,957,000 = \text{£}453,000$, giving a funded completion position of $2,300,000 + 3,410,000 = \text{£}5,710,000$ — consistent with the upper end of the EVM forecast range. The schedule simulation shows a P80 completion at month fourteen, so two months of schedule contingency are proposed. The practitioner reports the variances with causes, recommends the P80 funding position to the board, and initiates change control rather than quietly absorbing the overrun.

Common pitfalls

- **Confusing spend with progress.** AC or elapsed time is not EV; earned value must derive from the measurement technique assigned to each work package.
- **Trusting SPI late in the project.** SPI converges on 1.0 as work completes, however late; corroborate with critical path analysis of the schedule.
- **Anchoring the forecast to the budget.** Persistent CPI below 1.0 rarely self-corrects; "it will come back" is hope, not forecasting.
- **Unowned or uncosted risks.** A register without named owners, response actions and dates is a list of worries, not a management instrument.
- **Ignoring correlation in QRA.** Independent distributions produce unrealistically narrow ranges and understate the contingency required.
- **Treating contingency as a scope fund.** Drawdown must trace to realised risks, never to unapproved scope growth.
- **Rebaselining to erase variance.** Baselines change only through change control, with history preserved.
- **Accepting AI outputs unvalidated.** Machine-generated forecasts and risk scores enter reports only after a competent human has tested, understood and taken ownership of them.

How AI changes integrated performance management

Integrated cost, schedule and risk management is the most data-rich competency in the delivery leader's portfolio, and therefore the one most visibly reshaped by predictive tooling. Machine-learning models now generate estimates at completion by pattern-matching a project's performance trajectory against historical portfolios, rather than relying solely on index-based formulae. Anomaly detection flags implausible progress claims, out-of-sequence schedule logic and misbooked actual costs before period close. Natural-language processing mines risk registers, lessons-learned repositories, contracts and site reports to propose candidate risks the team has not yet articulated. Simulation engines run quantitative risk analysis continuously, so confidence levels and contingency adequacy can be re-examined at every reporting cycle instead of only at stage gates. The net effect is a shift from periodic, backward-looking variance reporting towards continuous, forward-looking inference — provided the leader governs what the machinery produces.

What the leader automates — and what the leader validates

The working rule is that computation is automated and judgement is validated. A competent leader will routinely automate data assembly and cleansing, the arithmetic of earned value, threshold-based variance flagging, first-pass risk identification prompts, Monte Carlo execution and the drafting of performance narratives. What the leader must personally validate is different in kind: the earning rules and progress claims behind every earned value figure; the completeness, ownership and response quality of the risk register; the probability distributions and correlation assumptions inside any quantitative model; and every AI-generated EAC before it reaches a governance forum. The discipline the examination expects is reproducibility — the candidate who cannot hand-calculate a CPI, or explain why a P80 sits where it does, cannot meaningfully validate a model that claims to have done it for them.

AI forecasting also carries specific, examinable failure modes. Models trained on historical projects inherit the optimism bias, scope patterns and delivery context of that history; applied to a materially different project, they can be confidently wrong. Earned value inputs are only as reliable as the progress measurement behind them, and automation amplifies poor data faster than it amplifies good data. Simulation tools that default to independence between risks understate the spread of outcomes, producing contingency figures that look precise and are systematically thin. Machine-generated forecasts quoted to spurious precision imply a confidence the underlying data cannot support. Finally, models drift as the project or portfolio changes character, so the leader must insist on periodic revalidation against realised outcomes.

Human accountability

Institute doctrine is explicit. Every AI output that enters a performance measurement baseline, a forecast or a report to a sponsor must have a named, competent human owner who can explain and defend it. Setting contingency at a stated confidence level is a management decision informed by analysis, never a number transcribed from a model. Drawdown of contingency, and any change to the baseline, passes through formal change control regardless of what a predictive tool recommends. Assurance functions should expect an audit trail showing what the model proposed, what the professional accepted, amended or rejected, and why. Any output that cannot be explained — its inputs, assumptions and sensitivity — must not be used to set a baseline, a forecast or a contingency figure. Candidates are examined on precisely this posture: interpreting AI-assisted outputs, challenging their assumptions, and owning the resulting decision.

AI proposes; the professional disposes. A model may propose an estimate at completion, a risk mined from project records, or a P80 contingency drawn from ten thousand simulations. The delivery leader disposes: verifying the data behind the earned value, interrogating the distributions and correlations, deciding the confidence level at which contingency is set, and signing the forecast that goes to the sponsor. No algorithm holds the baseline; the accountable professional does.

Key points

- Cost, schedule and risk form one system: the baseline predicts, risk explains why the prediction may fail, EVM measures, and integrated forecasting reconciles all three.
- EVM converts physical progress into money: $CV = EV - AC$, $SV = EV - PV$, $CPI = EV \div AC$, $SPI = EV \div PV$, and $EAC = BAC \div CPI$ when current cost efficiency is expected to persist.
- Contingency reserve funds identified residual risk under project authority; management reserve funds the unidentified and sits with the sponsor, outside the PMB.
- Quantitative risk analysis replaces deterministic estimates with distributions; contingency is set at a governance-mandated confidence level (for example P80) minus the base estimate.
- SPI loses meaning late in a project and must be corroborated with critical path analysis; persistent CPI shortfalls rarely self-correct.
- Correlation matters in simulation: assuming independence between risks narrows the spread and understates contingency.

- AI proposes; the professional disposes: every machine-generated forecast, risk or contingency figure must be explainable, validated against project evidence and owned by a competent human before it informs a decision.

Practice questions

1. At its data date a project reports BAC £2,000,000, PV £900,000, EV £800,000 and AC £1,000,000. Assuming current cost efficiency persists, what is the estimate at completion?

A. £2,000,000 B. £2,222,000 C. £2,250,000 D. £2,500,000

Answer: D. $CPI = 800,000 \div 1,000,000 = 0.80$, so $EAC = BAC \div CPI = 2,000,000 \div 0.80 = £2,500,000$; option C wrongly divides by SPI, and option A anchors to the budget.

2. A governance board mandates that cost contingency be funded at P80. This means the base estimate plus contingency gives a total cost with:

A. an 80% probability of not being exceeded B. an 80% probability of being exceeded
C. a value equal to 80% of the worst simulated outcome D. a guaranteed 20% margin above the P50

Answer: A. P80 is the value on the cumulative distribution with an 80% probability that the actual outcome will not exceed it; the other options misread the percentile.

3. An AI tool produces an EAC materially lower than the index-based forecast, but cannot show which inputs or assumptions drove its prediction. The delivery leader should:

A. adopt the AI forecast, since it draws on more historical data B. average the two forecasts to balance the evidence C. report the index-based EAC and exclude the AI figure until it can be explained and validated D. report both figures and let the sponsor choose

Answer: C. Under Institute doctrine an output that cannot be explained and validated must not inform a forecast — AI proposes; the professional disposes.

12. Predictive Delivery

The nature of the predictive approach

A predictive delivery approach — sometimes called plan-driven or, informally, waterfall — is one in which the scope, schedule and cost of a project are defined and approved substantially before execution begins, and in which delivery performance is thereafter measured against those approved definitions. The defining commitment is not sequence but predictability: the sponsoring organisation invests early effort in specification, estimation and design precisely because it expects the requirement to remain stable enough for that early plan to stay meaningful.

Three terms anchor everything that follows. A **baseline** is the approved, version-controlled statement of scope, schedule or cost against which performance is measured; it changes only through formal change control, never by silent revision. A **stage** (or phase) is a discrete, bounded portion of the project, planned in detail one stage at a time within an outline whole-project plan. A **stage gate** is a formal decision point at the boundary between stages at which a governance body reviews evidence of performance and viability before authorising further work. Predictive delivery is therefore best understood as a control system: baselines supply the reference, stages bound the commitment, and gates supply the decision mechanism.

Frameworks and standards context

The candidate should be able to locate these ideas within the recognised professional canon. The PMBOK® Guide describes the predictive life cycle and its planning, executing and monitoring-and-controlling work, including the scheduling techniques examined in this chapter. PRINCE2® contributes the most fully developed treatment of stage-based governance: its principles of manage by stages and manage by exception formalise stage-limited authorisation and tolerances — permitted deviations from baseline within which the project manager may act without escalation. ISO 21502 provides international-standard guidance on governance and phase-based direction, and the APM Body of Knowledge treats deployment approaches, scheduling and gated life cycles in comparable terms. The Institute's examination does not test allegiance to any single framework; it tests the candidate's command of the common conceptual machinery these frameworks share.

Suitability: when prediction is a rational bet

Because the predictive approach spends heavily before it delivers, its suitability turns on whether early definition is achievable and valuable. It is achievable where requirements are stable and specifiable, the solution technology is well understood, and comparable work provides a reliable estimating basis. It is valuable where the cost of late change is high (physical construction, embedded systems), where regulation or contract demands demonstrable, documented control (safety cases, fixed-price

procurement, licensed industries), or where multiple parties must coordinate against a common integrated plan. Conversely, where requirements are emergent, feedback is cheap and the product can be delivered incrementally, an adaptive or hybrid approach will usually dominate; Chapter 13 develops that comparison.

In applied form, the screen tests three criteria: requirement stability (can scope be specified substantially in full before build?), risk profile (is rework intolerably expensive or unsafe?) and regulatory context (must evidence of a complete, approved design exist before execution?). The judgement and its rationale are recorded and revisited at every gate. Suitability assessment is thus a risk judgement, not a preference: the candidate is examined on weighing these factors against one another for a given scenario.

The predictive method, step by step

Applied predictive delivery is a repeatable sequence, and the examination expects the candidate to execute every step, not merely name them.

1. **Screen for suitability**, as above, and record the rationale.
2. **Structure the work**. Decompose scope into a work breakdown structure, then group the work into management stages, each ending in a gate with defined entry evidence and exit criteria.
3. **Build the network**. Identify precedence relationships, estimate durations, and run the forward pass (earliest start ES, earliest finish EF) and backward pass (latest start LS, latest finish LF). The critical path is the longest path; total float is $LS - ES$.
4. **Baseline**. Freeze scope, schedule and cost as the approved baseline, version-controlled and altered only through formal change control.
5. **Measure, forecast, decide**. Each period, compare performance against baseline, forecast completion, and bring the evidence to the gate for a proceed, amend or stop decision.

The scheduling engine: critical path and critical chain

Within a stage, the schedule baseline rests on network logic. A **network diagram** (in modern practice, activity-on-node precedence form) represents activities and their dependencies — finish-to-start and its variants, with leads and lags. The **forward pass** yields each activity's earliest dates and hence the minimum project duration; the **backward pass** yields latest dates; the difference yields **float**, total float being the delay an activity can absorb without delaying project completion. The **critical path** is the longest path through the network — the sequence of activities with zero total float — and it, not the activity list, determines project duration.

Schedule compression operates on this logic: **crashing** adds resource to critical activities at increased cost, while **fast-tracking** overlaps activities at increased risk,

and both can shift which path is critical. **Critical chain** scheduling, derived from the theory of constraints, extends the model by recognising resource contention as well as logical dependency: it schedules the resource-constrained longest chain, strips padding from individual estimates, and concentrates protection in a project buffer and feeding buffers whose consumption is then the primary progress signal.

Worked example: compressing a regulated laboratory upgrade

Setup. A pharmaceutical company upgrades a quality-control laboratory. Requirements are stable and validation is mandatory, so a predictive approach is confirmed. Six activities, durations in weeks:

Activity	Description	Predecessors	Duration
A	Requirements and regulatory scoping	—	4
B	Detailed design	A	6
C	Long-lead equipment procurement	A	10
D	Build and configuration	B	8
E	Installation and commissioning	C, D	5
F	Validation and handover	E	4

Passes. The forward pass gives A: ES 0, EF 4; B: 4–10; C: 4–14; D: 10–18; E starts at the later of its predecessors' finishes, $\max(14, 18) = 18$, so E: 18–23; F: 23–27. The backward pass from week 27 returns:

Activity	ES	EF	LS	LF	Total float
A	0	4	0	4	0
B	4	10	4	10	0
C	4	14	8	18	4
D	10	18	10	18	0
E	18	23	18	23	0
F	23	27	23	27	0

Two paths exist: A–B–D–E–F = $4 + 6 + 8 + 5 + 4 = \mathbf{27 \text{ weeks}}$ (critical, zero float) and A–C–E–F = $4 + 10 + 5 + 4 = 23$ weeks, giving C total float of $8 - 4 = \mathbf{4 \text{ weeks}}$.

Compression. The sponsor requires 25 weeks to meet a site shutdown window. Crash options: B by 1 week at USD 14,000; D by up to 2 weeks at USD 9,000 per week; C by 2 weeks at USD 6,000 per week. C is not on the critical path — crashing it buys nothing. The cheapest critical-path option is D: crashing $8 \rightarrow 6$ weeks costs $2 \times 9,000 = \mathbf{USD 18,000}$ and yields $27 - 2 = \mathbf{25 \text{ weeks}}$. Always recheck for path shift: the

secondary path remains 23 weeks, still 2 weeks shorter, so the critical path is unchanged — but C's float falls from 4 to 2 weeks and must be re-flagged as a schedule risk.

Interpretation. The professional brings this to the design gate as an **amend** recommendation: rebaseline to 25 weeks, authorise USD 18,000 of crash cost, and record the reduced float as a new risk. The fast-track alternative — overlapping design and build — was rejected because rework risk in a validated environment outweighs the saving.

Measurement, forecasting and the gate

Progress is measured objectively against baseline — through milestone completion, physical per cent complete and earned-value measures — and converted into forecasts of completion date and cost. At each stage boundary the delivery leader presents evidence, not assertion: cumulative progress against planned value, forecast completion date and cost, risk exposure, and the health of the business case. That evidence is what makes a gate a genuine decision rather than a ceremony: the board confronts a choice on demonstrated performance and forecast viability, not intention. Three outcomes are open — **proceed** (performance and business case sound), **amend** (rebaseline under change control, as in the worked example) or **stop** (the business case no longer holds — a legitimate control outcome, not a failure of nerve). A gate without a genuinely available stop option is ceremony, not control.

AI in predictive delivery: automate the mechanics, validate the meaning

Predictive delivery is the most heavily instrumented of the delivery approaches, and therefore the one in which AI tooling has the most to offer — and the most to distort. Modern tools can draft a first-pass logic network from a scope statement, run schedule-quality diagnostics, recalculate the critical path and float continuously as progress is recorded, generate probabilistic completion forecasts, enumerate compression scenarios with their cost-time trade-offs, and assemble draft evidence packs for a stage gate. None of this changes the discipline itself; it changes its economics. Analyses that once justified a monthly reporting cycle can now run daily, and the delivery leader's competency shifts accordingly: from producing the analysis to interrogating it.

A defensible division of labour follows. **Automate the mechanics:** the forward and backward pass, float calculation, schedule-health checks (dangling activities, missing predecessors, excessive constraints and lags), variance arithmetic, first-draft reporting narrative, and the enumeration of crash and fast-track options. These are deterministic or near-deterministic tasks in which tooling is faster and less error-prone than manual work. **Validate the meaning:** whether AI-proposed logic reflects the real physical, contractual and resource dependencies of the project rather than statistical habit;

whether a recommended compression option is actually executable given resource, quality and safety constraints; whether the assumptions beneath a probabilistic forecast still hold after change; and — above all — the suitability judgement itself, which no tool can own.

Model risk and accountability at the gate

Forecasting models are only as trustworthy as the progress data feeding them. Subjective percentage-complete assessments, lagging actuals and optimistic status reporting propagate directly into machine-generated forecasts, now delivered with a confidence the underlying data does not warrant. Historical reference data may embed the optimism bias of past estimates, derive from projects that are not genuinely comparable, or rest on samples too small to support the probability statements displayed. Models also drift: a forecast calibrated against one baseline is not automatically valid after a rebaseline, and every approved change requires the leader to confirm that the model's reference frame has moved with the plan. Finally, an opaque output is inadmissible as gate evidence: a forecast whose data sources, assumptions and sensitivities cannot be explained to the project board cannot support a proceed, amend or stop decision, however accurate it may later prove to have been.

Stage gates exist precisely to concentrate accountability, and AI must not be allowed to dilute it. AI-generated analysis may inform a gate, but the decision is made and owned by named, competent humans — the sponsor and project board, advised by the delivery leader. The leader who presents machine-assisted evidence signs for it as their own: they must be able to state what the tool produced, what they changed or rejected, and why. That audit trail is good practice everywhere and an obligation in regulated environments, where the choice of a predictive approach is often driven by exactly this need for demonstrable control. The examination tests this division of responsibility directly: knowing which steps of critical-path analysis, forecasting and gate preparation may be automated is inseparable from knowing which judgements may never be delegated.

AI proposes; the professional disposes. In predictive delivery, AI may draft the network, recompute the path and forecast the finish; the delivery leader validates the logic against reality, owns the baseline and answers for the gate decision. A forecast nobody can explain is not evidence, and a proceed, amend or stop recommendation is never delegated to a model.

Common pitfalls

- **Spending float as if it were free.** Total float belongs to the path, not the activity; consuming it early strips protection from every successor.

- **Crashing non-critical activities.** Money spent off the critical path shortens nothing.
- **Compressing without rechecking the network.** Near-critical paths can become critical; recompute both passes after every compression step.
- **Confusing crashing with fast-tracking.** Crashing trades cost for time; fast-tracking trades risk for time. State which trade is being made.
- **Baseline drift.** Absorbing scope change informally destroys the meaning of every variance reported afterwards.
- **Gate theatre.** Gates that always return "proceed" measure optimism, not performance.
- **Unvalidated AI schedules.** A machine-generated network with plausible but wrong logic fails exactly when it matters; the planner who signs it owns it.

Key points

- Predictive delivery is a control system: baselines supply the reference, stages bound the commitment, and gates supply the decision mechanism. Baselines change only through formal change control.
- Suitability is a risk judgement, not a preference: stable, specifiable requirements, expensive or unsafe late change, and regulatory demand for demonstrable control all favour prediction.
- The critical path — the longest path through the network, with zero total float — determines project duration; total float ($LS - ES$) belongs to the path, not to individual activities.
- Crashing trades cost for time; fast-tracking trades risk for time. After every compression step, recompute both passes, because the critical path can shift.
- Critical chain scheduling adds resource dependency to logical dependency and uses buffer consumption as the primary progress signal.
- A gate is a genuine decision — proceed, amend or stop on evidence — and stop is a legitimate control outcome; a gate without a real stop option is ceremony.
- AI proposes; the professional disposes: automate the deterministic mechanics, validate the meaning, and never delegate gate decisions, baseline ownership or the suitability judgement to a tool.

Practice questions

1. In a precedence network, an activity has ES 6, EF 12, LS 9 and LF 15. What is its total float?

A. 0 weeks B. 3 weeks C. 6 weeks D. 9 weeks

Answer: B. Total float = $LS - ES$ (equivalently $LF - EF$) = $9 - 6 = 3$ weeks.

2. To recover two weeks of schedule, a project manager overlaps detailed design with build rather than adding resource to build. Which trade-off has been chosen?

A. Cost for time, by crashing B. Risk for time, by fast-tracking C. Scope for time, by de-scoping D. Float for time, by resource levelling

Answer: B. Fast-tracking overlaps logically sequential activities and accepts increased rework risk; adding resource at extra cost would have been crashing.

3. Under the Institute's doctrine on AI-enabled planning, which of the following may properly be delegated to a scheduling tool?

A. The proceed, amend or stop decision at a stage gate B. Ownership of the schedule baseline after a rebaseline C. Computation of the forward and backward passes and float D. The judgement that the project remains suitable for a predictive approach

Answer: C. Deterministic mechanics such as network calculation may be automated; gate decisions, baseline ownership and suitability judgements must remain with accountable humans — AI proposes, the professional disposes.

13. Agile Delivery

This chapter defines agile delivery, situates it within professional standards, sets out the practice sequence from suitability decision to governance forecast, and examines how AI-enabled tooling changes — and does not change — the delivery leader's accountabilities. Its sections form one control loop: suitability analysis determines whether, and how far, agile applies; backlog and timebox practice generates increments and evidence; roles and cadence institutionalise inspection and adaptation; and metrics turn the accumulated evidence into forecasts a governance body can rely on.

13.1 Core definitions and the empirical basis

Agile delivery, as this Body of Knowledge uses the term, is a family of **iterative and incremental** approaches in which requirements and solutions evolve through short, structured feedback cycles between an empowered delivery team and engaged stakeholders. An **iteration** is a fixed-duration cycle of planning, building and reviewing work; an **increment** is the usable, verified addition to the product that each cycle produces.

Agile is not an absence of discipline but a different mechanism of control. Instead of measuring conformance to a detailed upfront plan, the leader inspects working output frequently and adapts scope, priority and process on the evidence. This is **empirical process control**: because complex work cannot be fully specified in advance, control is achieved through **transparency** (all parties see the same real state of the work), **inspection** (frequent examination of output and progress) and **adaptation** (prompt adjustment when deviation appears). The project remains fully accountable for its business case, expenditure and benefits; only the evidence of progress changes.

13.2 Frameworks and standards context

The intellectual foundation is the Manifesto for Agile Software Development (2001), which records four value preferences — favouring individuals and interactions, working product, customer collaboration and responsiveness to change, without dismissing the items each is weighed against — supported by twelve principles emphasising early and continuous delivery of value, sustainable pace, technical excellence and regular reflection. The most widely used instantiations are Scrum, defined in the Scrum Guide, and the Kanban method, with Lean thinking supplying the underlying flow and waste-elimination logic.

Contemporary standards, including ISO 21502 and current editions of the PMBOK Guide, recognise predictive, adaptive and hybrid life cycles as equally legitimate. Agile therefore sits within professional project management, not outside it, and the PCI PML-AI examination assesses the candidate's ability to select, apply and govern these approaches, not allegiance to any one of them.

13.3 The delivery-approach decision

The delivery approach is an explicit leadership decision, recorded in the delivery approach statement, discussed with the sponsor and revisited at stage gates rather than treated as a one-off choice. Three determinants dominate.

- **Requirement volatility.** Where requirements are stable, well understood and contractually fixed, predictive planning is efficient; where they are emergent, or expected to change materially as users learn from working increments, iterative discovery reduces the risk of building the wrong thing.
- **Stakeholder availability.** Agile presumes that a person with authority to make value and priority decisions — an empowered product owner — and users who can attend reviews are genuinely available at every iteration boundary. Without that commitment, feedback loops decay into assumption.
- **Organisational readiness.** Funding models, procurement arrangements, audit expectations and culture must tolerate incremental releases and evolving scope within a fixed envelope of time, cost and quality.

The work must also decompose into increments that are individually verifiable and, ideally, valuable. Where the determinants point in different directions, **hybrid delivery** — a predictive governance envelope of business case, stage gates and budget containing agile execution — is the professionally defensible answer.

13.4 Backlogs, refinement and prioritisation

The **product backlog** is the single, ordered list of everything currently known to be needed from the product — the sole source of work, never complete while the product lives. Candidate scope is captured as user-centred backlog items, each with acceptance criteria. **Backlog refinement** is the continuing activity of decomposing, clarifying, estimating and re-ordering items — progressive elaboration under another name. Refinement is rolling: items near the top of the backlog are small, estimated and ready for an iteration; items further down remain deliberately coarse.

Prioritisation is value-driven. Examinable techniques range from categorical schemes — MoSCoW's must have, should have, could have and won't have — to economic ordering by cost of delay relative to duration. Effort is estimated in relative units (story points) calibrated by the team itself; the consequences of that team-local calibration for measurement are developed in section 13.6.

AI tools can usefully draft acceptance criteria, cluster duplicate items and flag dependencies. Consistent with Institute doctrine — AI proposes; the professional disposes — the product owner validates and owns every ordering decision a tool suggests.

13.5 Timeboxes, roles and cadence

A **timebox** fixes duration and quality while scope flexes; its output is an increment meeting an agreed **definition of done**, applied uniformly so that "complete" means the same thing in every report to governance. Iteration length — commonly two weeks — is fixed and never extended: stretching the timebox to "finish" work destroys the empirical basis of every metric derived from it. Each iteration is planned against demonstrated velocity, not aspiration, and coordinated through a short daily event.

Two events close each timebox. In the **review**, stakeholders inspect the working increment against acceptance criteria and give evidence-based feedback on the product. In the **retrospective**, the team inspects its own way of working and commits to at least one specific, concrete improvement. Reviews adapt the what; retrospectives adapt the how.

Three role archetypes recur across frameworks: a product-ownership role accountable for value and priority; a self-organising, cross-functional team accountable for the increment; and a facilitation role accountable for process health and impediment removal. **Cadence** — the fixed rhythm of planning, daily coordination, review and retrospective — converts these events into a reliable governance interface: sponsors receive demonstrated working output on a predictable schedule rather than narrative status reports.

13.6 Metrics, forecasting and scale

Agile measurement is empirical, extrapolating from observed delivery data — never intuition alone.

- **Velocity** is the amount of work, in relative estimation units, that a given team completes per timebox. It is a planning signal for that team, never a productivity comparator between teams.
- **Throughput** counts completed items per unit of time; **cycle time** measures elapsed time from starting an item to finishing it; **work in progress (WIP)** links the two, since in stable flow cycle time rises with WIP.

Forecasts are presented as ranges with stated confidence — optimistic, central and pessimistic outcomes — never a single date, and burn-up charts make both progress and scope change visible to governance bodies. Forecasts are revalidated every iteration as new data arrives.

At scale, multiple teams working from one backlog demand explicit dependency management, consistent definitions of done and aggregated metrics. Published scaling frameworks exist, but practices should scale only as far as coordination genuinely requires.

13.7 Worked example — velocity-based release forecasting

Setup. A team delivering a customer-portal release works in two-week iterations. The refined backlog for the release totals **264 story points**. Velocity over the last six iterations was 28, 24, 31, 26, 30 and 29 points.

Method and arithmetic.

1. Central velocity — the six-iteration mean: $28 + 24 + 31 + 26 + 30 + 29 = 168$;
 $168 \div 6 = \mathbf{28 \text{ points per iteration}}$.
2. Optimistic velocity — mean of the best three iterations: $(31 + 30 + 29) \div 3 = 90 \div 3 = \mathbf{30 \text{ points per iteration}}$.
3. Pessimistic velocity — mean of the worst three iterations: $(24 + 26 + 28) \div 3 = 78 \div 3 = \mathbf{26 \text{ points per iteration}}$.
4. Iterations required (always round up, since a partial iteration is a whole timebox):
 - Central: $264 \div 28 = 9.43 \rightarrow \mathbf{10 \text{ iterations}}$
 - Optimistic: $264 \div 30 = 8.80 \rightarrow \mathbf{9 \text{ iterations}}$
 - Pessimistic: $264 \div 26 = 10.15 \rightarrow \mathbf{11 \text{ iterations}}$
5. Calendar conversion at two weeks per iteration: $9 \times 2 = 18 \text{ weeks}$; $10 \times 2 = 20 \text{ weeks}$; $11 \times 2 = 22 \text{ weeks}$.

Result. The release is forecast at **18-22 weeks, with a central estimate of 20 weeks**.

Interpretation. The delivery leader reports the range, not the single figure, and states the assumptions: stable team composition, no material backlog growth and a consistent definition of done. If refinement typically adds scope, the pessimistic case should be re-run with an explicit growth allowance. Governance receives a defensible confidence statement — "very likely within 22 weeks, plausibly by 18" — that can be revalidated every iteration as new velocity data arrives.

13.8 AI and the agile competency

Agile delivery generates precisely the kind of data on which predictive tooling thrives: a continuous stream of small, dated, comparable records — backlog items, state transitions, cycle times, throughput counts. AI-enabled tooling now sits across the whole iterative cycle. Backlog assistants cluster duplicate items, flag ambiguous acceptance criteria and draft first-pass user stories from stakeholder inputs; flow-analytics engines detect ageing work in progress, growing queues and unstable arrival rates before they surface in a review. Most significantly, probabilistic forecasting has displaced single-point extrapolation: simulation over historical throughput yields a

distribution of completion dates ("85 per cent confidence of delivery by the June release") — a materially more honest instrument for governance communication than a burndown line drawn to a hoped-for endpoint.

What the leader automates, and what the leader validates. The dividing line follows judgement. Sensible candidates for automation include data capture from workflow tools, backlog hygiene (duplicates, stale items, missing fields), flow-metric dashboards, first drafts of iteration summaries and governance reports, and the mechanical generation of probabilistic forecasts. What must remain validated, and never delegated, includes: the suitability assessment, since requirement volatility, stakeholder availability and organisational readiness are contextual judgements no model observes directly; prioritisation, because value ordering is an accountability of the product owner however persuasive an AI-suggested ranking may appear; any commitment communicated to sponsors or steering groups; and the interpretation of metrics, where the leader must guard against Goodhart's law — velocity that is targeted, or optimised by a tool, ceases to measure anything. Retrospective analytics deserve particular caution: mining team discussions for sentiment or themes may be technically feasible, but deploying it without consent corrodes the psychological safety on which honest inspection depends.

Model risk and data caveats. Agile data flatters models and misleads the unwary. Story points are relative, team-local currency: any tool that compares velocity across teams, or treats points as absolute effort, produces analytically invalid output. Historical samples are small — a dozen iterations is thin ground for a confident distribution — and the history is perishable: changes in team composition, a tightened definition of done, scope rebaselining or process improvement all invalidate the baseline a model learned from. That tension is inherent, since improvement is the very purpose of empiricism; the leader must know when a forecast's foundations have moved. Casually maintained boards — batch closures, tickets left open across iterations, inconsistent workflow states — poison cycle-time and throughput data at source. A forecast is only as sound as the board discipline beneath it, and the leader is accountable for both.

Human accountability before governance. When an AI-derived forecast reaches a governance body, accountability is undiminished: the delivery leader must be able to explain what data the forecast rests on, what assumptions it embeds, why its confidence range is credible, and what would invalidate it. "The tool predicted it" is not a defence available to a certified professional. Examination candidates should expect to be tested on interpreting and challenging AI-generated agile metrics and forecasts — including recognising when an output is unreliable — not merely on operating the tooling.

AI proposes; the professional disposes. An AI forecast of a release date is a proposal, not a commitment; a suggested backlog ranking is an input, not a decision. The delivery leader validates the data and assumptions, the product

owner owns the priority order, the team owns its improvement actions — and the human professional owns every number presented to governance.

13.9 Common pitfalls

- **Treating velocity as a productivity target.** Velocity is a planning input local to one team; setting targets or comparing teams inflates estimates and corrupts the forecast.
- **Extending the timebox** to "finish" work, which destroys the empirical basis of every metric.
- **Forecasting from too few data points**, or from iterations affected by holidays and team changes without adjustment.
- **Presenting a single-date forecast** to governance instead of a range with stated assumptions.
- **A weak definition of done**, so increments carry hidden rework and reported progress overstates reality.
- **A proxy product owner** without authority, delaying decisions and hollowing out reviews.
- **Skipping retrospectives under schedule pressure**, forfeiting the improvement loop on which agile depends.
- **Accepting AI-generated backlogs or forecasts unexamined.** Every AI output must be explainable, validated and owned by an accountable professional before it informs a delivery commitment.

Key points

- Agile delivery is a disciplined family of iterative and incremental approaches controlled empirically — through transparency, inspection and adaptation — with accountability for business case, expenditure and benefits unchanged.
- The delivery approach is an explicit, revisited leadership decision driven by requirement volatility, stakeholder availability and organisational readiness; hybrid delivery is a professionally legitimate outcome.
- The ordered, continuously refined product backlog is the sole source of work; timeboxes fix duration and quality while scope flexes, closing with a review that adapts the product and a retrospective that adapts the process.
- Velocity, throughput and cycle time are team-local empirical signals for planning — never inter-team comparators, and never targets.
- Forecasts are ranges with stated assumptions and confidence, revalidated each iteration; partial iterations always round up to whole timeboxes.

- AI accelerates backlog hygiene, flow analytics and probabilistic forecasting, but the suitability decision, prioritisation, commitments to governance and the interpretation of metrics remain human accountabilities.
- AI proposes; the professional disposes: every AI output must be explainable, validated and owned before it reaches a governance body.

Practice questions

1. A team works in two-week iterations. The refined release backlog totals 180 story points and mean velocity over recent iterations is 24 points per iteration. What is the central estimate of release duration?

A. 14 weeks B. 15 weeks C. 16 weeks D. 7.5 weeks

Answer: C. $180 \div 24 = 7.5$, which rounds up to 8 whole timeboxes (a partial iteration is a whole timebox), and $8 \times 2 \text{ weeks} = 16 \text{ weeks}$.

2. A PMO proposes ranking all delivery teams by velocity and setting each team a velocity improvement target. What is the professionally correct response?

A. Agree, because velocity is the standard agile productivity measure B. Agree, provided story points are first normalised across teams C. Decline, because story points are relative, team-local units and a targeted velocity ceases to be a valid measure D. Agree, provided the targets are set by the teams themselves

Answer: C. Velocity is a planning signal local to one team; comparing or targeting it invokes Goodhart's law, inflating estimates and corrupting every forecast built on the data.

3. An AI tool has produced a probabilistic release forecast and a re-ordered product backlog. Which action is consistent with Institute doctrine?

A. Present the forecast to the steering group unchanged, citing the tool as the source B. Adopt the suggested backlog order automatically, since the tool has analysed more data than any individual C. Validate the forecast's data and assumptions before use, refer the suggested ordering to the product owner for decision, and own every number presented to governance D. Reject both outputs, because AI-generated analysis may not inform governance decisions

Answer: C. AI proposes; the professional disposes — AI outputs are inputs to accountable human decisions, neither adopted unexamined nor rejected outright.

14. Hybrid Delivery

14.1 What hybrid delivery is — and what it is not

Every project sits somewhere on a spectrum of development approaches. At one end lies the **predictive approach**, in which scope, schedule and cost are defined early, baselined and protected by formal change control; work proceeds through sequential phases and progress is measured against the approved plan. At the other end lies the **adaptive approach**, in which detailed requirements are deliberately allowed to emerge: work is organised into short, fixed-length iterations, scope is drawn from a continuously prioritised backlog, and the stable quantities are team capacity, cadence and quality rather than a fully specified scope. Between them sit **iterative** techniques (repeated refinement of an evolving product) and **incremental** techniques (delivery of usable functionality in successive slices), which either kind of project may employ.

Hybrid delivery is the deliberate, designed combination of predictive and adaptive approaches within a single project or programme, in which each component of the work is assigned the approach that best fits its characteristics, and the whole is bound together by one coherent governance framework. Three configurations recur in practice: component hybrids, where parallel workstreams run under different approaches — a construction package managed predictively alongside a software product built adaptively; life-cycle hybrids, where successive phases change approach — predictive definition and procurement, adaptive build, predictive commissioning and handover; and governance hybrids, where an adaptive delivery engine operates inside a predictive stage-gate wrapper demanded by the funding or regulatory environment. The defining test in every case is intent. A hybrid model is a designed configuration, justified component by component; an accidental mixture of methods, adopted through drift or fashion, is not hybrid delivery but the absence of a delivery decision.

14.2 Tailoring as the governing principle

Hybrid delivery is the most visible product of **tailoring**: the deliberate adaptation of the delivery approach, governance arrangements and management processes to the context of a specific project. Three principles govern it. First, tailoring is decided against assessable characteristics of the work — requirements stability, technical novelty, cost of change, regulatory or contractual constraint, criticality of safety or compliance, and the feasibility of releasing value incrementally — not against team preference. Second, tailoring is a governance act: the chosen configuration is recorded, justified and approved by the sponsoring authority, so that divergence from organisational method is visible rather than silent. Third, tailoring is revisited: the assignment of approaches is reviewed at stage boundaries and adjusted as uncertainty resolves.

The principal published sources of professional guidance converge on this position: international standards require the management framework to be tailored across the

predictive-to-adaptive range; the major bodies of knowledge treat the development approach as a design choice made per deliverable; the leading structured-governance methods combine stage-based direction and management by exception with the expectation that delivery within a stage may be iterative; and widely used agile frameworks supply the cadences, backlog disciplines and empirical review mechanisms on which adaptive components rely. References to such standards are contextual: the Institute does not claim accreditation by, affiliation with, or endorsement from their publishers, and the PML-AI examination assesses reasoning with these concepts, not recital of any proprietary framework.

14.3 Core vocabulary and the operating model

Four terms carry the analytical weight of this chapter. A **component** is any separable element of the project — a workstream, deliverable, sub-project or phase — to which a development approach can be assigned. A **cadence** is the rhythm at which a component plans, delivers and reviews: gate-to-gate for predictive components, iteration-by-iteration for adaptive ones. A **stage gate** is a governance checkpoint at which the sponsoring authority reviews evidence and authorises, redirects or halts continued investment. A **baseline** is the approved reference against which performance is measured: in predictive components, scope, schedule and cost are baselined in detail; in adaptive components, the baseline is properly set at the level of capacity, cadence, release milestones and outcome targets, while detailed scope remains intentionally fluid beneath it.

These pieces interlock as follows. The hybrid **operating model** maps components to approaches on justified grounds; governance then reconciles the resulting cadences by requiring each gate to consume evidence in the form each component naturally produces — earned value and milestone data from predictive components; throughput, burn-up and increment demonstrations from adaptive ones — while preserving a single point of accountability for the integrated result. Dependencies concentrate at the **interfaces** between components, where one approach must make firm commitments to another, and there the characteristic failure modes of hybrid delivery arise: cadences that never synchronise, controls duplicated across two management systems, and accountability that dissolves between them.

14.4 Designing the hybrid model: a step-by-step method

Hybrid delivery is designed, not declared. The Institute expects candidates to follow a repeatable method and defend every assignment decision on evidence.

Step 1 — Decompose and profile. Break the project into components at the level where a single delivery approach can genuinely apply, and profile each against the tailoring criteria of section 14.2 — in particular requirements volatility, cost of change, regulatory or contractual constraint, and integration criticality.

Step 2 — Assign an approach per component, with written justification. Stable requirements, high cost of change and hard external constraints indicate predictive control; volatile requirements and cheap, reversible change indicate adaptive delivery. Record the rationale — an unjustified assignment is indistinguishable from habit.

Step 3 — Establish the governance spine. Place stage gates on the predictive backbone and define, for each gate, what evidence the adaptive stream must present. Gates consume evidence of working product and validated forecasts, not documents produced for the gate's sake.

Step 4 — Baseline each stream appropriately. The predictive stream carries a conventional cost and schedule baseline measured by earned value. The adaptive stream carries a capacity-and-outcome baseline: a fixed team run rate, a prioritised backlog and a velocity-based completion forecast. Do not baseline the backlog item by item.

Step 5 — Synchronise cadences. Set the reporting period to a whole number of sprints and place every gate on a sprint boundary. Enter every cross-stream dependency in a register with a named owner, a need-by date and the sprint or stage that satisfies it.

Step 6 — Integrate the forecast and assign single accountability. Combine the earned value forecast and the velocity forecast into one programme position, reported side by side. Where AI-enabled tools propose completion dates or risk flags, the doctrine of section 14.7 applies without exception: the delivery leader validates the inputs, explains the arithmetic and owns the published forecast.

14.5 Worked example: reconciling a stage gate with an adaptive forecast

A logistics operator is delivering an automated distribution centre with a total budget of £8.40m. Component A — building fit-out and conveyor installation — is predictive (£6.00m, 40 weeks). Component B — the warehouse management software — is adaptive (£2.40m budget; a stable team costing £150,000 per two-week sprint; a backlog estimated at 480 story points). Gate 3, integrated commissioning, is fixed at week 32 by the mechanical completion of the conveyors.

At the week-20 review, six software sprints are complete, 216 points are delivered and £0.90m is spent. Velocity over the last three sprints was 38, 42 and 40 points, so the rolling mean is $(38 + 42 + 40) \div 3 = \mathbf{40 \text{ points per sprint}}$.

Adaptive forecast. Remaining backlog = $480 - 216 = 264$ points. Sprints required = $264 \div 40 = 6.6$, rounded up to **7 sprints = 14 weeks**, giving forecast completion at week $20 + 14 = \mathbf{\text{week 34}}$ — two weeks beyond the gate. The correct hybrid lever is scope, not schedule: 80 points of features not required for commissioning are moved to a post-gate release. Revised pre-gate backlog = $264 - 80 = 184$ points; $184 \div 40 =$

4.6, rounded up to **5 sprints = 10 weeks**; completion at week 30, restoring a two-week buffer before the gate.

Predictive forecast. Component A at week 20 shows planned value £3.00m, earned value £2.70m and actual cost £2.85m. $CPI = 2.70 \div 2.85 = \mathbf{0.947}$; $SPI = 2.70 \div 3.00 = \mathbf{0.90}$. Estimate at completion = $6.00 \div 0.947 = \mathbf{£6.33m}$, a £0.33m overrun.

Integrated position. Component B's forecast cost — including the post-gate release of the deferred 80 points, seven sprints in all — is $£0.90m + (7 \times £0.15m) = \mathbf{£1.95m}$, £0.45m under budget. The programme forecast is $£6.33m + £1.95m = \mathbf{£8.28m}$ against £8.40m. The delivery leader reports both components separately: the programme is within budget only because the software underspend masks the construction overrun, and netting the two without governance visibility would conceal a control problem. If an AI forecasting assistant produced the £8.28m figure, the leader must still reproduce this arithmetic, test the velocity assumption and own the number presented at the gate.

14.6 Common pitfalls

- **Hybrid by indecision** — attaching agile ceremonies to an unchanged predictive plan ("water-scrum-fall"), gaining the overhead of both approaches with the flexibility of neither.
- **Misaligned cadences** — gates or reporting cut-offs landing mid-sprint, forcing the adaptive team to fabricate percent-complete figures between demonstrations.
- **Duplicated control** — measuring the adaptive stream with both earned value task tracking and velocity, producing two contradictory forecasts and doubled reporting effort.
- **Freezing the backlog** — baselining adaptive scope item by item, which converts every reprioritisation into formal change control and defeats the purpose of the assignment.
- **Netting variances across streams** — presenting only the combined forecast, so one component's underspend hides another's overrun.
- **Interfaces with two owners or none** — dependencies logged without a single accountable name and a need-by date.
- **Treating rounded-up forecasts as certainties** — reporting "7 sprints" without the confidence range implied by velocity variation.
- **Unowned AI output** — publishing machine-generated completion dates or risk scores that no competent professional has validated or can explain.

14.7 AI in this competency: reading two data regimes as one project

Hybrid delivery is the setting in which AI-enabled project management offers the greatest assistance and the greatest scope for error, for the same reason: the project generates two dissimilar data regimes. Predictive components produce baseline-referenced data — planned value, milestone status, gate criteria, change-log entries — while adaptive components produce cadence-referenced telemetry such as throughput, velocity, cycle time and backlog movement. Translating between the two for governance purposes has traditionally been the delivery leader's burden; it is precisely what modern tooling now automates. Probabilistic forecasting models can convert throughput history into a distribution of completion dates expressed against the stage-gate plan; classification and language tools can assemble gate documentation from sprint records, backlog items and decision logs; anomaly detection can flag when reported gate status and underlying delivery telemetry diverge — often the earliest warning of the "watermelon" report, green outside and red within.

The candidate should distinguish mechanical translation from delivery judgement. Suitable for automation: consolidating adaptive metrics into the integrated reporting calendar; first-draft mapping of completed backlog items to baseline scope; monitoring dependencies across workstreams on different cadences; drafting gate submissions from the delivery record. Requiring validation before use: any probabilistic forecast presented to a gate or funding decision; every scope mapping the tooling proposes, because whether an increment genuinely discharges a baseline commitment is an acceptance judgement, not a data join; inferred cross-workstream dependencies, which models deduce from correlation rather than knowledge of the technical interface; and any suggested rebaseline. One decision is never delegated: the assignment of predictive or adaptive approaches to components. That is the tailoring judgement on which the whole operating model rests, and a model trained on past projects will reproduce past habits, not reason about this project's uncertainty profile.

Three model-risk caveats are examinable. First, adaptive telemetry is behavioural, not contractual: velocity is team-relative, unstable under team change and susceptible to gaming, and forecasts trained on it inherit every distortion, usually from a small sample of sprints. Second, the two regimes rarely share definitions — of "done", of dates, of scope granularity — and a model that fuses them without a governed data dictionary produces confident, precisely wrong forecasts; the integration work of section 14.4 is a data-governance precondition, not merely a reporting nicety. Third, generative summarisation introduces a seam-specific hazard: a fluent AI-drafted gate paper can smooth over exactly the ambiguity the gate exists to expose. Tooling must also not recreate the duplicated-control failure of section 14.6 in digital form: two dashboards, drawn from differently filtered data, telling the sponsor two different stories.

14.8 Human accountability

Every AI output entering the governance record — a forecast, a mapping, a drafted gate submission — must be explainable to the board that relies on it, validated against source data, and owned by a named, competent professional. Gate decisions remain human decisions taken by identified role-holders, and AI-assisted artefacts should be marked as such, with the reviewing owner recorded. Where accountability for a hybrid component is already contested, tooling amplifies the ambiguity rather than resolving it: assign the single accountable owner first, then instrument.

AI proposes; the professional disposes. In hybrid delivery, AI may translate adaptive telemetry into stage-gate language, forecast completion probabilistically and draft the governance record — but the professional decides which components are predictive and which adaptive, judges whether an increment truly discharges a baseline commitment, and answers to the gate. The seam between the two delivery worlds is held together by accountable human judgement, never by the tooling that reports across it.

Key points

- Hybrid delivery is a designed, justified combination of predictive and adaptive approaches under one governance framework; an accidental mixture of methods is the absence of a delivery decision, not a hybrid.
- Tailoring is decided per component against assessable characteristics — requirements volatility, cost of change, regulatory constraint, integration criticality — then recorded, approved and revisited at stage boundaries.
- Baseline each stream in its own currency: detailed scope, schedule and cost for predictive components; capacity, cadence, release milestones and outcomes for adaptive ones — never the backlog item by item.
- Synchronise cadences: set reporting periods to a whole number of sprints, place gates on sprint boundaries, and give every cross-stream dependency a single named owner and a need-by date.
- When an adaptive forecast misses a fixed gate, the correct lever is scope — defer features not needed for the gate — not schedule; and report the streams' variances side by side, never netted.
- AI may translate telemetry between the two regimes, forecast probabilistically and draft the governance record, but approach assignment is never delegated, and every AI output entering the record must be explainable, validated and owned by a named, competent human: AI proposes; the professional disposes.

Practice questions

Question 1. A programme runs a predictively managed construction package alongside an adaptively built software product under a single stage-gate framework. What makes this genuine hybrid delivery rather than an accidental mixture of methods?

A. The presence of both a critical-path schedule and a product backlog B. The deliberate, justified assignment of an approach to each component under one governance framework C. The use of sprints inside a formally gated stage D. The approval of a single combined budget by the sponsor

Correct answer: B. The defining test of hybrid delivery is designed intent — approaches assigned and justified component by component under coherent governance — not the mere co-existence of techniques.

Question 2. An adaptive stream has 264 story points remaining and a rolling mean velocity of 40 points per two-week sprint. How many weeks should the delivery leader forecast to complete the backlog?

A. 12 weeks B. 13.2 weeks C. 14 weeks D. 20 weeks

Correct answer: C. $264 \div 40 = 6.6$ sprints, rounded up to 7 whole two-week sprints because partial sprints cannot be planned.

Question 3. Which decision may never be delegated to an AI-enabled tool in hybrid delivery?

A. Drafting a first-cut gate submission from the delivery record B. Consolidating adaptive metrics into the integrated reporting calendar C. Flagging divergence between reported gate status and delivery telemetry D. Assigning predictive or adaptive approaches to components

Correct answer: D. Approach assignment is the tailoring judgement on which the operating model rests; a model trained on past projects reproduces past habits rather than reasoning about the current project's uncertainty.

15. Quality and Change Management

This chapter treats four disciplines as one system of control. Quality planning defines what a conforming output is; quality assurance and quality control demonstrate that conformance is being achieved; integrated change control protects the baselines that give "conformance" its meaning; and organisational change management carries the conforming output into use, where benefits are realised. AI-enabled tools now assist every strand, within the governance discipline that closes the chapter.

Quality as a defined, managed property of project outputs

Quality, in the project context, is the degree to which a project's outputs conform to stated requirements and satisfy the needs for which they were commissioned. It is not a synonym for excellence or high specification: a low-grade output can be of high quality if it meets its defined requirements reliably, and a high-grade output can be of poor quality if it does not. Candidates must hold this distinction between quality and grade precisely, because it anchors every subsequent decision about standards, acceptance criteria and cost.

Quality management comprises three related but distinct activities. **Quality planning** determines which standards and regulatory obligations apply to the project's outputs, translates requirements into measurable acceptance criteria, and specifies how conformance will be demonstrated, by whom and at what points in the delivery lifecycle; its principal product is the quality management plan. **Quality assurance** is process-facing and forward-looking: it provides independent confidence that the methods, competences and controls in use are capable of producing conforming outputs, through audits, peer reviews, method-compliance checks and gate reviews — most credible when performed independently of the work being assured. **Quality control** is product-facing and evidence-based: it inspects, tests and measures actual outputs against the defined acceptance criteria and identifies non-conformances for correction.

The governing principle across all three is prevention over inspection: building quality in almost always costs less than failure, a relationship the cost-of-quality model formalises by setting the costs of conformance (prevention and appraisal) against the costs of non-conformance (internal and external failure).

The governing frameworks

Project quality management sits within a wider standards landscape. The ISO 9000 family articulates the quality management principles — customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision-making and relationship management — from which project-level practice derives. ISO 10005 provides guidance on quality plans, ISO 10006 addresses quality management in projects specifically, and ISO 21502 treats quality as an integrated component of

project management guidance. The continuous-improvement tradition associated with Deming and Juran, expressed in the plan-do-check-act (PDCA) cycle, underpins both predictive and adaptive delivery: in agile contexts it surfaces as built-in quality, a maintained definition of done, and retrospectives as a structured improvement mechanism. The PML-AI examination tests the candidate's ability to apply these frameworks to delivery situations, not to recite them.

Producing the quality management plan

A quality management plan is built deliverable by deliverable, not written as a general essay. The Institute expects candidates to follow six steps. First, identify the standards that apply to each output — contractual specifications, regulatory obligations and the organisation's own quality management system. Second, convert each standard into measurable acceptance criteria: quantified thresholds a competent reviewer could test, never adjectives. Third, define assurance activities — peer reviews, stage-gate reviews and independent audits — specifying timing, ownership and independence from the delivery team. Fourth, define control measures: the inspections, tests and sampling regimes applied to outputs, with the metrics and tolerances that trigger corrective action. Fifth, estimate the cost of quality, separating prevention and appraisal costs from internal and external failure costs, and carry the figures in the budget. Sixth, assign responsibilities and the escalation route for non-conformances.

Continuous improvement runs on the same data. Plot recurring metrics such as defect escape rate on a control chart to distinguish common-cause variation from genuine signals; use Pareto analysis to isolate the vital few causes; test remedies through the PDCA cycle; and return confirmed lessons to the quality management system.

Integrated change control

A **baseline** is an approved reference — for scope, schedule, cost or a product's configuration — against which performance is measured and from which no departure is made without authorisation. A **change request** is a formal proposal to depart from a baseline, whatever its origin: a stakeholder requirement, a defect, a regulatory development or a corrective action. **Integrated change control** is the governed process by which such proposals are captured, assessed, dispositioned, implemented and communicated. The word integrated is load-bearing: before disposition, each proposal is assessed for its impact across scope, cost, schedule, quality, risk, resources and — critically — the benefits case, because a change that is affordable in cost and schedule terms may still erode the outcome the project exists to deliver.

Disposition authority rests with a defined change authority, commonly a change control board, operating within tolerances delegated by the project's governance tier; changes exceeding those tolerances are escalated. The available dispositions are approval, rejection or deferral, and every disposition is recorded with its rationale. Configuration management is the companion record-keeping discipline: it keeps the approved version

and status of every product known at all times, so the baseline being changed is unambiguous and each approved change is traceable.

The method has seven steps: capture the request on the change log; screen it for completeness; assess impact on scope, cost, schedule, risk and benefits; recommend a disposition; route the decision to the correct authority level; update baselines, registers and the benefits case on approval; and communicate the outcome. The worked example below shows the assessment arithmetic.

Worked example: assessing change request CR-041

A 40-week systems project has an approved budget of £1,200,000 and a benefits case worth £10,000 per week from go-live, under a contract whose benefit window ends on a fixed date. Change request CR-041 proposes extending the compliance module to meet newly published data-retention rules.

Cost. The estimate is 280 hours of design, build and test at £95 per hour = £26,600, plus £6,000 of additional test-environment costs = £32,600. A 10 per cent estimating allowance adds £3,260, giving a total of £35,860 — 3.0 per cent of the approved budget ($£35,860 \div £1,200,000 = 2.99$ per cent).

Schedule. The work adds four weeks to the build activity, which holds one week of float, so the critical path extends by $4 - 1 = 3$ weeks.

Benefits. Because the benefit window closes on a fixed date, a 3-week deferral of go-live permanently forgoes $3 \times £10,000 = £30,000$.

Risk. Approval retires the non-compliance risk, assessed at 40 per cent probability of a £250,000 regulatory penalty — an expected monetary value of $0.40 \times £250,000 = £100,000$. It introduces a new integration risk of $0.20 \times £25,000 = £5,000$. Net risk exposure falls by $£100,000 - £5,000 = £95,000$.

Net appraisal and disposition. Expected value of approving = $£95,000 - (£35,860 + £30,000) = +£29,140$, so the assessment recommends approval. Because £35,860 exceeds the project manager's £25,000 delegated tolerance, the disposition belongs to the project board, which approves the change funded from the change budget. The cost and schedule baselines are re-versioned from v1.2 to v1.3, the risk register and benefits profile are updated, and future performance is measured against the new baseline. An AI assistant may draft this assessment from the schedule model and risk register, but the professional validates every figure and owns the decision presented to the board.

Organisational change and adoption

Integrated change control governs changes to the project; organisational change management governs the change the project causes in its host organisation. The distinction matters because outputs generate benefits only when their intended users

adopt them into working practice, so adoption is planned, resourced work — led visibly by the sponsor, supported by business change roles close to the affected users — never an assumed consequence of delivery.

Recognised models give this work structure, and the adoption plan applies one of them rather than a communications calendar alone. Using ADKAR: assess each user group against Awareness, Desire, Knowledge, Ability and Reinforcement; target interventions at the weakest element first (sponsor-led messaging for awareness and desire; training for knowledge; coached practice and floor-walking for ability; measurement and recognition for reinforcement); then sequence these against the release plan. Kotter's eight-step model, building urgency, coalition, vision and reinforcement, and Lewin's unfreeze-change-refreeze serve equally where the examination scenario suits them. Define adoption metrics in advance — for example, percentage of active users at week four and error rates against the pre-change baseline — and track them in the benefits realisation plan.

How AI changes quality and change management

AI reaches this competency along all three of the chapter's threads. In quality management, machine learning extends the classical toolset: models trained on inspection and test histories predict where defects will concentrate, so assurance effort can be risk-weighted rather than spread evenly; anomaly detection watches process data continuously instead of at sampling intervals; and language models check requirements and acceptance criteria for ambiguity, contradiction and untestability before they are baselined — a new capability, since poor criteria historically surfaced only at acceptance. In integrated change control, AI accelerates the analytical core of the process: tracing a proposed change through the scope, schedule, cost, risk and benefits baselines, retrieving comparable past changes and their outturns, and triaging the change log so trivial requests do not consume board attention. In organisational change, adoption analytics — usage telemetry, training completion, support-ticket themes — give early, quantified sight of whether outputs are actually being taken up by their intended users.

The delivery leader may appropriately automate the clerical layer: administration of the change log; first-draft impact assessments assembled from the linked baselines; consistency checks between a change request and the artefacts it touches; control-chart signal detection and non-conformance trending; and first drafts of adoption communications and training materials — each removing latency and transcription error from a process whose value lies elsewhere.

What is never automated is disposition. A machine-drafted impact assessment is a hypothesis: before it informs a decision, a competent professional must trace its cost and schedule figures to their assumptions, confirm that the affected baselines were correctly identified, and test whether the retrieved "comparable" changes are genuinely comparable. Acceptance of deliverables remains a human judgement against

the criteria in the quality management plan — an AI verdict that outputs "pass" is evidence to be weighed, not a decision taken. Likewise, adoption analytics inform the change plan but do not replace conversation with the people expected to change.

Model risk, data caveats and human accountability

Quality data carries a specific inheritance problem: defect predictors learn where defects were found, which reflects where inspection was directed, not necessarily where failure lives. Sparse or inconsistently coded non-conformance records, screening thresholds tuned on one supplier or technology and reused on another, and impact estimates drawn from analogues that no longer apply can all be systematically wrong while appearing precise. Two behavioural hazards follow. Automation bias tempts a change authority to wave through a machine-scored "low impact" change without scrutiny — precisely the unexamined change that integrated change control exists to prevent. And adoption analytics process data about identifiable individuals, so privacy, proportionality and works-council or equivalent obligations apply before any such monitoring is deployed. Models, prompts, thresholds and training data belong under version control as governed artefacts, and the quality assurance function should audit the AI pipeline with the same independence it applies to any other process.

Accountability is untouched by the tooling. The change authority named in the governance framework — individual, board or sponsor — owns every disposition, and the record must show a named person approving, rejecting or deferring each change, whatever analysis supported them. The delivery leader owns the quality management plan and answers for the fitness of accepted outputs; sponsors and line managers own adoption. Where an AI-derived assessment informed a disposition or an acceptance, the record should disclose it, and someone competent must be able to explain it.

AI proposes; the professional disposes. AI may draft the impact assessment, flag the anomaly, predict the defect and chart the adoption curve; only an accountable person approves the change, accepts the deliverable and answers for the baseline. A change dispositioned on an explanation nobody can give is an uncontrolled change — however sophisticated the tool that scored it.

Common pitfalls

- Acceptance criteria written as adjectives ("robust", "intuitive") that no reviewer can objectively test.
- Confusing quality assurance (process-focused, preventive) with quality control (product-focused, detective), and resourcing only one of them.
- Impact assessments that price cost and schedule but omit risk and benefits — in CR-041, the £95,000 exposure reduction that justified approval would have been invisible.

- Dispositions taken at the wrong authority level, or approved changes never re-baselined, so performance is measured against a superseded plan.
- Scope creep through "minor" changes accepted outside the change log, and gold-plating that adds unrequested refinement at the project's cost.
- Treating training as the whole adoption plan while neglecting desire and reinforcement, the elements where adoption most often fails.
- Presenting an AI-drafted impact assessment to a change authority unvalidated; the professional who signs the recommendation owns its figures.

Key points

- Quality is conformance to stated requirements, not grade; quality planning, assurance (process-facing) and control (product-facing) are distinct activities, governed by prevention over inspection and priced by the cost-of-quality model.
- The quality management plan is built deliverable by deliverable in six steps, converting applicable standards into measurable acceptance criteria with named assurance and control activities, costs and escalation routes.
- Integrated change control protects approved baselines: every change request is impact-assessed across scope, cost, schedule, quality, risk and benefits before a recorded disposition by the correct authority, with configuration management keeping baselines unambiguous.
- Approved changes are re-baselined and carried into the risk register and benefits case; as CR-041 shows, a change affordable in cost terms can still destroy value through lost benefits.
- Organisational change management (Lewin, Kotter, ADKAR) turns delivered outputs into adopted outputs; adoption is planned, sponsored and measured work, because unadopted outputs yield no benefits.
- AI legitimately automates the clerical layer — drafting assessments, detecting anomalies, triaging the change log — but disposition and acceptance are never automated, and AI models, prompts and thresholds are governed, auditable artefacts.
- AI proposes; the professional disposes: every AI-generated assessment must be explainable, validated against evidence and owned by a named, accountable person before it informs a decision.

Practice questions

1. During execution, an independent reviewer audits whether the project team is actually following its documented peer-review procedure. This activity is best described as:

A. Quality control B. Quality planning C. Quality assurance D. Configuration management

Correct answer: C. Quality assurance is process-facing and preventive — it independently confirms that the methods in use can produce conforming outputs, whereas quality control (A) inspects the outputs themselves.

2. A change request costs £18,000 and adds three weeks to an activity holding one week of float, on a project earning £8,000 per week of benefits within a fixed benefit window. Retiring an existing risk reduces expected exposure by £40,000 with no new risk introduced. The expected value of approving the change is:

A. +£6,000 B. -£2,000 C. +£22,000 D. +£14,000

Correct answer: A. The critical path extends by $3 - 1 = 2$ weeks, forgoing $2 \times £8,000 = £16,000$ of benefits, so the appraisal is $£40,000 - (£18,000 + £16,000) = +£6,000$.

3. An AI triage tool scores a change request as "low impact" and recommends fast-track approval. Under the Institute's doctrine, the change authority should:

A. Approve the change, since the tool was validated at deployment B. Treat the score as a proposal, validate the assessment against the baselines and evidence, and record a named human disposition C. Reject the change, because AI-scored requests cannot be trusted D. Defer the change until the tool's confidence exceeds a set threshold

Correct answer: B. AI proposes and the professional disposes — machine output is evidence to be validated and owned by a named, accountable person, and waving it through unexamined (A) is automation bias.

16. Benefits Realisation

A project that delivers on time, on budget and to specification can still fail its investors, because schedules and budgets measure delivery, not worth. Benefits realisation is the discipline that closes this gap. This chapter sets out its concepts, the artefacts that operationalise them, the tracking regime that must outlive the project, and how AI-enabled tooling changes the economics of the discipline without changing where accountability sits.

From outputs to benefits: the concepts, precisely

The vocabulary of benefits realisation must be exact, because the terms form a causal chain, and confusion between links in that chain is the single most common source of unrealised business cases.

An **output** is a deliverable the project produces — a system, an asset, a process, a trained team. A **capability** is what the organisation can now do because the outputs exist and have been adopted: outputs assembled, commissioned and embedded in working practice. An **outcome** is the changed operational state that results when the capability is actually used — faster order processing, fewer defects, shorter patient waits. A **benefit** is the measurable improvement resulting from an outcome that is perceived as an advantage by one or more stakeholders and that contributes to a strategic objective — typically expressed in money, time, quality, risk reduction or stakeholder value. A **disbenefit** is a measurable outcome perceived as negative by a stakeholder — for example, redundancy costs or increased workload elsewhere — and must be identified and managed with the same rigour, not hidden. The **strategic objective** is the organisational goal, stated outside the project, that the benefit serves.

The chain — outputs enable capabilities, capabilities enable outcomes, outcomes yield benefits, benefits serve objectives — is the logical backbone of this chapter. Each link is a dependency, not an automatic consequence: an output that is delivered but not adopted creates no capability; a capability that is not exercised produces no outcome; an outcome that no one measures yields no demonstrable benefit.

The governing frameworks

The candidate should recognise the framework landscape in which this discipline sits. Programme-management practice — notably the Managing Successful Programmes (MSP) tradition — established benefits management as a core theme, including the benefits map, the benefit profile and the principle that programmes exist to deliver outcomes and benefits rather than outputs. PMI has published dedicated guidance on benefits realisation management, and the PMBOK Guide embeds benefits within the business case and project charter. The APM Body of Knowledge treats benefits management as integral to both project and programme domains, ISO 21502 frames benefits as the rationale that justifies initiating and continuing a project, and public-

investment traditions such as the UK Treasury's Green Book add the discipline of appraising and then evaluating benefits against the original case. This Body of Knowledge draws on the shared logic of these sources in its own words and does not claim conformance with, or accreditation against, any of them; the PML-AI examination tests the principles as defined here.

Two framework principles recur everywhere. First, **the business case is the anchor**: benefits are forecast in the business case to justify investment, and realisation is always reported against that forecast, not against a conveniently revised one. Second, **benefits outlive the project**: most benefits accrue after handover, so accountability for them cannot end at project closure.

The benefits map: built from right to left, read from left to right

Three artefacts operationalise the concepts — the benefits map, the benefit profile and the benefits realisation plan — and each answers a distinct question. The **benefits map** answers "why does this work matter?" It is a directed diagram reading left to right from outputs, through capabilities and outcomes, to benefits and strategic objectives. Read forwards, it demonstrates the investment logic; read backwards, it tests scope.

Although the finished map is read from outputs towards strategy, it is constructed backwards from strategy, not forwards from deliverables. The candidate should apply the following steps.

1. **State the strategic objective** the investment serves, in the sponsor's own words (e.g. "grow service revenue without growing headcount").
2. **Identify the measurable benefits** that would evidence progress against that objective, and any expected **disbenefits** to be managed openly.
3. **Trace each benefit back to the outcomes** — changes in the way the organisation operates — that must occur for the benefit to arise.
4. **Trace each outcome back to the capabilities** created when project outputs are combined with business change (training, revised processes, new roles).
5. **Trace capabilities back to project outputs**, then test every link for causality: "if this output is delivered and adopted, will that outcome plausibly follow?" Delete orphan outputs that support no benefit, and flag orphan benefits that no output supports — both are early warnings of a weak business case.

Where AI-assisted tools draft candidate maps or mine historical benefit data, the Institute's doctrine applies in full: every causal link must be explainable and owned by a named individual, not inferred by a model and left unexamined.

The benefit profile: what exactly is this benefit, and how will we know?

For every benefit on the map, complete a one-page **benefit profile** recording: a precise description; a classification (cashable, non-cashable, strategic); the accountable **benefit owner** — a business manager, never the project manager; the **measure**, its unit and its data source; the **baseline** value, captured before delivery alters it; the **target** value and realisation timescale; the **realisation trajectory** (ramp-up and full run-rate dates); measurement frequency; attribution and assumptions; dependencies on business change; associated disbenefits; and links to the benefits map and business case.

Two elements are non-negotiable. Without a baseline there is no evidence of improvement — a baseline reconstructed after go-live is an estimate, not evidence, so baselining must precede delivery. Without a named owner there is no one answerable for the benefit; owners sit in the business, not the delivery team, because benefits are harvested in operations after the project team has disbanded.

The benefits realisation plan: tracking beyond closure

The **benefits realisation plan** answers "when, and by whose hand?" It schedules measurement points, assigns tracking responsibility during delivery and — critically — beyond closure, and defines the reporting line through which realisation is compared with the business-case forecast. The candidate should design the regime as follows.

1. **Write the plan during delivery**, naming review points that extend past project closure (typically 6, 12, 24 and 36 months).
2. **Transfer accountability at handover**: each profile passes formally from the project to its benefit owner in operations, with the measurement method and data source tested and working. This transition is a transfer of live accountability, not an administrative filing.
3. **Report realisation against the business case**, benefit by benefit, at each review — actual versus profile, forecast at completion, and corrective actions where adoption is lagging.
4. **Give the regime a home**: a portfolio office, sponsor board or equivalent must own post-closure reviews, because the project board no longer exists to demand them.
5. **Close the loop**: feed realised-versus-promised results back into estimating assumptions for future business cases.

Worked example: field-service scheduling capability

Setup. A project delivers a dynamic scheduling system plus retrained dispatch processes for a maintenance business with 120 technicians working 220 days per year. Baseline productivity is 3.9 completed jobs per technician per day; the target is 4.5. Contribution per additional job is USD 38. Incremental running costs (licences, support, model maintenance) are USD 150,000 per year. The profile assumes 40% of the productivity uplift is realised in year 1 and 100% in years 2 and 3. The business case commits to a net benefit of USD 950,000 over three years.

Full-run-rate gross benefit. $\text{Uplift} = 4.5 - 3.9 = 0.6 \text{ jobs/technician/day}$. Additional jobs per year $= 120 \times 0.6 \times 220 = \mathbf{15,840 \text{ jobs}}$. Gross annual benefit $= 15,840 \times \text{USD } 38 = \mathbf{\text{USD } 601,920}$.

Net benefit by year. Year 1: $(0.40 \times 601,920) - 150,000 = 240,768 - 150,000 = \mathbf{\text{USD } 90,768}$. Years 2 and 3: $601,920 - 150,000 = \mathbf{\text{USD } 451,920}$ each. Three-year net benefit $= 90,768 + 451,920 + 451,920 = \mathbf{\text{USD } 994,608}$.

Comparison with the business case. $994,608 - 950,000 = \text{USD } 44,608$ of headroom (about 4.7%) — an acceptable but slim margin, so the tracking regime must detect slippage early.

Tracking snapshot. At the six-month post-closure review, the 13-week rolling average is 4.2 jobs/technician/day, an uplift of 0.3 against the profiled year-1 expectation of $0.40 \times 0.6 = 0.24$. Annualised realised gross benefit $= 120 \times 0.3 \times 220 \times 38 = \mathbf{\text{USD } 300,960}$, versus the profiled 240,768 — realisation is running at $300,960 \div 240,768 = \mathbf{125\% \text{ of profile}}$. The benefit owner reports this to the sponsor with the evidence trail, and the forecast at completion is revised upwards only after the owner confirms the uplift is attributable to the new capability and not to seasonal workload mix.

How AI changes benefits realisation — and what it cannot change

Benefits realisation has always suffered a structural weakness: measurement effort peaks at business-case approval and collapses after handover — precisely when benefits begin to accrue. AI-enabled tooling changes the economics of that measurement. Analytics platforms can ingest operational data continuously and report movement against benefit baselines long after closure. Predictive models can forecast realisation trajectories from early adoption signals, flagging benefits at risk of shortfall months before a scheduled review would detect them. Natural-language tools can interrogate a benefits map, testing whether every strategic benefit traces back to a delivered output and whether claimed contributions are double-counted across a portfolio. None of this changes what a benefit is; it changes how early, how cheaply and how persistently the organisation can see whether one is emerging.

What the leader automates. Automation is appropriate where the work is mechanical and the data is owned: harvesting benefit measures from source systems; scheduled reporting of actuals against the benefit profile; alerting when a tracked

measure breaches tolerance; consolidating realisation data across a portfolio; and drafting candidate benefits, leading indicators and measure definitions for professional refinement. These high-volume, low-judgement tasks can be delegated, freeing the leader for the work that matters.

What the leader validates. Validation cannot be delegated. The leader personally tests the **causal logic** of the benefits map, because correlation in operational data is not evidence that the project caused the movement; the **attribution** of benefit to this investment rather than to concurrent initiatives, market conditions or seasonality; the **integrity of baselines**, because a model that infers its baseline from contaminated pre-project data will misstate realisation permanently; and every **forecast of future realisation** before it reaches a sponsor or board. A predicted benefit is not a realised benefit, and governance reporting must never conflate the two.

Model-risk and data caveats. Benefits data is among the noisiest an organisation handles, and models inherit its defects. Business cases are systematically optimistic, so a model trained on historical cases and their outturns may normalise overstatement or learn from survivorship bias — organisations rarely keep honest records of benefits that quietly failed. Measure definitions drift after handover, silently breaking comparability with the baseline, and external confounders — price changes, regulation, competitor behaviour — move the very indicators a benefit is measured by. The discipline that answers these risks is explainability: every machine-produced realisation figure must be traceable to its source data, method and assumptions, and a figure that cannot be explained must not be reported. Disbenefits deserve equal instrumentation; a tracking regime that automates only the favourable measures is a governance failure, not a technical one.

Human accountability. Ownership is the non-negotiable centre of this competency. Every benefit profile names a benefit owner — a human accountable for realisation against the business case — and no tool, dashboard or model can occupy that role. Where AI output feeds a realisation report, a competent individual validates it and stands behind it; where tracking continues beyond closure, the tooling, its data pipelines and its known limitations are handed over to operational ownership as deliberately as any other product. The examination expects candidates to state plainly who answers for a benefit at every point in its life — during delivery, at transition and in sustained operation.

AI proposes; the professional disposes. An analytics platform may surface the trend, forecast the shortfall and draft the realisation report — but only a named benefit owner can attest that a benefit is real, attribute it honestly to the investment, and account to the sponsor for the business case. Machine measurement informs the judgement of realisation; it never constitutes it.

Common pitfalls

- **Baselining after go-live**, leaving no defensible "before" measure.
- **Making the project manager the benefit owner** — benefits arise in operations, after the project team has disbanded.
- **Counting gross benefits** and ignoring incremental running costs and disbenefits.
- **Double-counting** the same saving across two projects' business cases.
- **Assuming full run-rate from day one** instead of profiling a realistic ramp-up.
- **Stopping measurement at closure**, when most benefit value typically arises afterwards.
- **Claiming unattributable benefits** — movement in the measure caused by external factors, not the new capability.
- **Accepting AI-generated benefit forecasts unvalidated** — projections drafted by a model must be checked, explained and owned by the accountable professional before they enter the business case or any realisation report.

Key points

- Outputs enable capabilities, capabilities enable outcomes, outcomes yield measurable benefits, and benefits serve strategic objectives; every link is a dependency, not an automatic consequence.
- The business case is the anchor: realisation is always reported against the original forecast, never against a conveniently revised one, and disbenefits are managed with the same rigour as benefits.
- The benefits map is built backwards from strategy and tested for causality; orphan outputs are descoping candidates and orphan benefits signal a weak business case.
- Every benefit carries a profile with a measure, a baseline captured before delivery, a target, a realisation trajectory and a named owner drawn from the business, never the delivery team.
- Because most benefits accrue after handover, the benefits realisation plan schedules post-closure reviews, transfers live accountability to operations, and lodges the regime with a body that outlives the project board.
- AI-enabled tooling makes measurement continuous and cheap — harvesting actuals, forecasting trajectories, flagging shortfalls — but causal logic, attribution, baseline integrity and every forward forecast remain the professional's to validate.
- AI proposes; the professional disposes: only a named human benefit owner can attest that a benefit is real and account to the sponsor for the business case.

Practice questions

1. A project delivers a new rostering system and trains all shift supervisors to use it. Six months later, average overtime spend per depot has fallen by 12%. In benefits-realisation terms, the reduction in overtime spend is best described as:

A. An output B. A capability C. A benefit D. A strategic objective

Answer: C. The 12% reduction is a measurable improvement, perceived as an advantage and expressed against a baseline — a benefit; the system is the output and the trained, embedded rostering practice is the capability.

2. A benefit profile assumes 80 engineers working 200 days per year, a productivity uplift of 0.5 jobs per engineer per day at USD 40 contribution per job, incremental running costs of USD 100,000 per year, and 50% realisation of the uplift in year 1. The profiled net benefit for year 1 is:

A. USD 60,000 B. USD 160,000 C. USD 220,000 D. USD 320,000

Answer: A. Full-run-rate gross benefit is $80 \times 0.5 \times 200 \times 40 = \text{USD } 320,000$; year 1 gross is 50% of that (USD 160,000), minus USD 100,000 running costs = USD 60,000.

3. An AI analytics platform tracking a benefit after handover forecasts a significant realisation shortfall and drafts the report for the next post-implementation review. Under the Institute's doctrine, what must happen before the report reaches the sponsor?

A. The platform submits the report automatically, since its data feed is authoritative B. The project manager, as benefit owner, signs off the forecast C. The named benefit owner validates the figures, confirms attribution and stands behind the report D. The forecast is excluded, because AI output may never appear in governance reporting

Answer: C. Machine output may inform the report only after the accountable benefit owner — a business manager, not the project manager — has validated, explained and owned it; attestation is not delegable.

17. Sustainability in Project Delivery

What sustainability means in project delivery

Sustainability, as the Institute uses the term, derives from the definition of sustainable development popularised by the Brundtland Commission's 1987 report *Our Common Future*: meeting the needs of the present without compromising the ability of future generations to meet their own. Applied to project delivery, the concept operates at two levels that the candidate must keep distinct:

- **Sustainability of the output.** Whether the asset, product, service or change the project delivers is environmentally, socially and economically viable across its whole life — construction or build, operation, maintenance, and eventual decommissioning, disposal or reuse.
- **Sustainability in delivery.** Whether the temporary endeavour itself is conducted responsibly — its energy and material consumption, emissions and waste, supply-chain conduct, treatment of workers, and effects on the communities around it.

The distinction matters because of a structural asymmetry: delivery typically lasts months or a few years, while outputs endure for decades. Decisions taken during initiation and design lock in the great majority of an asset's whole-life impacts long before those impacts are felt. Two further terms anchor the chapter. **Whole-life perspective** means appraising an option over its entire life cycle rather than to handover, using techniques such as whole-life costing and life cycle assessment. **Project legacy** means the enduring consequences — beneficial and adverse — that persist after closure: benefits realised, capabilities and infrastructure left behind, environmental residues, and changes to livelihoods and communities.

The triple bottom line

The **triple bottom line**, a framing introduced by John Elkington in the 1990s, holds that performance must be assessed on three dimensions — **environmental** (planet), **social** (people) and **economic** (prosperity) — rather than on financial results alone. The economic dimension is deliberately wider than project cost: it encompasses whole-life value to the owner and to society, including resilience and avoided future cost. The discipline it imposes is not that every project optimise all three dimensions equally, but that all three are appraised explicitly and trade-offs surfaced for the sponsor to decide, rather than resolved silently in favour of capital cost or schedule.

Two related constructs must not be conflated with it. **ESG** (environmental, social and governance) is the investor-facing disclosure lens through which organisations report; the triple bottom line is the decision lens through which delivery leaders appraise options. **Materiality** determines which sustainability topics warrant measurement and reporting: financial materiality asks what affects the organisation's value, while double

materiality — the basis of current European reporting regulation — also asks what impact the organisation has on people and the environment.

Governing frameworks and standards

Several published instruments give the field its common vocabulary. The **United Nations Sustainable Development Goals** (the seventeen SDGs of the 2030 Agenda) provide a widely used taxonomy for social and environmental outcomes to which many owners map project benefits. The **Greenhouse Gas Protocol** classifies emissions as Scope 1 (direct), Scope 2 (purchased energy) and Scope 3 (the value chain); on most projects, Scope 3 — embodied in purchased materials and subcontracted work — dominates. Management-system and method standards include **ISO 14001** (environmental management systems), **ISO 20400** (sustainable procurement), **ISO 14040/14044** (life cycle assessment) and **PAS 2080** (carbon management in buildings and infrastructure, with its hierarchy of avoid, switch and improve). On the reporting side sit the **GRI Standards** (impact-oriented disclosure), the **IFRS Sustainability Disclosure Standards** IFRS S1 and S2 issued by the ISSB (investor-oriented), and the EU's **Corporate Sustainability Reporting Directive** with its European Sustainability Reporting Standards (double materiality). Profession-specific treatments, such as GPM's **P5 Standard**, extend the triple bottom line to the delivery process and product. These instruments are cited for orientation and study; the PML-AI credential is examination-based, and the Institute claims no accreditation by, or affiliation with, the bodies that publish them.

Together these pieces form a single chain of practice: triple-bottom-line principles supply the criteria for options appraisal; the selected option's commitments become measures baselined, reported and flowed to suppliers, so that corporate disclosures reconcile to verifiable project-level data rather than assertion; and whole-life and legacy assessment closes the loop after handover.

A step-by-step method for sustainability-weighted option appraisal

Sustainability becomes real when it enters the option appraisal, the baseline and the contract. The Institute expects candidates to apply a repeatable method, not to append sustainability commentary to a business case.

Step 1 — Frame the decision with the sponsor. Agree the environmental, social and economic criteria that matter, the appraisal period (whole life, not the delivery phase alone) and any organisational commitments — a net-zero target, a social-value policy — that constrain the choice.

Step 2 — Establish whole-life economic cost. For each option, sum capital, operating, maintenance and end-of-life costs over the appraisal period.

Step 3 — Quantify environmental impact. Convert activity data (energy, fuel, materials, waste) into tonnes of carbon-dioxide equivalent (tCO₂e) using published

emission factors, covering both embodied carbon (construction) and operational carbon (use).

Step 4 — Monetise what can credibly be monetised. Apply the organisation's shadow carbon price to the tCO₂e figures so carbon competes with cash in the same units. Keep factors that resist credible monetisation — biodiversity, community disruption, local employment — as scored criteria with defined scales and recorded evidence.

Step 5 — Compare and recommend. Present the monetised whole-life comparison alongside the scored criteria, state assumptions and sensitivities, and record the recommendation and its rationale in the decision log.

Step 6 — Validate any AI contribution. Where an AI tool proposes emission factors, energy forecasts or supplier sustainability ratings, the professional verifies source, vintage and applicability before the numbers enter the appraisal.

Worked example: whole-life appraisal of two depot designs

A sponsor must choose between two designs for a regional distribution depot, appraised over 20 years. The organisation's shadow carbon price is USD 80 per tCO₂e; electricity costs USD 0.14 per kWh; the grid emission factor is 0.40 kgCO₂e per kWh. Figures are undiscounted for clarity; in practice, apply the organisation's stated discount rate to future costs and test sensitivity.

Input	Option A (conventional)	Option B (low-carbon)
Capital cost	USD 4,200,000	USD 4,650,000
Annual electricity	620,000 kWh	380,000 kWh
Embodied carbon	2,900 tCO ₂ e	2,450 tCO ₂ e

Operational energy cost (20 years). Option A: $620,000 \times 0.14 = \text{USD } 86,800$ per year; $\times 20 = \text{USD } 1,736,000$. Option B: $380,000 \times 0.14 = \text{USD } 53,200$ per year; $\times 20 = \text{USD } 1,064,000$.

Operational carbon. Option A: $620,000 \times 0.40 = 248,000 \text{ kg} = 248 \text{ tCO}_2\text{e}$ per year; $\times 20 = 4,960 \text{ tCO}_2\text{e}$. Option B: $380,000 \times 0.40 = 152,000 \text{ kg} = 152 \text{ tCO}_2\text{e}$ per year; $\times 20 = 3,040 \text{ tCO}_2\text{e}$.

Monetised carbon at USD 80 per tCO₂e. Operational: A = $4,960 \times 80 = \text{USD } 396,800$; B = $3,040 \times 80 = \text{USD } 243,200$. Embodied: A = $2,900 \times 80 = \text{USD } 232,000$; B = $2,450 \times 80 = \text{USD } 196,000$.

Whole-life comparison. Option A: $4,200,000 + 1,736,000 + 396,800 + 232,000 = \text{USD } 6,564,800$. Option B: $4,650,000 + 1,064,000 + 243,200 + 196,000 = \text{USD } 6,153,200$.

Option B is USD 411,600 (6.3 %) cheaper on a whole-life basis and avoids 7,860 – 5,490 = 2,370 tCO₂e. Note the shape of the result: on cash alone, B's extra USD 450,000 of capital buys energy savings of only 86,800 – 53,200 = USD 33,600 a year — a 13.4-year simple payback (450,000 ÷ 33,600) that many sponsors would reject. It is the monetised carbon, together with B's stronger scored social criteria (apprenticeship and local-labour commitments), that carries the recommendation. This is precisely why criteria, carbon price and appraisal period must be agreed with the sponsor before scoring, never fitted afterwards to a preferred answer.

Embedding the measures in baselines, reporting and the supply chain

An approved appraisal becomes delivery discipline in three moves. First, **baseline the measures**: Option B's 2,450 tCO₂e embodied-carbon figure becomes a carbon budget alongside the cost baseline, decomposed to work packages and placed under change control — a design change that adds carbon is a change, not a footnote. Second, **report against them**: the routine performance report shows carbon and agreed social measures (for example, local-labour hours) as planned versus actual, with variance and forecast at completion, on the same cadence as cost and schedule. Third, **flow them to suppliers**: tender evaluation carries a stated sustainability weighting; contracts carry the KPIs and — critically — an obligation to supply the underlying data (material quantities, delivery mileage, metered energy) in a defined format, without which no KPI is verifiable. At handover, the whole-life measures transfer to the benefits and legacy register with a named operational owner, so the 20-year case remains accountable after the project closes.

Common pitfalls

- **Capital-only appraisal.** The cheapest option to build is often the dearest to run and to decommission.
- **Double-counting.** Monetising carbon and then scoring it again as a weighted criterion.
- **Unstated assumptions.** Emission factors, carbon price and appraisal period left implicit, making the appraisal unrepeatable and unauditable.
- **Discount-rate abuse.** A high rate quietly extinguishes long-term benefits; state it, justify it and sensitivity-test it.
- **Measures without baselines.** Objectives never baselined or reported — brochure, not baseline, and a greenwashing risk.
- **KPIs without data clauses.** Supplier targets with no contractual right to the evidence needed to verify them.
- **Unvalidated AI inputs.** AI-proposed emission factors, forecasts or supplier ratings adopted without a competent human confirming source, vintage and applicability.

- **Orphaned legacy.** Closing the project with no named owner for whole-life measures, so the benefits case evaporates at handover.

How AI changes the sustainability competency

Sustainability has historically suffered from a data problem: environmental and social impacts are diffuse, accumulate across long supply chains and asset lifetimes, and are recorded — if at all — in systems never designed for project reporting. AI and predictive tooling change this in three practical ways. First, **estimation**: machine-learning models linked to design data, materials libraries and emissions-factor databases can generate carbon, energy, water and waste estimates for competing options far earlier in the life cycle than manual assessment allows — when the leader's influence is greatest. Second, **monitoring**: analytics platforms can ingest metered consumption, logistics records and site data to track sustainability performance continuously against the baseline, flagging variances in the same cadence as cost and schedule. Third, **disclosure support**: language models can screen supplier declarations, certifications and sustainability reports at a scale no project team could match, and can draft the project's own sustainability reporting.

None of this changes what the competency is. The triple-bottom-line trade-offs remain judgements, and AI's contribution is to make those judgements better informed, faster and more consistently evidenced.

What the leader automates — and what the leader validates

The delivery leader may sensibly automate data aggregation, emissions-factor lookups, option-comparison calculations, supplier-document screening and the first draft of routine sustainability reporting. The leader must personally validate: the system boundary and scope of any footprint calculation; the emission factors and assumptions applied; any AI-derived ranking used in a recommendation to the sponsor; any social-impact conclusion, which rests on stakeholder evidence no model can substitute; and every sustainability claim that leaves the project. Whole-life and legacy assessments deserve particular scrutiny: their outputs are dominated by assumptions (asset life, usage profiles, discount treatment) the model inherits rather than tests.

Model risk, data quality and human accountability

Sustainability data is among the weakest an AI system will encounter on a project. Scope 3 and supply-chain figures are frequently estimated, self-reported or missing; emission factors vary by region and vintage; third-party ESG ratings of the same supplier can disagree materially. Models trained on such data can present fragile numbers with unwarranted precision, and generative tools can fabricate plausible figures, factors or references outright. Bias is a live risk: datasets over-represent large, well-documented suppliers and under-represent local and social impacts, skewing

comparisons against exactly the options the appraisal exists to surface. The competent leader therefore demands provenance for every input, an uncertainty range on every estimate, and reproducibility for any figure that enters a baseline or report.

Sustainability disclosure is increasingly regulated, and overstated environmental claims — greenwashing — carry legal, commercial and reputational consequences. Accountability for them cannot be delegated to a tool or its vendor. Every AI-assisted sustainability figure used in governance must be explainable to the sponsor and to an auditor, validated by a competent person, and formally owned by a named individual — normally the project delivery leader. "The model calculated it" is not a defence the Institute recognises, in the examination or in practice.

AI proposes; the professional disposes. AI can estimate a footprint, rank the options and draft the disclosure — but it cannot weigh an environmental gain against a social cost, and it cannot answer for a claim. The delivery leader who signs the recommendation, the baseline and the sustainability report owns every number in them, and must be able to explain, evidence and defend each one without reference to the tool that produced it.

Key points

- Sustainability operates at two distinct levels — sustainability of the output across its whole life, and sustainability in the conduct of delivery — and early design decisions lock in most whole-life impacts.
- The triple bottom line (environmental, social, economic) is a decision lens for appraising options; ESG is a disclosure lens; double materiality adds the organisation's impact on people and planet to financial materiality.
- Scope 3 (value-chain) emissions dominate most project footprints; ISO 14001, ISO 20400, ISO 14040/14044 and PAS 2080 supply method, while GRI, IFRS S1/S2 and CSRD govern disclosure.
- Appraise options with sponsor-agreed criteria fixed in advance: cost the whole life, quantify carbon, monetise credibly at the shadow carbon price, score the rest, and state every assumption.
- Approved measures must be baselined under change control, reported on the same cadence as cost and schedule, and flowed to suppliers with contractual rights to the underlying data.
- AI accelerates estimation, monitoring and disclosure drafting, but sustainability data is weak and bias-prone: every AI figure needs provenance, an uncertainty range and reproducibility before it enters governance.
- Accountability for sustainability claims rests with a named professional, never with a tool: AI proposes; the professional disposes.

Practice questions

1. A low-carbon option carries USD 450,000 more capital cost and a 13.4-year simple payback on energy savings alone, yet is correctly recommended. Which feature of the appraisal most directly justifies this?

A. Applying a lower discount rate to the low-carbon option only B. Monetising carbon at the organisation's shadow price within the agreed whole-life comparison C. Adjusting the criteria weightings after the results were known D. Excluding embodied carbon so that operational savings dominate

Correct answer: B. Pricing carbon into the whole-life comparison, on criteria agreed with the sponsor before scoring, is what legitimately overturns the capital-only view; A, C and D are appraisal abuses.

2. Under the Greenhouse Gas Protocol, the emissions embodied in a project's purchased materials and subcontracted work are classified as:

A. Scope 1, because the project directly causes them B. Scope 2, because they are procured C. Scope 3, because they arise in the value chain D. Outside the project boundary and excluded from its footprint

Correct answer: C. Value-chain emissions are Scope 3 and typically dominate a project's footprint — hence the contractual data obligations on suppliers.

3. An AI tool supplied the emission factors behind a project sustainability report that an auditor later challenges. Under Institute doctrine, accountability for the figures rests with:

A. The tool vendor, under the software licence B. The named professional who validated and owns the figures C. No one, provided the tool's methodology was published D. The auditor, once the report has been accepted

Correct answer: B. AI proposes; the professional disposes — every AI-assisted figure in governance must be explainable, validated and owned by a competent human, and "the model calculated it" is no defence.

Chapter 18: Project Closure

The nature and purpose of closure

Project closure is the final, governed phase of the project life cycle, in which the project is formally concluded: deliverables are accepted, obligations are discharged, resources are released, records are completed and archived, and the learning generated by the work is captured for the organisation. Two principles define the concept precisely. First, closure is a process, not an event — a structured sequence of verifications and authorisations, typically executed against a closure checklist. Second, projects end by decision, not by attrition. A project that simply fades away, with staff drifting to other work while contracts, cost accounts and access rights remain open, exposes the organisation to unmanaged commercial, financial and security risk. Formal closure completes the accountability loop opened at initiation: the sponsor who authorised the investment against a business case now confirms how it has been concluded.

Planned closure and premature termination

Closure takes two forms. **Planned closure** occurs when the project has delivered its agreed scope and its outputs have been accepted. **Premature termination** occurs when governance decides, before completion, that continued investment is no longer justified — because the business case has become invalid, funding or strategic priorities have changed, a key party has failed, or a technical obstacle has proved unrecoverable. This Body of Knowledge treats termination as a legitimate governance outcome, not as managerial failure: continuing an unjustifiable project in order to avoid acknowledging sunk costs destroys more value than stopping it.

A controlled termination has three defining obligations: protect the value already created (part-finished deliverables, data, intellectual property, reusable designs); honour contractual, legal and ethical commitments to suppliers, staff and other stakeholders; and document the rationale for the decision so that it is auditable and instructive. Both forms of closure use substantially the same close-out machinery — termination compresses it and adds negotiation of early-exit terms.

Administrative, contractual and financial close-out

Close-out comprises three interlocking strands. **Administrative close-out** completes and archives the project record, releases people, facilities and equipment in a planned manner, revokes system access, and formally disbands the temporary organisation. **Contractual close-out** verifies that every obligation under each contract has been discharged or formally settled: final claims and variations resolved, warranties and defect-liability arrangements activated, retentions released on the agreed terms, and closure confirmed in writing with each party. **Financial close-out** reconciles final cost against the approved baseline, clears outstanding commitments and accruals, closes

the cost accounts so that no further expenditure can be booked to the project, and produces the final financial report. The strands are sequenced, not parallel: financial figures cannot be final until contractual liabilities are settled, and the administrative archive is complete only when it contains the evidence of both. Executing a closure checklist — this chapter's learning objective — rests on understanding these dependencies.

Handover, acceptance and transition to operations

Acceptance is the formal confirmation, by the authorised acceptor, that a deliverable meets its pre-agreed acceptance criteria. It is distinct from mere delivery and is the pivotal control of closure, since it triggers contractual settlement, warranty commencement and transfer of ownership. **Handover** is the transfer of the accepted deliverable to its operational owner, together with everything required to operate and sustain it: documentation, training, support arrangements and known-issue registers. **Transition** covers the managed period that follows — hypercare or early-life support — and, critically, the transfer of the benefits realisation plan to a named business owner, because most benefits accrue after the project has closed. Where deliverables include AI-enabled components, handover must additionally transfer model documentation, monitoring and retraining responsibilities, and the named human ownership that must survive the project team's departure.

Executing the closure checklist

Closure is executed, not declared. The delivery leader works through a controlled sequence and evidences each step.

1. **Confirm completion of scope.** Reconcile the deliverable register against the baselined scope statement; every deliverable is either accepted, formally descoped through change control, or listed as an agreed exclusion.
2. **Obtain formal acceptance.** Test each deliverable against its pre-agreed acceptance criteria and secure written sign-off from the authorised acceptor — never rely on implied acceptance through use.
3. **Close the contracts.** Verify that all supplier obligations are discharged; agree final accounts; resolve or formally register outstanding claims; confirm warranty terms, defect-liability arrangements and retention release dates.
4. **Close the finances.** Process final invoices and accruals, close cost codes and control accounts to prevent late charges, reconcile the final cost outturn against the approved budget, and return unused contingency.
5. **Release resources deliberately.** Plan a staged demobilisation, complete performance feedback for team members, and transfer key-person knowledge before departure, not after.

6. **Archive and report.** Index the project records for retention, issue the closure report with final performance figures, and formally stand down the governance bodies.

Where AI drafts the closure report or reconciles records, its output is a proposal only: the delivery leader validates every figure and owns the submission.

Managing a controlled premature termination

A premature termination is run as a project in its own right, with the same discipline as planned closure plus additional steps.

1. **Revalidate the business case** with current forecasts of cost to complete and expected benefits.
2. **Frame the decision on forward-looking figures only.** Costs already incurred are sunk in every option and must be excluded from the comparison.
3. **Present options to the sponsor or board** — complete, descope, pause or terminate — and record the rationale, the figures relied upon and the decision-maker in the decision log.
4. **Protect value.** Secure completed deliverables, intellectual property, data and licences that other parts of the organisation can reuse.
5. **Honour obligations.** Apply contractual termination clauses correctly, give proper notice, settle non-cancellable commitments and negotiate cancellation charges.
6. **Then execute the full closure checklist above,** including lessons learned — terminated projects are the richest source of them.

Worked example: the termination decision

Setup. A systems project has a Budget at Completion (BAC) of USD 2,400,000. At the review gate, Earned Value (EV) is USD 1,080,000 and Actual Cost (AC) is USD 1,200,000. Benefits have been revalidated at USD 1,300,000 (down from USD 3,000,000 after a market change). Outstanding purchase orders total USD 300,000: USD 120,000 non-cancellable, USD 180,000 cancellable at a 10% charge. Demobilisation and close-out would cost USD 60,000; reusable design assets and licences are valued at USD 90,000.

Formulae. $CPI = EV \div AC$; $EAC = BAC \div CPI$; $ETC = EAC - AC$. Each option is compared on future cash flows only.

Substitution and arithmetic.

- $CPI = 1,080,000 \div 1,200,000 = 0.90$
- $EAC = 2,400,000 \div 0.90 = 2,666,667$
- $ETC = 2,666,667 - 1,200,000 = 1,466,667$

- **Option A — complete:** future benefit 1,300,000 – future cost 1,466,667 = **(166,667)**
- **Option B — terminate:** future cost = 120,000 + (10% × 180,000) + 60,000 = 120,000 + 18,000 + 60,000 = 198,000; less recovered value 90,000 = net **(108,000)**

Result. Termination is the better forward-looking outcome by 166,667 – 108,000 = **USD 58,667**.

Interpretation. Both options lose money, but the USD 1,200,000 already spent is sunk and irrelevant to the choice. The organisation should terminate, protect the USD 90,000 of reusable value, settle the USD 198,000 of obligations honourably, and record this rationale so the decision is defensible to audit and governance.

Lessons learned and knowledge retention

A **lesson identified** is an observation about what helped or hindered performance; a lesson is **learned** only when it is embedded in changed organisational practice — an updated method, template, estimating dataset or behaviour. The lessons-learned review is therefore a structured, no-blame, evidence-based examination of the project record, and its output is a set of owned improvement actions, not a filed report. In practice:

1. **Prepare with evidence.** Assemble variance analyses, the risk register, the decision log and stakeholder feedback before the session, so discussion rests on data rather than recollection.
2. **Facilitate without blame.** Work through what was planned, what actually happened, why it differed, and what should be repeated or changed; attribute causes to process, not persons.
3. **Convert findings into actions.** Each lesson becomes a specific improvement action with a named owner, a deadline and a target artefact — an estimating norm, a checklist item, a template or a training change. A lesson without an owner is an anecdote.
4. **Retain the knowledge.** File lessons in a searchable organisational repository, tagged by theme, and route systemic items to the PMO or process owners for verification that the change was made.

Knowledge retention is the organisational discipline that converts these actions into durable assets. AI tools can mine project records, change logs and review narratives to cluster recurring patterns across an entire portfolio, converting anecdote into organisational evidence — but every proposed lesson must be explainable, validated against the evidence and tested with the people who lived the project, and adopted or rejected by a competent professional. A fluent summary of the wrong lesson is worse than no lesson at all.

AI and predictive tooling in closure

Closure has historically been the most neglected and least instrumented phase of the life cycle; AI-enabled delivery environments change that. Predictive completion analytics can forecast the true end date of the punch list, flag deliverables at risk of failing acceptance, and estimate the cost-to-close with a discipline that manual trackers rarely achieve. Natural-language tooling can read the contract suite and extract outstanding obligations — warranties, retentions, licences, insurances, data-return clauses — into a close-out register, and can reconcile final accounts against commitment records to surface unmatched purchase orders, unreturned assets and dormant access rights that a human review would take weeks to find. For premature termination, portfolio analytics can provide early, dispassionate evidence that a benefits case has collapsed — often before political momentum allows anyone to say so aloud.

The certified professional therefore automates the mechanical substrate of closure: assembly and indexing of the documentation set, reconciliation of financial and commitment data, tracing of open contractual obligations, first drafts of close-out reports and lessons summaries, and archiving with searchable metadata. What is never automated is the judgement layer. Acceptance is a human decision made against agreed criteria, not a model score. The recommendation to terminate may originate in an algorithm; the decision, its rationale and its documentation belong to the sponsor and the delivery leader, who must be able to explain it without reference to an opaque output. Final accounts, claims positions, release of retentions and the closure certificate itself carry contractual and financial consequence and require validated human sign-off.

Closure-stage models also carry distinctive weaknesses that the examination expects candidates to recognise. Training data suffers from survivorship bias: organisations archive completed projects far more faithfully than terminated ones, so models systematically understate the signals that precede failure and can lend false comfort to a project that should be stopped. Cost-to-close forecasts degrade precisely when they matter most, because late-stage data is sparse, disputed and politically shaped. Obligation-extraction tools trained on standard forms can miss bespoke or negotiated clauses — and a missed clause at close-out is a liability, not an inconvenience. Generative summarisation can introduce plausible but unfounded statements into formal records; nothing machine-drafted enters a contractual close-out document, termination rationale or lessons register unverified. Finally, closure concentrates sensitive material — settlements, personal data, dispute correspondence — so feeding it into analytical tooling without data-governance controls creates a confidentiality exposure just as oversight winds down.

Accountability does not close when the project does. A named, competent individual must own every accepted deliverable, every financial settlement, every retained record and — above all — any decision to terminate. Where AI analysis contributed to that

decision, its inputs, assumptions and limitations are documented in the closure record so that an auditor or a successor can reconstruct the reasoning years later. The delivery leader who signs the closure certificate is attesting to the whole of it, including every machine-produced element within it.

AI proposes; the professional disposes. In closure, AI may forecast the cost-to-close, extract the obligations, draft the report and cluster the lessons — but acceptance is granted, termination is decided, accounts are settled and lessons are declared learned only by an accountable human who can explain and defend each judgement. The algorithm surfaces the evidence; the professional closes the project.

Governing frameworks

ISO 21502 addresses closing or terminating a project or phase as an integrated management practice; the PMBOK Guide treats closure within its project performance and life-cycle guidance; PRINCE2 defines a dedicated Closing a Project process with explicit acceptance and handover products; and the APM Body of Knowledge situates closure within governance and benefits management. Across all of them the architecture is the same, and acceptance is its hinge: acceptance triggers contractual and financial settlement; settlement enables administrative completion; handover carries the deliverable and the benefits plan into operations; and the lessons-learned review converts the whole experience into organisational capability. Closure done well is how a temporary endeavour leaves a permanent asset.

Common pitfalls

- Treating handover to operations as acceptance — formal sign-off against agreed criteria is still required.
- Including sunk costs in a termination decision, biasing the board towards continuing ("we've spent too much to stop").
- Leaving cost codes open after closure, allowing unrelated charges to distort the final outturn.
- Releasing key people before knowledge transfer, so the transition to operations fails quietly later.
- Terminating abruptly without honouring notice periods and settlement clauses, converting a controlled exit into a dispute.
- Recording lessons "identified" but never assigning owners, so nothing is actually learned — and skipping the lessons review on terminated projects, precisely where the organisation has most to learn.

- Accepting an AI-drafted closure report or lessons summary unvalidated; every figure and finding must be explainable and owned by the accountable professional.

Key points

- Closure is a governed process, not an event: projects end by decision, and the sponsor confirms how the investment concluded.
- Planned closure and premature termination use the same close-out machinery; a controlled termination additionally protects value created, honours obligations and documents its rationale.
- Close-out is sequenced: contractual settlement precedes final financial figures, and the administrative archive must evidence both.
- Formal acceptance against pre-agreed criteria is the pivotal control — it triggers settlement, warranty commencement and transfer of ownership, and is never implied through use.
- Termination decisions compare forward-looking cash flows only; sunk costs are excluded from every option.
- A lesson is learned only when embedded in changed practice, with a named owner, a deadline and a target artefact.
- AI may forecast, extract, reconcile and draft throughout closure, but acceptance, termination, settlement and lessons remain explainable, validated human decisions — AI proposes; the professional disposes.

Practice questions

1. A project has a Budget at Completion of USD 2,400,000. At the closure review gate, Earned Value is USD 1,080,000 and Actual Cost is USD 1,200,000. Using performance-based forecasting, what is the Estimate at Completion?

A. USD 2,400,000 B. USD 2,666,667 C. USD 2,160,000 D. USD 1,466,667

Answer: B. $CPI = 1,080,000 \div 1,200,000 = 0.90$, so $EAC = BAC \div CPI = 2,400,000 \div 0.90 = \text{USD } 2,666,667$; option D is the ETC, not the EAC.

2. When a board evaluates whether to terminate a project early, which figure must be excluded from the comparison of options?

A. The estimated cost to complete the remaining work B. Cancellation charges on cancellable purchase orders C. Costs already incurred on the project to date D. The recoverable value of reusable assets and licences

Answer: C. Costs already incurred are sunk in every option and cannot be changed by the decision; only forward-looking cash flows are compared.

3. A closure-stage AI model trained predominantly on the organisation's archived completed projects is used to advise on whether a struggling project should be stopped. What is the most significant risk?

A. The model will overstate the cost of demobilisation B. Survivorship bias understates the signals that precede failure, lending false comfort to a project that should stop C. The model will automatically breach contract confidentiality clauses D. The forecasts will be too pessimistic because terminated projects dominate the data

Answer: B. Completed projects are archived far more faithfully than terminated ones, so the training data under-represents failure precursors and biases the model towards recommending continuation.

19. AI-Enabled Project Management

What AI-enabled project management is — and is not

AI-enabled project management is the disciplined use of artificial intelligence systems to support the work of initiating, planning, executing, monitoring and closing projects. The term covers a spectrum of capability, and precision matters: each class carries a different failure mode and therefore a different governance obligation. **Descriptive analytics** summarises what has happened in delivery data. **Predictive analytics and machine learning** infer patterns from historical data to produce forecasts, estimates or classifications — a completion-date probability, a cost-at-completion range, a risk likelihood score. **Generative AI**, most commonly delivered through large language models, produces drafts: schedules, risk registers, status narratives, stakeholder communications, lessons-learned syntheses. What none of these systems does is decide. However capable, an AI system produces a proposal — an output whose quality depends on the data it learned from, the prompt or parameters it received, and the fit between its training and the project's actual context.

Two boundaries define the discipline. First, AI-enabled project management is assistive, not autonomous: accountability for every plan, baseline, forecast, report and decision remains with a named, competent professional, exactly as it does when the supporting tool is a spreadsheet or a scheduling engine. Second, it is evidence-bound: an AI output enters project use only after it has been tested against independent evidence — never because it is fluent, fast or confident.

Core concepts: outputs, explainability, validation and ownership

Four concepts recur throughout this chapter and are defined here once. An **AI output** is any artefact or judgement produced wholly or partly by an AI system and considered for use in delivery — a draft plan, an estimate, a forecast, a risk score, a summarised report, a recommended action. **Explainability** is the ability of the accountable professional to state, in terms a sponsor, governance board or auditor can follow, what the output is, what data and assumptions it rests on, and why it is credible; an output the owner cannot explain is not usable, whatever its apparent quality. **Validation** is the act of checking an AI output against independent evidence before adoption — reference-class data, parametric benchmarks, expert judgement, first-principles calculation or back-testing against known outcomes — and it must precede, never follow, incorporation into a baseline. **Ownership** is the assignment of each adopted output to a competent human who answers for it in governance forums and who carries the authority to reject it: the owner signs for the output as if they had produced it themselves, because in governance terms they have.

The governing doctrine: AI proposes; the professional disposes

The PCI doctrine that runs through this Body of Knowledge applies here in its strongest form.

AI proposes; the professional disposes. AI may draft the plan, sharpen the estimate and sense the risk — but only a competent, named professional may commit a baseline, sign a report or accept a risk response. Every AI output used in delivery must be explainable, validated against independent evidence, and owned by the human who answers for it.

The doctrine is not a restraint on the use of AI; it is the condition that makes its use professionally defensible. It converts a probabilistic tool into a governable one by inserting the same control that project management has always applied to expert input from any source: independent scrutiny before reliance. A forecast from a model is treated exactly as a forecast from a consultant would be — interrogated for its basis, tested against other evidence, and adopted only under a named owner. In practice the doctrine imposes three tests on every AI output, corresponding to the concepts defined above: explainable, validated proportionately to its consequence, and owned. An output failing any test is advisory material at most. Accountability for scope, schedule, cost, risk and benefits cannot be delegated to a model, a tool or a vendor; a governance body should be able to trace every AI-influenced decision to the professional who disposed of it.

Frameworks and standards context

No single standard yet codifies AI-enabled project management, so the professional draws on two converging bodies of guidance. From the project management side, established references — ISO 21502's guidance on project management, the PMBOK's treatment of tailoring, estimating and performance measurement, and ISO 31000 on risk management — supply the control framework into which AI outputs must fit: baselines are version-controlled, changes are governed, forecasts are evidenced, risks are owned. From the AI governance side, emerging instruments — ISO/IEC 42001 on AI management systems, ISO/IEC 23894 on AI risk management, the NIST AI Risk Management Framework, and risk-tiered regulation such as the EU AI Act — establish organisational expectations for transparency, human oversight, data governance and accountability. The delivery leader is rarely the author of the organisation's AI policy, but is always accountable for complying with it within the project. This chapter describes these frameworks generically, as context for professional judgement; the PCI credential is examination-based, and PCI claims no accreditation against, or endorsement by, any of the bodies named. Equally, this Body of Knowledge discusses tool categories — scheduling assistants, estimating models, risk-sensing analytics, generative drafting tools — never particular vendors.

AI across the life cycle: use cases and triage

Artificial intelligence now touches every stage of the project life cycle. At initiation, natural-language tools summarise business cases, draft charters and surface comparable historical projects. In planning, machine-learning models propose work breakdown structures, dependency networks and reference-class estimates. During execution, predictive analytics flag schedule slippage, cost variance and emerging risk clusters before they appear in conventional reports; at closure, AI accelerates lessons-learned synthesis and benefits tracking. The delivery leader's first competence is therefore triage: identifying which activities genuinely benefit from AI support and appraising the value, limitations and risks of each use case. Pattern-rich, data-heavy, repetitive tasks — status aggregation, document drafting, anomaly detection — are strong candidates. Judgement-heavy tasks — stakeholder negotiation, trade-off decisions, benefits ownership — are not. The examination expects candidates to make this distinction explicitly, use case by use case, rather than treating "AI adoption" as a single decision.

Appraising an AI use case: the five-step method

Before any AI capability enters the delivery environment, the delivery leader works through five steps. The sequence is examinable and should be applied exactly as given.

1. **Inventory candidate activities** across the life cycle — estimating, schedule development, risk identification, progress narrative drafting, forecasting, benefits tracking — and state, for each, the decision the AI output would inform.
2. **Appraise value against limitation.** For each candidate, record the expected value (time saved, wider option sets, earlier warning) alongside the specific limitations and risks: data quality, bias in the historical record, hallucination, confidentiality exposure and erosion of accountability.
3. **Confirm the data foundation.** An estimating or forecasting model is only as good as the organisation's project history; sparse, unrepresentative or uncleaned data disqualifies the use case until remediated.
4. **Assign ownership and explainability up front.** Name a competent human owner for every output category and set an explainability requirement proportionate to the decision at stake — a forecast feeding a governance gate demands a fuller account of drivers than a draft meeting summary.
5. **Pilot in parallel.** Run the AI method alongside the established method for a defined period; only demonstrated, documented performance earns the tool a place in live delivery.

AI support for planning, estimating and forecasting

AI-generated plans and estimates are proposals, not baselines. Models trained on historical delivery data can counter optimism bias, generate probabilistic ranges rather than single-point estimates, and expose assumptions human planners leave tacit. But they inherit the defects of their training data: unrepresentative project histories, survivorship bias, and silent changes in delivery context that make the past a poor guide. Where a model cannot explain which factors drove its output, the output cannot be defended before a sponsor or steering committee, and it must not become a commitment. The rule is simple: automate the generation of options; never automate the acceptance of a baseline.

Validating an AI output before adoption: the standing routine

When an AI tool proposes a plan, estimate or forecast, the owner validates it before it touches a baseline.

1. **Obtain the output with its drivers** — the inputs used, the assumptions made and the factors weighted most heavily. An output that cannot be explained cannot be validated.
2. **Verify the inputs** against the project's authoritative records (ledger, schedule status, risk register).
3. **Build an independent cross-check** using an established technique — parametric, three-point or earned-value based — from the same verified inputs.
4. **Reconcile.** Where the AI figure and the independent figure diverge, the divergence is diagnostic: interrogate what the model assumed that the professional would not, or vice versa.
5. **Dispose.** Adopt, adjust or reject; record the proposal, the validation evidence, the decision and the named owner in the decision log.

Worked example: validating an AI-proposed estimate at completion

Setup. A workplace-technology programme has a budget at completion (BAC) of USD 2,400,000. At the end of month 8 the authoritative records show earned value EV = USD 960,000, actual cost AC = USD 1,090,000 and planned value PV = USD 1,120,000. An AI forecasting assistant proposes an estimate at completion (EAC) of USD 2,580,000 and a finish six weeks beyond baseline. The programme manager owns the forecast.

Step 1 — verify inputs. EV, AC and PV are reconciled to the cost ledger and schedule status; they agree.

Step 2 — independent cross-check (earned value).

- $CPI = EV \div AC = 960,000 \div 1,090,000 = 0.88$

- $SPI = EV \div PV = 960,000 \div 1,120,000 = 0.86$
- Performance-based EAC = $BAC \div CPI = 2,400,000 \times 1,090,000 \div 960,000 =$
USD 2,725,000 (unrounded CPI used)
- Budget-rate EAC = $AC + (BAC - EV) = 1,090,000 + 1,440,000 =$ **USD 2,530,000**

The defensible bracket is therefore USD 2,530,000–2,725,000.

Step 3 — reconcile. The AI's USD 2,580,000 sits inside the bracket but near its optimistic end: it implies the remaining work will run at close to budgeted efficiency. The tool's explanation shows it weighted the last two months heavily, when a supplier delay was easing. The programme manager tests that assumption against independent evidence — the revised supplier schedule confirms only partial recovery. The six-week slip is separately checked against a critical-path reforecast rather than SPI alone, since SPI loses reliability late in a project.

Step 4 — dispose. The point figure is rejected. The reported forecast is EAC = USD 2,725,000, giving VAC = $BAC - EAC = 2,400,000 - 2,725,000 =$ **(USD 325,000)**, a projected overrun of 13.5% of BAC, with the USD 2,530,000–2,725,000 range and the rationale disclosed to the board.

Interpretation. The AI added genuine value — it assembled the data and surfaced the schedule signal early — but its optimism was visible only because an independent cross-check existed. AI proposed; the professional disposed.

AI support for monitoring, reporting and risk sensing

In execution, AI shifts the leader's posture from lagging to leading indicators. Predictive tooling can scan schedule networks, cost actuals, change logs and even communication metadata to sense risk earlier than periodic reporting cycles allow. The governance caveats are real. False positives erode attention; false negatives breed complacency; and models degrade as the project's context drifts from the data on which they were trained. Model risk must therefore be managed like any other delivery risk: known limitations recorded, performance monitored against outcomes, thresholds for human escalation defined, and data quality — completeness, timeliness, provenance — treated as a precondition rather than an afterthought. AI-drafted status reports become the reporting owner's words the moment they are issued: the professional who signs the report is accountable for every figure and every narrative claim in it, however it was drafted.

Common pitfalls

- **Automation bias.** Adopting an output because it is fluent and precise-looking; confidence of presentation is not evidence of accuracy.

- **Baseline contamination.** Letting an unvalidated AI figure into the performance measurement baseline, after which every variance is measured against a fiction.
- **Validate-once syndrome.** Data drifts and models are updated; revalidation is required at defined intervals and on every material change.
- **Orphaned outputs.** No named owner — "the tool produced it" is never an acceptable answer to a governance board or an auditor.
- **Confidentiality leakage.** Feeding commercially sensitive or personal project data into tools outside the organisation's approved environment.
- **Unfit history.** Applying AI where the underlying record is sparse or biased and expecting the tool to compensate; it will instead reproduce the defect.
- **Cherry-picking.** Quoting whichever of several AI figures best supports the message the reporter wishes to send.
- **Disproportionate explainability — in either direction.** Explainability is scaled to the consequence of the decision, but it is never absent.

Key points

- AI-enabled project management is assistive and evidence-bound: descriptive, predictive and generative capabilities produce proposals; none of them decides.
- The governing doctrine is "AI proposes; the professional disposes": every AI output used in delivery must be explainable, validated against independent evidence, and owned by a named, competent human.
- Appraise use cases one by one with the five-step method: inventory, value-versus-limitation appraisal, data-foundation check, up-front ownership and explainability, and a parallel pilot.
- No AI-generated plan, estimate or forecast enters a baseline until an independent cross-check (parametric, three-point or earned-value based) has been built, reconciled and recorded with a named owner.
- Divergence between the AI figure and the independent figure is diagnostic, not a nuisance: it exposes assumptions that must be tested against evidence before disposal.
- In monitoring and risk sensing, AI enables leading indicators, but model risk is managed like any delivery risk — limitations recorded, performance tracked, escalation thresholds defined, data quality assured.
- Accountability is never delegable: the professional who commits a baseline, signs a report or accepts a risk answers for it, however it was drafted.

Practice questions

1. A project has BAC = USD 1,200,000. At the status date, EV = USD 480,000 and AC = USD 600,000. An AI assistant proposes an EAC of USD 1,250,000. Using the cost-performance-index method, what independent cross-check figure should the owner build before disposing of the proposal?

A. USD 1,200,000 B. USD 1,320,000 C. USD 1,500,000 D. USD 1,680,000

Correct answer: C. $CPI = 480,000 \div 600,000 = 0.80$, so $EAC = BAC \div CPI = 1,200,000 \div 0.80 = \text{USD } 1,500,000$, exposing the AI proposal as materially optimistic; B is the budget-rate formula, not the CPI method asked for.

2. An owner's earned-value cross-check yields an EAC of USD 2,700,000; the AI tool proposes USD 2,550,000. What is the professionally correct next step?

A. Adopt the AI figure, because the model has processed more data than the owner can
B. Adopt the cross-check figure automatically, because human calculations always override machine output
C. Report the average of the two figures to the board as a balanced compromise
D. Treat the divergence as diagnostic and interrogate the differing assumptions against independent evidence before disposing

Correct answer: D. Divergence is information, not a tie to be split or a contest to be pre-judged; the owner tests the competing assumptions against evidence and then adopts, adjusts or rejects on the record.

3. A generative AI tool drafts the monthly status report and the project manager issues it unchanged. A material figure in the report later proves wrong. Who is accountable to the governance board?

A. The tool vendor, since the model generated the erroneous figure
B. The organisation's AI governance function, since it approved the tool
C. The project manager who issued the report
D. No one, because the error originated in an automated system

Correct answer: C. An AI-drafted report becomes the signing owner's words on issue; "the tool produced it" is never an acceptable answer, so accountability rests with the professional who disposed.

20. Decision Intelligence

Every stage of the delivery life cycle — the business case, the gate review, the baseline change, the supplier award, the decision to stop — converges on a moment of choice. Decision intelligence is the discipline of improving the quality of project decisions by combining structured decision-making methods, an evidence-based understanding of human judgement and its biases, and the governed use of data, analytics and artificial intelligence — while keeping a named, competent human accountable for every decision taken.

What decision intelligence means

A **decision**, for the purposes of this Body of Knowledge, is a deliberate commitment of resources or direction that is costly to reverse: approving a business case, passing a gate, selecting a supplier, rebaselining a schedule, or stopping a workstream. The Institute distinguishes **decision quality** from **outcome quality**. Under uncertainty, a well-made decision can produce a poor outcome, and a careless decision can be rescued by luck. The examination therefore tests the candidate's command of decision process — framing, options, criteria, evidence, accountability and record — because process is what a professional can control at the moment of choice.

Decision intelligence sits inside the wider governance frameworks of this Body of Knowledge. ISO 31000 positions risk management as an input to decision-making, requiring decisions to reflect the best available information and its limitations. Organisational governance assigns **decision rights** — delegated authority levels, gate authorities and a single accountable decision-maker for each decision, however many parties are consulted.

Decision-making models

Three families of model frame the chapter. **Normative (prescriptive) models** describe how an ideally rational actor should choose: expected-value reasoning, decision trees and multi-criteria decision analysis (MCDA), in which options are scored against weighted criteria. **Descriptive models** describe how people actually choose. Herbert Simon's **bounded rationality** holds that decision-makers satisfice — accept the first adequate option — because time, information and cognitive capacity are limited; the heuristics-and-biases programme of Tversky and Kahneman shows that intuitive judgement relies on mental shortcuts that are efficient but systematically error-prone. **Naturalistic models**, notably Gary Klein's recognition-primed decision model, explain how experienced practitioners decide rapidly by pattern-matching a situation to prior cases — powerful under time pressure, unreliable in genuinely novel conditions.

No family is "correct". The competent project delivery leader selects the approach the situation warrants — high-stakes, hard-to-reverse decisions justify formal structure;

routine, reversible ones justify delegated, faster paths — deliberately matching method to stakes, reversibility and available time.

Cognitive bias and debiasing

A **cognitive bias** is a systematic, predictable deviation of judgement from the standard a normative model would set; it differs from **noise**, which is random variability between judgements. The biases most consequential in project delivery are: optimism bias and the planning fallacy (underestimating cost and duration); anchoring (over-weighting an initial figure); confirmation bias (seeking evidence that supports the preferred option); availability (over-weighting vivid or recent events); sunk-cost reasoning and escalation of commitment (persisting because of past investment); groupthink; and strategic misrepresentation (deliberate optimism to win approval).

Debiasing techniques are countermeasures applied to estimates, forecasts and gate decisions: reference-class forecasting, which takes the "outside view" from the actual outcomes of comparable past projects and is recognised in public-sector appraisal guidance such as HM Treasury's Green Book treatment of optimism bias; the premortem, which imagines the decision has failed and works backwards to causes; independent estimate challenge and structured devil's-advocacy; calibration of estimators against realised results; and eliciting judgements independently before group discussion.

Structuring a decision: the seven-step method

The Institute expects candidates to be able to run a structured decision process from a standing start. The steps are:

1. **Frame the question.** State the decision in one sentence, including what is not being decided, the decision deadline, and the accountable decision-maker. A mis-framed question invalidates everything downstream: a poor frame ("should we crash the schedule?") presupposes an option, whereas a sound frame asks "how do we recover the committed benefits date within the approved tolerance?"
2. **Generate options.** Produce at least three genuinely distinct options plus the do-nothing (or deferral) baseline. A single option presented for endorsement is not a decision but a ratification. AI assistants may be used to broaden the option set; every AI-generated option must be reviewed for feasibility by the professional.
3. **Set criteria and weights.** Agree criteria — benefit, cost, risk, deliverability, alignment — with the decision-maker before scoring, and weight them to sum to 1.00. Weighting before evaluation prevents weights being reverse-engineered to fit a favourite; the weights are a governance artefact expressing the sponsor's priorities, not the analyst's.
4. **Assemble evidence.** Attach a stated source and confidence to every score — a forecast, a benchmark, a consultation record. Unevidenced scores are opinions.

5. **Score and compute.** Score each option against each criterion (a 1–5 scale is standard), multiply by the weights, and sum. The result is a transparent, challengeable comparison rather than an unexplained preference.
6. **Test sensitivity and debias.** Vary the weights and the least-evidenced scores; if the ranking flips within plausible ranges, the decision hinges on that input and more evidence is needed. Apply at least one formal debiasing technique before committing.
7. **Decide, record, and set a review trigger.** The accountable individual decides — the matrix informs, it does not decide — and the record is completed the same day.

Worked example: recovering a delivery date

A customer-onboarding programme is forecast to miss its contractual go-live by ten weeks. The delivery leader frames the question — "How do we protect the January benefits stream within a £250,000 change budget?" — and tables three options: **A**, crash the critical path with additional suppliers (£180,000); **B**, descope two non-contractual features to a later release (£40,000 rework provision); **C**, negotiate a three-month extension (£0 direct cost, deferred benefits of roughly £310,000).

The sponsor approves four weighted criteria: time to benefit (0.30), whole-life cost (0.25), delivery risk (0.25), stakeholder acceptance (0.20). Scores (1 = poor, 5 = strong) are evidenced by the cost model, a schedule risk analysis, and minuted customer soundings. An AI-supported schedule model proposed the P80 completion dates used in the time scores; the planner validated its logic links and calendars before use, and the model's role is noted in the record.

Criterion (weight)	A: Crash	B: Descope	C: Extend
Time to benefit (0.30)	5	4	1
Whole-life cost (0.25)	2	4	3
Delivery risk (0.25)	3	4	5
Stakeholder acceptance (0.20)	4	2	3

The arithmetic:

- **A** = $(5 \times 0.30) + (2 \times 0.25) + (3 \times 0.25) + (4 \times 0.20) = 1.50 + 0.50 + 0.75 + 0.80 = \mathbf{3.55}$
- **B** = $(4 \times 0.30) + (4 \times 0.25) + (4 \times 0.25) + (2 \times 0.20) = 1.20 + 1.00 + 1.00 + 0.40 = \mathbf{3.60}$
- **C** = $(1 \times 0.30) + (3 \times 0.25) + (5 \times 0.25) + (3 \times 0.20) = 0.30 + 0.75 + 1.25 + 0.60 = \mathbf{2.90}$

B leads A by only 0.05, so the leader runs a sensitivity test. Raising stakeholder acceptance to 0.30 and lowering time to benefit to 0.20 gives $A = 1.00 + 0.50 + 0.75 + 1.20 = 3.45$ against $B = 0.80 + 1.00 + 1.00 + 0.60 = 3.40$ — the ranking flips. The decision therefore hinges on how much the sponsor truly values acceptance, and on the evidence behind B's acceptance score of 2. The leader commissions a two-day customer consultation, which confirms the descoped features are low-usage; the score is revised to 3, B rises to 3.80, and the sponsor selects Option B. The margin, the flip condition and the consultation evidence all go into the decision record.

Two debiasing techniques were applied before scoring was finalised. First, **reference-class forecasting**: the team's ten-week recovery estimate for Option A was compared with the last six comparable interventions, which overran their recovery plans by a median of 30 per cent; the time score therefore used thirteen weeks, not ten. Second, a **premortem**: the team assumed Option B had failed twelve months on and listed causes, surfacing a contractual notification requirement that became a mitigation action. For gate decisions, add an independent challenger who did not build the analysis, and record their dissent verbatim.

AI in the decision cycle: automate the mechanics, validate the judgement

Decision intelligence is the competency in which AI's influence on project leadership is most direct, because predictive tooling does not merely inform decisions — it increasingly pre-shapes them. Machine-learning forecasts, anomaly detection, scenario simulation and generative assistants can assemble the evidence base for a gate review in a fraction of the traditional time: reference-class comparisons from historical portfolio data, probabilistic cost and schedule ranges, early-warning signals, and drafted options papers complete with weighted-criteria scoring. Used well, these capabilities counter several of the biases examined above — optimism bias and anchoring in particular — because a model's outside view is indifferent to a sponsor's enthusiasm. Used carelessly, they introduce a bias of their own: **automation bias**, the tendency to accept a machine-generated recommendation because it arrives quantified, formatted and confident.

The delivery leader may legitimately automate the mechanical layers of the decision cycle: gathering and normalising evidence, generating candidate options, running sensitivity analysis across weighted criteria, stress-testing estimates against reference classes, and drafting the decision record itself. What must be validated, never delegated, is the judgement layer: whether the decision has been framed correctly, whether the criteria and their weights reflect the organisation's genuine priorities, whether the option set is complete, and whether the model's evidence is fit for this decision. A leader presented with an AI-scored options matrix should be able to answer three questions before relying on it: what data the model drew upon, why it ranks the options as it does, and under what conditions its reasoning would break down. If no adequate explanation can be obtained, the output may inform exploration, but it must

not determine the decision. Management-system thinking for AI, exemplified by ISO/IEC 42001, reinforces the same expectation of documented human oversight.

AI proposes; the professional disposes. AI may frame options, score criteria, forecast outcomes and draft the decision record — but it cannot own a decision. Every model-supported recommendation must be explainable, validated against independent evidence, and disposed of — accepted, adapted or rejected — by a named, competent decision-maker who remains accountable for the outcome.

Model risk and data caveats

Model-supported decision-making carries risks the professional is expected to recognise. Models trained on historical portfolios inherit the biases of those portfolios: where past estimates were systematically optimistic, an uncorrected model will faithfully reproduce that optimism at scale. Data drawn from dissimilar projects, small samples or poorly governed sources yields confident-looking scores with wide, unstated error bars. Generative assistants can produce fluent options papers containing fabricated or unverifiable supporting claims, and models degrade silently as delivery context drifts away from the conditions on which they were trained. The leader therefore treats every model-supported input as an estimate with provenance — source data, known limitations, validation performed, and the model version and date — rather than as a fact. For significant decisions — a gate, a baseline change, a termination — model output is triangulated against independent evidence and expert judgement, never treated as sole arbiter.

Accountability and the decision record

Governance requires that accountability for a decision is never transferred to a tool. The accountable decision-maker named in the decision record remains a competent human, however extensive the analytical or AI support employed.

The **decision record** binds the discipline together. It captures: the framed question; the options considered, including those rejected and why; the criteria, weights and scores with their evidence sources; the analytical and AI support used, the validation performed, and any point where human judgement departed from a model's recommendation and why; the sensitivity result; the accountable decision-maker, date and rationale; and a review trigger (in the worked example: "revisit if consultation feedback deteriorates or rework provision exceeds £40,000"). Records are reviewed at the next gate, and annually for pattern analysis across the portfolio.

This transparency serves three purposes. It makes decisions auditable and reviewable; it protects the organisation and the individual when decisions are later challenged; and it builds the institutional evidence base from which future models — and future decision-makers — learn, exposing recurring bias so that debiasing can be targeted. An

unexplained model output is not evidence; it is an unverified claim, and the examination expects candidates to treat it as such.

Common pitfalls

- **Retro-fitted weights** — adjusting weights after scoring until the preferred option wins; agree and freeze weights first.
- **Scoring without evidence** — matrices full of unsourced 3s and 4s that launder intuition as analysis.
- **Option scarcity** — evaluating one real option against two straw men.
- **Ignoring near-ties** — treating a 0.05 margin as decisive instead of triggering sensitivity analysis.
- **Anchoring on the first estimate** — failing to apply a reference class to single-point team estimates.
- **Unvalidated AI outputs** — pasting model-generated scores or forecasts into the matrix without checking assumptions, data currency and explainability.
- **Missing ownership** — records that name no accountable decision-maker, making later review impossible.

Key points

- Decision quality is judged by process at the moment of choice — framing, options, criteria, evidence, accountability, record — not by outcome, which uncertainty can turn either way.
- Match the method to the decision: normative structure (MCDA, decision trees) for high-stakes, hard-to-reverse choices; recognition-primed and delegated paths for routine, reversible ones.
- Bias is systematic and predictable; counter it with reference-class forecasting, premortems, independent challenge, calibration and independent elicitation before group discussion.
- Criteria and weights are agreed with the decision-maker and frozen before scoring; every score carries an evidence source; a near-tie triggers sensitivity analysis, not a coin toss.
- AI may automate the mechanics — evidence gathering, option generation, sensitivity runs, record drafting — but framing, criteria, option completeness and evidence fitness are validated by the human.
- Every model-supported input is an estimate with provenance (data, limitations, validation, version), triangulated against independent evidence for significant decisions; automation bias is the failure mode to watch.

- The decision record names one accountable human, states what was AI-assisted and how it was validated, and sets a review trigger — making decisions auditable and organisational learning possible.

Practice questions

1. A delivery leader runs a rigorous decision process — framed question, three evidenced options, pre-agreed weights, sensitivity test, completed record — yet the chosen option fails when a key supplier collapses without warning. How should the decision be judged?

A. As a poor decision, because outcome is the only objective test
 B. As a good decision, because decision quality is assessed by the process followed at the moment of choice
 C. As unjudgeable until the programme closes
 D. As a poor decision unless the record is amended retrospectively

Answer: B. Under uncertainty the Institute assesses decision quality by process, not outcome; a well-made decision can still produce a poor result.

2. A weighted-criteria evaluation scores Option X at 3.60 and Option Y at 3.55. What should the leader do next?

A. Select X, because the highest weighted score wins
 B. Re-weight the criteria until a clearer winner emerges
 C. Test the sensitivity of the ranking to plausible changes in the weights and the least-evidenced scores
 D. Discard the matrix and decide on judgement alone

Answer: C. A near-tie means the ranking may hinge on a single input; sensitivity analysis identifies which, and where further evidence is needed — re-weighting after scoring is the retro-fitted-weights pitfall.

3. An AI assistant produces a scored options matrix recommending termination of a workstream, but the team cannot obtain an explanation of how the ranking was derived. Institute doctrine requires the leader to:

A. Accept the recommendation, since the model has processed more data than any individual could
 B. Use the output to inform exploration only, and not allow it to determine the decision
 C. Adopt the recommendation provided the model exceeds a validated accuracy threshold
 D. Transfer accountability for the decision to the owner of the model

Answer: B. An unexplained model output is an unverified claim, not evidence: AI proposes, but only a named, accountable professional disposes.

21. Digital Delivery

This chapter examines how the delivery professional specifies, governs and leads through connected digital environments: the project management information system (PMIS) and its specification, the single source of truth and data quality, the leadership of distributed teams, the governing disciplines of security and information management, and the admission of AI into the toolchain. Candidates are examined on judgement in scenario, not on knowledge of any product.

Digital delivery and the project management information system

Digital delivery is the conduct of project initiation, planning, execution and control through connected digital environments, such that the project's plans, records, communications and performance data exist primarily as structured, shareable information rather than as documents held by individuals. Its central instrument is the **PMIS**: the combination of tools, data stores, integrations, procedures and people that captures, validates, integrates, analyses and distributes delivery information across the lifecycle.

The Institute defines the PMIS **functionally, not as a product**. A candidate is examined on the ability to specify what a PMIS must do — the data it must hold, the integrations it must support, the controls it must enforce and the reporting it must enable — independently of any vendor's software. Guidance such as ISO 21502 treats the PMIS in the same way: as an organisational capability supporting governance and decision-making, configured to the project's scale, contracting environment and information-security obligations. A **toolchain** is the set of interoperating applications — scheduling, cost management, risk registers, collaboration, document control — that together realise the PMIS; the leader's concern is the coherence of the chain, not the merits of any single link.

Specifying the PMIS

A PMIS specification is written from decisions backwards, never from a product feature list forwards. The examinable method has six steps:

1. **List the decisions the system must support** — baseline change approval, monthly forecasting, risk response drawdown, benefits tracking — and the questions each decision asks of the data.
2. **Define the data objects and their owners**. For each object (activity, control account, risk, change, benefit), name one accountable owner and the system in which it is mastered. Every object is mastered exactly once; all other systems consume it.
3. **Specify integrations, not re-keying**. State which interfaces move data between the schedule, cost, risk and document environments, at what frequency, and with what reconciliation control.

4. **Set data standards as requirements** — coding structures (WBS, CBS, OBS), naming conventions, update cadence and validation rules — because standards, not software, create the single source of truth.
5. **State the non-functional requirements:** least-privilege access control, audit trail, retention, availability, and exportability of the organisation's own data at contract end.
6. **Write acceptance criteria as tests** — "a change approved on day 1 appears in the period-end report of the same cycle" — so conformance can be verified whatever product is eventually procured.

The single source of truth and data quality

A **single source of truth** exists when every controlled data item — a milestone date, a budget figure, a risk exposure, an approved change — has one authoritative location from which all reporting derives, so that two reports can never legitimately disagree. This is a property of governance and data architecture, not of buying one system: it is achieved by designating authoritative sources, integrating systems so data flows without re-keying, and version-controlling baselines. The concept parallels the **common data environment** formalised in the ISO 19650 series for built-asset information management; the Institute applies its logic generically to all delivery data.

Its foundation is **data quality**, assessed against recognised dimensions drawn from frameworks such as ISO 8000 and ISO/IEC 25012: accuracy, completeness, timeliness, consistency, validity and uniqueness. These dimensions matter because delivery data is consumed downstream — by status reporting, by cost and schedule forecasting, and increasingly by AI-supported analysis. Deficient data degrades each in turn: inaccurate progress data corrupts earned value and forecasts; incomplete risk data understates exposure; and models trained or prompted on poor data produce confidently wrong outputs.

Auditing data quality before reporting and AI-supported analysis

Before each reporting cycle — and before any AI model is run against delivery data — the leader has the PMO score the data cut on four of the dimensions above: **completeness** (required fields populated), **validity** (values pass business rules), **timeliness** (records updated within the cut-off) and **consistency** (values reconcile across systems). Each dimension is measured as passing records divided by total records, combined into a weighted **data quality index (DQI)** and compared with a published threshold. Below-threshold data is remediated or the affected outputs caveated; an AI-generated forecast built on failed data is not released.

Worked example. A programme PMO audits the month-end cut of 1,200 live activities before an AI-supported estimate-at-completion run. Weights: completeness 0.30, validity 0.30, timeliness 0.25, consistency 0.15; release threshold 95.0%.

- Completeness: $1,146 \div 1,200 = 95.5\%$
- Validity: $1,092 \div 1,200 = 91.0\%$
- Timeliness: $1,020 \div 1,200 = 85.0\%$
- Consistency: $1,164 \div 1,200 = 97.0\%$

$DQI = (0.30 \times 95.5) + (0.30 \times 91.0) + (0.25 \times 85.0) + (0.15 \times 97.0) = 28.65 + 27.30 + 21.25 + 14.55 = 91.75$, i.e. **91.8%** to one decimal place — below the 95.0% threshold.

The materiality of the gap is then demonstrated. The late records conceal 1,740 unposted labour hours at a blended USD 110 per hour: $1,740 \times 110 = \text{USD } 191,400$ of understated actual cost. With earned value $EV = \text{USD } 6,240,000$ and recorded actual cost $AC = \text{USD } 6,050,000$, the reported cost performance index $CPI = 6,240,000 \div 6,050,000 = 1.031$ (1.03 to two decimal places). Correcting AC to $6,050,000 + 191,400 = \text{USD } 6,241,400$ gives $CPI = 6,240,000 \div 6,241,400 = 0.9998$ (1.00). On a budget at completion of $\text{USD } 15,600,000$, the CPI-based estimate at completion ($EAC = BAC \div CPI$, calculated on unrounded values) moves from $\text{USD } 15,125,000$ to $\text{USD } 15,603,500$ — $\text{USD } 478,500$ of optimism created entirely by a data deficiency. The leader defers the forecast run until the timesheets post, and minutes the DQI, the correction and the decision.

Leading distributed and virtual teams

A **distributed team** is one whose members work from different locations; a **virtual team** collaborates primarily through digital channels; most contemporary delivery teams are **hybrid**, mixing co-located and remote working across time zones and organisational boundaries. Distributed working changes the medium of leadership, not its substance: direction, motivation, conflict resolution and performance management remain the delivery leader's accountabilities, but they must be exercised deliberately through explicit **collaboration norms**. In practice the operating rhythm is:

1. **Agree a team charter:** working hours by time zone, expected response times, and which channel carries which class of traffic.
2. **Default to asynchronous; escalate to synchronous.** Reserve live meetings for decisions and problem-solving, and record every decision in the designated system of record — never only in a chat thread.
3. **Protect a core overlap window** and rotate recurring meeting times so the same location does not always absorb the unsociable hours.
4. **Apply information security by norm:** least-privilege access, approved platforms only, and no project data in personal channels or in public AI tools the organisation has not sanctioned.

5. **Check inclusion deliberately** — invite contribution from remote and quieter participants — because digital settings amplify, rather than remove, participation imbalances.

Two obligations underpin this rhythm. First, **inclusive digital working**: norms must not systematically disadvantage members by time zone, language, accessibility need or home circumstance, because exclusion degrades both fairness and the quality of project information. Second, **information discipline**: in a distributed team the written record in the PMIS is frequently the only shared reality, so insisting that decisions, assumptions and changes are recorded at source is a leadership behaviour, not an administrative preference.

Security, information management and digital capability

Digital delivery is bounded by three governing disciplines. **Information security** protects the confidentiality, integrity and availability of project information; the reference framework is ISO/IEC 27001, which situates security controls within a managed system of risk assessment, policy and continual improvement rather than as isolated technical measures. **Information management** governs the lifecycle of project records — classification, retention, access control and disposal — including compliance with applicable data-protection legislation when personal data is processed. **Digital capability** is the demonstrated ability of the team and its stakeholders to use the toolchain competently and securely; a technically sound PMIS operated by an unprepared team still fails. These disciplines are the delivery leader's responsibility in their project context, exercised with specialist support, and they apply equally to AI-enabled tools: any AI capability admitted to the toolchain must satisfy the same security, records and explainability requirements as every other component.

How AI changes digital delivery

AI is now a native layer of the digital delivery environment rather than an add-on to it. Within a well-architected PMIS, machine learning models can flag schedule slippage earlier than periodic reporting, detect anomalies in cost and progress data, classify and route documents, summarise meeting records and generate first-draft status narratives. Predictive tooling turns the single source of truth from a repository into an early-warning system — but only when the underlying data merits that trust. Every such capability is a projection from historical and current project data; where that data is incomplete, stale or inconsistently coded, AI does not fail loudly. It produces fluent, confident and wrong outputs. The leader's competency therefore shifts: less effort assembling information, more effort interrogating it.

A defensible division of labour treats automation and validation as complementary, not interchangeable.

- **Appropriate to automate**: data capture and consolidation from source systems; consistency and completeness checks; routine collation of dashboards and

progress packs; document classification, version control and search; transcription and first-draft summaries of distributed-team meetings; alerting on threshold breaches.

- **Requiring human validation:** any forecast of cost, schedule or benefits before it informs a decision or a report to governance bodies; anomaly flags before escalation (a flagged anomaly may be a data defect, not a delivery problem); AI-generated narratives before issue to sponsors or clients; any recommendation affecting people — workload analysis, performance inferences or team-sentiment signals — which raises both ethical and legal obligations.

The rule is simple: automate the assembly of evidence; never automate the judgement, and never allow an AI-generated figure to reach a decision-maker without a named, competent owner who can explain it.

Model risk, data caveats and human accountability

Model risk in digital delivery arises predictably. Models trained on past projects inherit the biases and estimating errors of those projects; a portfolio that habitually underestimated will train a model that normalises underestimation. Predictions degrade when the current project differs materially from the training population, and gradual drift in data definitions — a recoded activity structure, a changed progress-measurement rule — silently invalidates trained relationships. Leaders should require of any predictive tool, from any vendor: transparency about input data and its provenance; stated confidence ranges rather than single-point outputs; evidence of back-testing against known outcomes; and a documented boundary of applicability. Data caveats belong on the outputs themselves — a forecast built on unassured data should say so wherever it appears. Distributed working adds a security dimension: uncontrolled use of external AI services to summarise or draft project material can leak commercially sensitive or personal data outside the information-management boundary, so the acceptable-use position for AI tools is part of the information security regime, not an afterthought.

Accountability is not transferable to a model. The leader who presents an AI-supported forecast owns it exactly as if it had been produced manually, and must be able to explain its basis, assumptions and limitations to a sponsor, an auditor or a client. Governance arrangements should record which AI tools are in use, for what purpose, on what data, and who owns each class of output. An output that no competent person can explain has no place in a delivery decision, whatever its apparent precision.

AI proposes; the professional disposes. In the digital delivery environment, AI may assemble, reconcile, flag and forecast from the single source of truth — but every prediction, anomaly and generated narrative is a proposal. The delivery leader validates it against the quality of the data beneath it, explains it in their

own words, and remains personally accountable for every decision and report that rests upon it.

Common pitfalls

- Specifying the PMIS by copying a vendor's feature list instead of the organisation's decisions, data objects and standards.
- Tolerating multiple "sources of truth" — the same figure mastered in the schedule tool, the cost tool and a spreadsheet, reconciled by hand each month.
- Treating data quality as an IT concern, so deficiencies surface only when reports, forecasts and AI outputs are already wrong.
- Running AI analysis over an unaudited data cut and presenting the output with unwarranted precision.
- Measuring distributed team members by online presence rather than outputs, corroding trust and encouraging performative availability.
- Allowing decisions to live in chat threads: unretrievable, unauditable and invisible across time zones.
- Applying security controls to systems but not behaviours, leaving authorised platforms surrounded by unofficial workarounds.

Key points

- The PMIS is defined functionally — by the decisions, data, integrations, controls and reporting it must support — and specified from decisions backwards, never from a vendor's feature list.
- A single source of truth is a property of governance and data architecture: every controlled data item is mastered exactly once, and all reporting derives from that source.
- Data quality is assessed on recognised dimensions (accuracy, completeness, timeliness, consistency, validity, uniqueness) and audited before each reporting cycle via a weighted DQI against a published threshold.
- Below-threshold data is remediated or its outputs caveated; an AI forecast built on failed data is deferred, and the audit, correction and decision are minuted.
- Distributed leadership works through explicit collaboration norms: a team charter, asynchronous-by-default working, protected overlap windows, deliberate inclusion and decisions recorded in the system of record.
- Information security (ISO/IEC 27001), information management and digital capability bound the whole environment and apply equally to any AI tool admitted to the toolchain.

- AI proposes; the professional disposes: automate the assembly of evidence, never the judgement, and keep a named, competent owner accountable for every AI-supported output.

Practice questions

1. A programme is procuring a new PMIS. Which starting point best reflects the Institute's examinable method for writing the specification?

- A. A comparison matrix of the leading scheduling and cost tools on the market
- B. The list of delivery decisions the system must support and the questions each asks of the data
- C. The feature set of the incumbent tool, extended with stakeholder wish-list items
- D. The IT department's standard infrastructure and licensing catalogue

Answer: B. The specification is written from decisions backwards — decisions, then data objects and owners, integrations, standards, non-functional requirements and acceptance tests — never from product features forwards.

2. A month-end data cut scores a DQI of 92.1% against a published release threshold of 95.0%, largely due to late timesheet postings. An AI-supported estimate at completion is scheduled to run tonight and the sponsor expects the forecast tomorrow. What should the delivery leader do?

- A. Run the forecast and present it, since the model will smooth out data gaps
- B. Run the forecast but round the outputs to reflect reduced confidence
- C. Defer the run until the data is remediated, minute the DQI and the decision, and tell the sponsor why
- D. Lower the threshold to 92.0% for this cycle so the run can proceed

Answer: C. Below-threshold data is remediated or the affected outputs withheld or caveated; releasing a forecast on failed data — or bending the threshold to fit — presents unwarranted precision the leader cannot validate.

3. An AI tool in the PMIS generates a schedule forecast that the delivery leader includes in a report to the project board. Who is accountable for the forecast, and on what condition may it be presented?

- A. The tool vendor, provided the model has been back-tested
- B. The PMO analyst who ran the model, provided the DQI passed
- C. No one, because a statistical output cannot carry personal accountability
- D. The delivery leader, provided they can explain its basis, assumptions and limitations

Answer: D. Accountability is not transferable to a model: the leader owns an AI-supported forecast exactly as if produced manually, and an output no competent person can explain has no place in a delivery decision.

22. Automation in Project Delivery

This chapter sets out the PML-AI competency of automation: what it is, when it is justified, how it is designed, and how it is governed once AI enters the workflow. Its concepts form one causal chain: the selection triad determines what may be automated; the checkpoint requirement, how it must be designed; testing discipline, whether it can be trusted; and monitoring with fallback, how long trust remains justified — with a named professional accountable for every link.

What Automation Means in Project Delivery

Automation in project delivery is the execution of a defined project management process, or a step within one, by a machine rather than a person, according to rules established and owned by a competent professional. The Institute distinguishes automation from mere digitisation. Digitisation moves a manual activity onto an electronic medium — a spreadsheet replacing a paper log — while the judgement and the labour remain human. Automation transfers the labour itself: the system detects a trigger, applies a rule and produces an outcome without a person performing the intermediate steps. A status report compiled and distributed on schedule, or a risk register entry escalated when its exposure score crosses a threshold, is an automation because the process runs to completion, or to a designated pause point, without human effort.

The Institute recognises a continuum. At one end sits deterministic automation: rule-based workflows fully specified in advance, with predictable, repeatable output. At the other sits AI-enabled automation, in which a model classifies, summarises, drafts or predicts, and the output is probabilistic rather than guaranteed. The distinction matters for governance: deterministic automations fail in ways testing can largely anticipate; AI-enabled automations can fail plausibly, producing outputs that look correct while being wrong. The PML-AI doctrine — AI proposes; the professional disposes — therefore binds most tightly at the AI-enabled end, though the accountability principle beneath it applies across the whole range: every automated output must be explainable, validated and owned by a named, competent human.

The Selection Triad: Volume, Repeatability and Rule Clarity

Automation is justified by the character of the process, not the availability of the technology. The Institute's selection triad holds that a project process is a candidate for automation when it exhibits, together, sufficient **volume** (it occurs often enough to repay the cost of building and maintaining the automation), **repeatability** (successive occurrences follow substantially the same path, so one design covers all cases) and **rule clarity** (the decision logic can be stated completely and unambiguously, so the machine's behaviour can be specified rather than hoped for).

The triad is conjunctive. A high-volume process with unclear rules automates error at scale; a rule-clear process performed twice a year rarely repays its build cost; and a repeatable process embedded in a consequential judgement — approving a change, communicating with a client — may satisfy all three criteria mechanically yet still demand a human decision at its core.

Checkpoints, Fallback and the Architecture of Control

A **human checkpoint** is a designed pause at which an automated workflow halts, presents its proposed action and reasoning, and proceeds only on the authorisation of an accountable person. The Institute's standard is categorical: a checkpoint is required at every consequential decision — one that commits resources, alters baselines, accepts risk or affects people — and at every external communication, because a message issued to a stakeholder cannot be recalled. Checkpoints are the workflow-level expression of the disposal doctrine; they convert "the professional disposes" from aspiration into architecture.

A **fallback arrangement** is the pre-defined route by which a process returns safely to human control when the automation fails, degrades or encounters a case outside its rules. Fallback presupposes **monitoring**: instrumented evidence that the automation is running and producing outputs within expected tolerances, with alerts when it is not. An automation without monitoring is invisible in failure; an automation without fallback leaves the underlying process ownerless when it stops. Both must exist before an automation is entrusted with live project work.

Frameworks and Standards Context

These concepts sit within an established governance landscape. International project management guidance, including ISO 21502, treats tailoring and process discipline as the project manager's responsibility — automation is a tailoring decision and inherits that accountability. AI management system standards, notably ISO/IEC 42001, and emerging AI regulation converge on requirements the Institute's doctrine already embodies: human oversight, transparency of logic, documented accountability. Internal control traditions add the segregation principle that an automation's designer should not be its sole reviewer. The examination tests these sources as context only; candidates are assessed against the Institute's own doctrine, and PCI claims no accreditation by, or endorsement from, the bodies publishing them.

A Three-Stage Method for Justified Automation

Candidates are examined on moving from intuition ("this feels automatable") to a defensible, quantified decision. The Institute's method proceeds in three stages: screen and score, design with checkpoints, then implement with monitoring and fallback.

Stage 1 — Screen and score candidates.

1. Inventory the recurring project management processes in scope (status consolidation, minutes distribution, register updates, timesheet chasing).
2. Score each against the triad on a 1–5 scale: **volume**, **repeatability** and **rule clarity**, the last tested by whether a competent professional could write the rules down unambiguously.
3. Apply agreed weights — a common convention is 0.4 for volume, 0.3 for repeatability, 0.3 for rule clarity — and set an admission threshold (for example, 3.5) below which candidates are rejected or deferred.
4. For candidates above the threshold, quantify baseline effort, expected automated effort, build cost and recurring maintenance, and compute payback before committing resources.

Stage 2 — Design with human checkpoints. Map the workflow end to end and classify every step as mechanical (data collection, formatting, calculation) or consequential (anything that commits the organisation: approvals, escalations, external communications, baseline changes). Mechanical steps may run unattended; every consequential step receives a named human checkpoint with authority to amend or reject the proposed output. No automated output leaves the project boundary without human disposition.

Stage 3 — Implement, monitor, fall back. Build in a controlled environment, test against historical cases with known correct outcomes, and run in parallel with the manual process for at least one full cycle. Define in advance the monitoring signals (exception rates, data-freshness checks, reviewer rejection rates) and the triggers on which the process reverts entirely to human execution.

Worked Example: Automating Weekly Portfolio Status Consolidation

A PMO supports 15 projects. Each week an analyst collects status data, reformats it and assembles a portfolio pack.

Step 1 — Score the candidate. Volume scores 5 (15 instances weekly, every week), repeatability 4 (identical structure, occasional format drift), rule clarity 4 (consolidation rules are documented; RAG thresholds are defined numerically).

Weighted score = $(5 \times 0.4) + (4 \times 0.3) + (4 \times 0.3) = 2.0 + 1.2 + 1.2 = \mathbf{4.4}$, above the 3.5 threshold. The candidate is admitted to business-case analysis.

Step 2 — Quantify the baseline. Collection and reformatting take 30 minutes per project: $15 \times 30 = 450$ minutes = 7.5 hours. Assembling the portfolio pack adds 1.5 hours. Total manual effort = **9.0 hours per week**. Over 46 working weeks, annual effort = $46 \times 9.0 = \mathbf{414 \text{ hours}}$. At a fully loaded rate of £52 per hour, the baseline cost = $414 \times £52 = \mathbf{£21,528 \text{ per annum}}$.

Step 3 — Quantify the automated state. The automation extracts data from the scheduling and cost tools and drafts each project summary and the portfolio pack. Human checkpoints remain: the analyst reviews each draft summary (5 minutes $\times$ 15 = 75 minutes) and performs a final portfolio review before issue (45 minutes). Automated-state effort = 120 minutes = **2.0 hours per week**; annually $46 \times 2.0 = 92$ hours = $92 \times \text{£}52 = \text{£}4,784$.

Step 4 — Compute the business case. Gross annual saving = $414 - 92 = 322$ hours = $322 \times \text{£}52 = \text{£}16,744$. Build and testing require 120 hours at $\text{£}60$ per hour = **£7,200**. Annual maintenance is estimated at 15% of build cost = **£1,080**. Net annual saving = $\text{£}16,744 - \text{£}1,080 = \text{£}15,664$. Payback period = $\text{£}7,200 \div \text{£}15,664 = 0.46$ **years, roughly 5.5 months** — a justified automation, provided the checkpoints in Step 3 are preserved. Note that the review hours retained are not inefficiency; they are the cost of professional ownership of every issued report.

How AI Changes the Automation Competency

AI-enabled automation extends the deterministic model in two consequential ways. First, predictive and generative components let workflows handle inputs previously too unstructured to automate — classifying incoming change requests, drafting status narratives from schedule and cost data, triaging risk register entries, routing correspondence. Second, agentic tooling can chain steps together, deciding at run time which action to take next. Both widen the pool of candidates but change the risk profile fundamentally: the triad's tests remain necessary yet are no longer sufficient, because an AI component may produce a plausible but wrong output on an input it has never seen. The PML-AI professional therefore evaluates AI-enabled candidates against an additional criterion — the cost of an undetected error — and reserves full automation for processes where that cost is demonstrably low.

What the Leader Automates and What the Leader Validates

The defensible division of labour: AI may automate the assembly, transformation and first-draft interpretation of project information — data collection, log consolidation, draft minutes, candidate schedule updates, anomaly flags. The professional validates everything that carries consequence: baseline changes, forecasts communicated to sponsors, contractual or financial actions, escalations, and any message leaving the project boundary. The checkpoint standard applies with greater force where AI is in the loop, because AI failure modes are silent: a broken deterministic script usually stops, while a degraded model keeps producing confident output. Checkpoints for AI-enabled steps must therefore involve genuine review by a competent person able to explain why the output is correct; the examination treats perfunctory approval clicks as a governance failure, not a control.

Model Risk and Data Caveats

AI-enabled automations inherit every weakness of the data and models beneath them. Predictive components trained on historical project data embed the biases and reporting habits of those projects; they degrade when the portfolio, methodology or economic context shifts (model drift), and they perform worst precisely on the novel situations where projects are most exposed. Generative components can fabricate detail that appears authoritative. For each AI-enabled automation the professional documents the data sources and their limitations, the validation conditions, the boundaries outside which output must not be trusted, and the cadence at which performance is re-tested against actuals. Monitoring must detect quiet degradation as well as hard failure — rising exception rates, drifting error distributions, outputs reviewers increasingly override — and fallback must return the process to fully manual operation without loss of records or audit trail.

Human Accountability

No delegation of execution is a delegation of accountability. The leader who deploys an AI-enabled automation owns its outputs as personally as if they had produced them by hand. Every automated decision must be explainable to a sponsor, an auditor or a regulator; a named, competent individual must own each automation with authority to suspend it; and governance records must show who approved it, on what evidence, and when it was last validated. Where an output cannot be explained, it must not be relied upon. These expectations are examined behaviourally: the assessment tests whether the candidate places accountability, monitoring and fallback at the centre of automation practice, not at its periphery.

AI proposes; the professional disposes. An AI-enabled automation may gather, draft, classify and recommend — but every consequential action it initiates is released, and owned, by a competent human who can explain it. When the automation's behaviour can no longer be explained, validated and monitored, the professional's duty is to stop it and take the process back into human hands.

Common Pitfalls

- **Automating a broken process.** Automation amplifies whatever it is given; standardise and stabilise first.
- **Counting gross rather than net savings.** Omitting maintenance, licence and checkpoint-review effort overstates the case at governance boards.
- **Removing checkpoints to "realise the full benefit".** Eliminating human review of consequential outputs breaches the disposition doctrine and converts efficiency into unmanaged risk.

- **Rule clarity assumed rather than tested.** If two experienced practitioners would decide an instance differently, the rules are not clear enough to encode.
- **Silent failure.** Monitoring must detect absence of output and degraded inputs, not merely errors — stopped or stale automations fail invisibly.
- **No rehearsed fallback.** A fallback that exists only on paper fails under pressure; the manual procedure must be documented, current and exercised.
- **Orphaned ownership.** Every automation requires a named owner accountable for its outputs, maintenance and retirement — an unowned automation is an unowned risk.

Key Points

- Automation executes a process by machine under rules owned by a competent professional; digitisation merely changes the medium while the labour stays human.
- The selection triad — volume, repeatability, rule clarity — is conjunctive: a candidate must satisfy all three, and passing the screen admits it only to business-case analysis, not to build.
- Every consequential decision and every external communication requires a named human checkpoint; only mechanical steps may run unattended.
- A justified business case counts net savings — maintenance, licences and retained checkpoint-review effort included; the review hours are the cost of professional ownership.
- No automation goes live without monitoring (to make failure visible) and a rehearsed fallback (to return the process safely to human hands).
- AI widens the candidate pool but fails silently and plausibly, so candidates face a further test — the cost of an undetected error — and checkpoints must involve genuine review.
- Accountability is never delegated: a named owner holds authority to suspend each automation and must be able to explain every decision it releases.

Practice Questions

1. A PMO identifies a daily, highly repeatable process, but two experienced practitioners handling the same instance would reach different decisions. Under the selection triad, the process is:

A. A strong candidate, because two of the three criteria are satisfied
 B. A candidate, provided an AI model is used to infer the missing rules
 C. Not a candidate until its rules are clarified and standardised, because the triad is conjunctive
 D. A candidate, provided the payback period is under one year

Correct answer: C. The triad is conjunctive — automating unclear rules automates error at scale, and disagreement between competent practitioners is precisely the test rule clarity fails.

2. Using weights of 0.4 (volume), 0.3 (repeatability) and 0.3 (rule clarity) with a 3.5 threshold, a process scores volume 3, repeatability 5 and rule clarity 4. The weighted score and correct next step are:

A. 3.9; admit to business-case analysis B. 4.0; admit to business-case analysis C. 3.9; proceed directly to build D. 3.0; reject, because the lowest criterion governs

Correct answer: A. The weighted sum is 3.9, clearing the threshold and admitting the candidate to business-case analysis, not directly to build; B is the unweighted mean and D misapplies the conjunctive principle to the arithmetic.

3. An AI model drafting weekly status narratives has degraded because the portfolio's context has shifted since validation. The most characteristic symptom, and its design implication, is:

A. The workflow halts with an error, so alerting on stopped jobs is sufficient B. The model keeps producing confident, plausible narratives that are wrong, so checkpoints need genuine review and monitoring must detect quiet degradation C. The model fails only in high-volume weeks, so capacity monitoring is the priority D. The model refuses to generate output outside its validated boundary, so no further control is needed

Correct answer: B. A degraded model fails silently and plausibly rather than stopping; the control response is substantive review at checkpoints plus monitoring for drift signals such as rising reviewer override rates.

23. Responsible AI

Responsible AI as a delivery discipline

Responsible AI is the discipline of using artificial intelligence in ways that realise its benefits without transferring unmanaged harm to individuals, to the organisation or to the project itself. For the Project Management Leader – AI it is not an abstract ethical debate but a delivery competence. Every AI use on a project — an estimating model, a schedule-risk predictor, a generative assistant drafting communications, a tool that shortlists suppliers or allocates people to work — creates obligations that must be identified, assessed and controlled with the same rigour the profession applies to cost, schedule and risk. The doctrine that governs this entire Body of Knowledge applies here in its strongest form: **AI proposes; the professional disposes**. An AI output remains a proposal until a competent, named human has understood it, validated it and taken ownership of the decision it informs. Responsible AI is the framework that makes that ownership real rather than rhetorical.

The chapter follows a single control loop: principles supply the criteria; regulation converts several of them into binding obligations; risk assessment determines what proportionate control looks like for a given use; and explainability, validation and assurance are how that control is exercised in practice.

The four principles, precisely defined

Four principles anchor the discipline, and each has a precise meaning.

Fairness is the property that an AI system's outputs do not produce unjustified differential outcomes for individuals or groups. Unfairness enters through biased or unrepresentative training data, through proxy variables that stand in for protected characteristics, and through the context of deployment. On projects, fairness questions arise wherever AI touches people: resourcing and task allocation, performance assessment, recruitment into teams, and supplier or bid evaluation.

Transparency is openness about the system: that AI is being used, for what purpose, on what data, and with what known limitations. It is distinct from **explainability**, which is the ability to give an intelligible account of how a particular output was produced — a distinction this chapter maintains throughout, because a project may be transparent about using a tool whose individual outputs it still cannot explain.

Privacy is the protection of personal data throughout the AI lifecycle: a lawful basis for processing, purpose limitation, data minimisation, and control over personal data entering training sets, prompts and outputs. Privacy exposure on projects is often inadvertent — team data pasted into a generative tool, or personal details embedded in documents used as model context.

Accountability is the ability to attribute every AI-informed decision to a named person or governance body with the authority and competence to answer for it. Accountability can never be delegated to a system, a vendor or an algorithm; it can only be assigned to humans and evidenced through records.

The four principles interlock. Transparency is the precondition of accountability: no one can answer for a system whose use is undisclosed. Explainability is the precondition of validation: an output that cannot be accounted for cannot be meaningfully checked. Privacy constrains what data may lawfully fuel a system, and fairness can only be demonstrated — not merely asserted — where transparency permits testing.

The governing frameworks

The principles are not free-floating; a body of law, regulation and standards gives them force. Data protection law — the UK GDPR and Data Protection Act 2018, the EU GDPR, and their counterparts in other jurisdictions — imposes binding obligations wherever AI processes personal data: lawful basis, data protection impact assessment for high-risk processing, data subject rights, and specific restrictions on decisions based solely on automated processing that significantly affect individuals. The EU AI Act establishes the first comprehensive AI-specific regulatory regime, classifying AI uses by risk tier: prohibited practices, high-risk systems subject to conformity and oversight requirements, limited-risk systems subject to transparency duties, and minimal-risk uses. Delivery leaders should note that employment-related AI, including workforce allocation and evaluation, falls within the high-risk tier.

Alongside binding law sit voluntary frameworks that shape professional expectations: the OECD AI Principles, the NIST AI Risk Management Framework with its govern-map-measure-manage functions, ISO/IEC 42001 for AI management systems, and ISO/IEC 23894, which extends ISO 31000 risk-management logic to AI. This Body of Knowledge draws on these frameworks as context for examination purposes; the credential does not confer, and PCI does not claim, certification or accreditation against any of them. The delivery leader's obligation is to recognise when a proposed AI use engages these regimes and to escalate to legal, data-protection or information-security specialists — not to substitute for them.

Risk assessment, validation and assurance

Three further terms complete the working vocabulary. **AI risk assessment** is the structured evaluation of the harms a specific AI use could cause — bias, hallucination, data leakage, model drift, over-reliance and automation bias — and of their likelihood and impact, so that controls can be **proportionate**: calibrated to consequence, not applied uniformly. **Validation** is the point-in-time activity of establishing that an AI output is accurate and fit for the decision it will inform, through testing against known results, benchmarking, back-testing and structured human review. **Assurance** is the

through-life counterpart: ongoing, suitably independent confirmation that controls continue to operate — monitoring, audit trails and periodic revalidation as data, models and usage change.

The responsible AI assessment, step by step

Responsible AI on a project is not a policy statement; it is a repeatable assessment that the delivery leader runs before any AI output is allowed to inform a decision. The Institute's method has seven steps, applied per use — not per tool, since the same tool can be benign in one use and high-risk in another.

1. **Specify the use and its decision consequence.** State what the AI produces, which project decision it informs, who is affected, and what happens if the output is wrong — a drafting aid and a tool that influences payments or people carry very different risk profiles.
2. **Screen against the four principles.** For each of fairness, transparency, privacy and accountability, score inherent risk as likelihood (1-5) × impact (1-5). Any dimension scoring 10 or more triggers enhanced controls; any processing of personal data triggers a data protection assessment.
3. **Map the regulatory touchpoints.** Identify whether personal data is processed and whether AI regulation classifies the use as higher-risk; recognise the trigger and obtain specialist data-protection or legal advice rather than improvising a legal conclusion.
4. **Assign accountability.** Name the competent individual who owns every decision the AI informs. The professional must be identified before deployment, not after an incident.
5. **Design proportionate controls.** Match control depth to the screening score: disclosure to affected parties, defined human review points, data handling rules, and logging sufficient to reconstruct how any output influenced a decision.
6. **Validate before reliance.** Back-test the AI against historical cases with known outcomes. Compute error rates overall and by relevant subgroup, and test explainability: the owner must be able to explain any individual output in plain language.
7. **Decide, document and monitor.** Approve, reject or approve with conditions; set a re-validation cadence and the drift triggers that force early re-testing.

Worked example: validating a delivery-risk flagging tool

Setup. A programme office proposes an AI tool that reads supplier progress data and flags work packages at risk of late delivery. Flagged packages receive escalated oversight, and milestone certification may be delayed pending review. Screening scores fairness at $4 \times 3 = 12$ and transparency at $3 \times 4 = 12$, so the use enters the

enhanced-control tier and must be validated before reliance. The back-test uses 60 closed work packages from the previous year, of which 21 finished late — a base rate of $21 \div 60 = 35.0\%$.

Overall validation. The tool flags 25 of the 60 packages: 17 were genuinely late (true positives), 8 were on time (false alarms), and 4 late packages were missed.

- Precision = $17 \div 25 = 68.0\%$ — roughly two of every three escalations are warranted.
- Recall = $17 \div 21 = 81.0\%$ — the tool catches four of every five genuinely late packages.

For an attention-directing use, this performance is workable: 68.0% precision far exceeds the 35.0% base rate a random escalation would achieve.

Fairness check by subgroup. The 60 packages split into 40 delivered by tier-1 contractors (12 late) and 20 by SME suppliers (9 late). The tool flagged 12 tier-1 packages ($12 \div 40 = 30.0\%$) but 13 SME packages ($13 \div 20 = 65.0\%$). The decisive test is the false-positive rate — the chance that an on-time package is wrongly escalated:

- Tier-1: $3 \text{ false alarms} \div 28 \text{ on-time packages} = 10.7\%$
- SME: $5 \text{ false alarms} \div 11 \text{ on-time packages} = 45.5\%$

An on-time SME package is therefore $45.5 \div 10.7 \approx 4.2$ **times** more likely to be wrongly flagged than an on-time tier-1 package — a material fairness defect, since a wrongful flag delays certification and harms the supplier commercially.

Disposition. The owner approves with conditions: flags are advisory only; the named owner reviews the evidence behind every flag before escalation; no payment consequence may follow from a flag alone; the use is disclosed to suppliers; and the tool is recalibrated and re-tested, with the SME false-positive rate to fall below an agreed 20% before the human review step may be lightened. Quarterly re-validation is scheduled. The AI proposed; the professional disposed.

Common pitfalls

- **Judging on aggregate accuracy alone.** The headline figures concealed a fourfold subgroup disparity; always cut error rates by the groups the decision affects.
- **Assessing the tool, not the use.** A screening done once "for the tool" misses the risk added when the same output starts driving payments or appraisals.
- **Treating validation as a one-off.** Data and behaviour drift; a regime without re-validation cadence and drift triggers decays silently.
- **Confusing transparency with disclosure of the algorithm.** Stakeholders need a plain-language explanation of each output, not source code.

- **Accepting vendor assurances as assurance.** Supplier claims inform, but never discharge, the project's own validation and data protection obligations.
- **Automation bias.** Letting the AI's confidence substitute for evidence at the review point turns "human oversight" into a rubber stamp.
- **Uniform maximal controls.** Applying high-risk controls to trivial uses breeds workarounds; proportionality is itself a control objective.
- **The accountability gap.** "The model decided" is never an acceptable answer; if no named professional owns the output, the use is not ready for deployment.

AI-enabled governance: automate the mechanics, validate the judgement

Responsible AI is the one competency in this Body of Knowledge that AI tooling transforms reflexively: the tools of governance are increasingly AI-enabled themselves. Contemporary platforms can maintain a live inventory of AI use across a project, run automated bias and data-quality scans, monitor deployed models for drift, and draft the documentation — impact assessments, model descriptions, transparency notices — that responsible-AI frameworks require. This shifts the leader's work from periodic, manual review towards continuous, instrumented oversight, but it does not change the object of the competency: a dashboard that reports a fairness metric is evidence to be interrogated, not a judgement already made.

The division of labour follows the Institute's standing doctrine. Suitable for automation are the mechanical substrates of governance: logging and audit trails of AI-assisted decisions, recurring data-quality and drift checks, first-pass triage of proposed uses against a risk taxonomy, and the assembly of draft assurance documents. Reserved for a competent human are the judgements those mechanics feed: the risk classification assigned to a use, any conclusion that a system is sufficiently fair or private, the adequacy of an impact assessment, the explanation offered to an affected stakeholder, and the decision to rely on an AI output at all. The test is simple: automation may gather and arrange evidence; it may never discharge an obligation.

Using AI to assure AI also creates second-order model risk that the examination expects candidates to recognise. Automated fairness scores depend on which metric was chosen and whether the reference data represents the affected population — a favourable score on unrepresentative data is false comfort. Post-hoc explanation tools produce accounts that are plausible rather than guaranteed faithful to the model's behaviour, and must be treated as hypotheses. Generative tools drafting compliance documents can fabricate regulatory citations, so every legal reference is verified against the source or referred to specialist advice. Project data entered into external tools carries its own confidentiality exposure, and monitoring thresholds tuned at deployment decay as context shifts. Proportionality governs throughout: control depth should match the consequence of the use, not the enthusiasm of the tooling.

The non-delegable core: human accountability

No configuration of tooling transfers accountability. Each AI use has a named, competent owner; escalation routes to data-protection, legal and technical specialists are defined before they are needed; and the delivery leader can explain, in their own words and without reference to the tool, why each AI-informed decision was justified. Where an output cannot be explained, validated and owned, it is not used — however efficient it promises to be. Governance records must show a human decision at every consequential point, because that is what a sponsor, an auditor or a regulator will ask to see.

AI proposes; the professional disposes. Applied to responsible AI, the doctrine becomes self-referential and stricter for it: AI may scan for bias, monitor for drift and draft the impact assessment, but only a competent, named human may declare an AI use fair, lawful, transparent and fit to inform a project decision. The tools of assurance are themselves models — so the final assurance is never a tool's output, but a professional's owned, explainable judgement.

Key points

- Responsible AI is a delivery competence, not an ethical abstraction: every AI use creates obligations managed with the same rigour as cost, schedule and risk, under the doctrine AI proposes; the professional disposes.
- Four principles anchor the discipline — fairness, transparency, privacy and accountability — and transparency (openness about the system) is distinct from explainability (accounting for a particular output).
- Binding law (data protection regimes, the EU AI Act's risk tiers) and voluntary frameworks (OECD, NIST AI RMF, ISO/IEC 42001 and 23894) give the principles force; the leader's duty is to recognise triggers and escalate to specialists.
- The seven-step assessment is applied per use, not per tool, and calibrates controls proportionately to decision consequence.
- Validation must cut error rates by affected subgroup: aggregate precision and recall can conceal material fairness defects, as the worked example's fourfold false-positive disparity shows.
- Assurance is through-life, not point-in-time: re-validation cadence, drift triggers and audit trails keep controls operating as data, models and usage change.
- AI may automate the mechanics of governance — logging, scanning, monitoring, drafting — but every judgement, and all accountability, remains with a named, competent human.

Practice questions

1. A back-test of an AI escalation tool shows 68% precision and 81% recall overall. Which additional check is most likely to reveal a fairness defect before the tool is relied upon?

- A. Comparing precision with the historical base rate
- B. Re-running the back-test on a larger aggregate sample
- C. Comparing false-positive rates across affected subgroups
- D. Obtaining the vendor's fairness certification for the model

Answer: C. Aggregate metrics can conceal differential error rates; the subgroup false-positive comparison exposed the 4.2-fold disparity in the worked example.

2. A project openly discloses that a generative tool drafts its stakeholder communications, but no one can account for how a specific draft was produced. Which statement is correct?

- A. The use is transparent but not explainable
- B. The use is explainable but not transparent
- C. The use is neither transparent nor explainable
- D. Disclosure makes the use both transparent and explainable

Answer: A. Transparency is openness about the system's use; explainability is the ability to account for a particular output — disclosure alone provides only the former.

3. Under the Institute's doctrine for AI-enabled governance tooling, which task may be automated without a competent human validating the result before it takes effect?

- A. Classifying the risk tier of a proposed AI use
- B. Maintaining the audit log of AI-assisted decisions
- C. Concluding that a system is sufficiently fair to deploy
- D. Approving the adequacy of a data protection impact assessment

Answer: B. Automation may gather and arrange evidence, such as logging and audit trails, but it may never discharge an obligation — classification, fairness conclusions and adequacy judgements are reserved for a named human.

24. Human Accountability

This chapter sets out the discipline that underpins every other competency in the PML-AI Body of Knowledge: however capable the tooling, every decision in AI-assisted delivery must be owned, explained and answered for by a named, competent human.

Accountability and responsibility: two distinct obligations

The Institute draws a hard distinction between two terms that everyday usage blurs. **Responsibility** is the obligation to perform work: to produce an estimate, draft a schedule, analyse a risk exposure, prepare a stage-gate submission. Responsibility can be delegated, shared among team members and — critically for this chapter — assisted or partially automated by AI systems. **Accountability** is the obligation to answer for a decision and its consequences: to explain why it was taken, on what evidence, under what authority, and to accept the professional and organisational consequences that follow. Accountability can never be delegated, shared, diffused across a committee, or transferred to a tool.

From this distinction follows the doctrine that anchors the whole Body of Knowledge: **for every decision in AI-assisted delivery, accountability rests with a named, competent human**. An AI system may carry out the mechanical work that responsibility denotes — generating a forecast, drafting a change assessment, flagging a schedule anomaly — but it cannot bear responsibility in the professional sense, and it can hold accountability for nothing. In assignment-matrix terms (RACI and its variants), an AI system may contribute to the work behind the "R", but the "A" is always a single, identifiable person. A decision log entry that names a tool, a vendor or "the model" as the deciding agent records an accountability vacuum, not an assignment.

AI concentrates accountability; it does not dilute it

When forecasts, risk scores, schedule diagnostics and drafted recommendations arrive machine-generated, the temptation is to let ownership blur — to treat "the model indicated" as though it were a decision-maker. The examination treats this as the central hazard of AI-assisted delivery. Predictive tooling changes the competency in two ways: it multiplies the decision inputs whose provenance the leader must be able to trace, and it creates a new failure mode — the **accountability gap** — in which an outcome has no named human owner because everyone deferred to the system. The competent leader closes that gap deliberately: every AI-assisted decision is owned, in advance and by name, by a human competent to challenge the recommendation they are adopting.

The governing doctrine: AI proposes; the professional disposes

The Institute's doctrine — AI proposes; the professional disposes — operationalises the principle. An AI output, however sophisticated, is a **proposal**: an input to professional judgement, of the same governance status as a junior analyst's draft or a consultant's recommendation. The **disposal** — accepting, amending, overriding or rejecting the proposal — is a human act, performed by a professional with the competence and the authority to make it. Three tests attach to every AI contribution before it may inform a project decision:

- **Explainable** — the accountable professional can articulate, in their own words, what the output says, what data and assumptions it rests on, and its material limitations. "The system said so" is never an explanation.
- **Validated** — the output has been checked against independent evidence, professional experience, or documented review appropriate to its consequence: proportionate scrutiny for routine outputs, rigorous challenge for decisions affecting baselines, contracts, safety or benefits.
- **Owned** — a named individual has adopted the output as their own professional position and is recorded as having done so. Ownership converts a machine proposal into a human decision; without it, no disposal has occurred.

The doctrine also settles what may legitimately be automated. The leader may automate the clerical substrate of accountability: capturing decision records as they are made, logging which inputs were machine-generated, assembling gate evidence packs, drafting minutes and actions, and flagging decisions that lack a recorded owner. These automations strengthen governance by making ownership visible at scale. What is never automated is the disposition itself, nor the three tests that precede it. Where a leader cannot obtain an adequate explanation, or lacks the competence to evaluate one, the obligation is escalation, not acceptance: signing what one cannot explain is an abdication, not a delegation.

Frameworks and standards context

This doctrine reflects a broad convergence across governance and AI-assurance frameworks. Project governance standards — ISO 21502 and the governance chapters of the leading professional bodies of knowledge — locate decision authority in defined roles (sponsor, board, project manager) exercising delegated authority within tolerances; nothing in that architecture contemplates a non-human decision-maker. AI-specific instruments reinforce the point: ISO/IEC 42001 requires organisations to define human roles, competence and oversight for AI use; ISO/IEC 23894 and ISO 31000 treat AI as a risk to be owned and managed like any other; the NIST AI Risk Management Framework places a "Govern" function — accountability structures, defined roles, documented oversight — at the centre of trustworthy AI use; and the EU AI Act's human-oversight requirements make effective human supervision of consequential AI systems a legal expectation, not merely good practice. Professional

codes of conduct, including the Institute's own Code of Professional Conduct, complete the picture: a professional remains answerable for work products issued under their name regardless of the tools used to produce them. These frameworks converge on one proposition: the human in the loop is not a courtesy; it is the locus of accountability.

Applying the accountability method step by step

The Institute expects candidates to operate, not merely describe, this discipline. The method below applies to any delivery decision to which an AI system contributes analysis, a forecast or a recommendation.

Step 1 — Register the decision. Before any AI output is consulted, enter the decision in the project decision log: what is to be decided, by when, and against which delegated authority limit.

Step 2 — Name the accountable individual. Assign accountability to one named, competent person — never to a team, a committee or the tool. Responsibility for preparatory work may be distributed; accountability may not. Confirm that the named individual is competent to explain the method the AI has used; if not, either develop that competence or reassign before proceeding.

Step 3 — Classify the AI contribution. Record whether the AI input is advisory (an option set), analytical (a computed forecast) or generative (a drafted deliverable), and note the model or service version, the input data set and its cut-off date. This provenance record is created at the moment of use, not reconstructed later.

Step 4 — Validate proportionately. Set validation effort in proportion to consequence. As a minimum, re-perform the central calculation independently, test the inputs for completeness and currency, and challenge the assumption most capable of changing the answer.

Step 5 — Dispose. The accountable professional accepts, amends or rejects the recommendation, recording the disposition and a rationale in their own words. An unexplained acceptance is treated as no review at all.

Step 6 — Complete the audit trail entry. Each entry records: decision reference; AI contribution and provenance; validation performed; disposition (accepted, amended, rejected); rationale; the named accountable individual; and date. Rejections and amendments are recorded with the same rigour as acceptances.

Step 7 — Close the loop. At governance reviews, sample entries, compare AI recommendations with human dispositions and eventual outcomes, and apply consequences — corrective, developmental or disciplinary — to the named human, exactly as for any other professional decision.

Worked example: positioning an AI schedule-recovery recommendation

A programme with a budget at completion (BAC) of £24,000,000 and a planned duration of 92 weeks is at its week-40 review. Planned value (PV) is £10,500,000 and earned value (EV) is £9,240,000. An AI scheduling assistant proposes a recovery package. The project manager, the named accountable individual, works through the method.

The AI's proposal (Step 3, analytical contribution) rests on this arithmetic:

- $SPI = EV \div PV = £9,240,000 \div £10,500,000 = \mathbf{0.88}$
- Forecast duration = $92 \div 0.88 = \mathbf{104.5 \text{ weeks}}$, a forecast delay of 12.5 weeks
- Liquidated damages exposure at £45,000 per week = $12.5 \times £45,000 = \mathbf{£562,500}$
- Recommended action: a £340,000 acceleration package recovering 10 weeks, for a claimed net benefit of $(10 \times £45,000) - £340,000 = £450,000 - £340,000 = \mathbf{£110,000}$

Validation (Step 4): the project manager re-performs the SPI calculation and confirms it, then challenges the governing assumption — that SPI-based extrapolation reflects the critical path. Decomposing the schedule variance, $SV = EV - PV = -£1,260,000$, shows £890,000 of the shortfall sits in a fittings procurement package holding 15 weeks of float; only £370,000 lies on the critical path. A critical-path reforecast shows true slippage of **4 weeks**, not 12.5.

Disposition (Step 5): **amended**. Crashing two critical activities costs £128,000 and recovers the 4 genuine weeks; the benefit is $4 \times £45,000 = £180,000$, giving a net benefit of $£180,000 - £128,000 = \mathbf{£52,000}$. The AI's package would have spent £340,000 to "recover" float that needed no recovery. The audit trail entry (Step 6) records the model version, data cut-off, the independent SV decomposition, the amended decision, the rationale and the project manager's name. The non-critical package is placed on watch, closing the loop at the next governance review (Step 7).

The examination presents scenarios of this kind and expects candidates to show both the arithmetic and the point at which judgement must depart from the AI's proposal.

Model risk, data caveats and the audit trail

Model risk translates directly into accountability risk: the human owner answers for the model's failures. Models trained on unrepresentative portfolios reproduce historical bias; outputs degrade silently as delivery context drifts; generative assistants produce fluent text containing unverifiable claims; and opaque systems offer confidence without reasons. Because the accountable human cannot outsource these weaknesses, the audit trail must make them examinable. A defensible trail — the Step 6 entry, kept for every material decision — shows what the AI proposed and on what

data, what the human examined against what independent evidence, and who disposed, how and why, particularly where judgement overrode the model. The trail serves governance in both directions: it protects the professional who exercised diligent judgement and exposes the professional who did not, and consequences — commendation, remediation or sanction — attach to the named human, never to the tool.

Common pitfalls

- **Accountability diffusion** — attributing an outcome to "the model" or "the team"; every decision must trace to one named human.
- **Rubber-stamping** — a recorded review that involved no independent re-performance or challenge; automation bias is the most heavily examined failure mode.
- **Committee accountability** — naming a board where an individual is required; boards oversee, individuals account.
- **Provenance gaps** — logging the AI's answer but not its version, inputs and data cut-off, making the decision unexplainable later.
- **Selective trails** — recording acceptances but not amendments and rejections, which destroys the evidential value of the log.
- **Competence mismatch** — assigning accountability to someone who cannot explain the AI's method, as with SPI extrapolation above.
- **Retrospective reconstruction** — building the audit trail after the outcome is known rather than at the moment of decision.
- **One-off validation** — treating a model validated once as validated forever, ignoring drift in data and context.

The standard the Institute examines

The elements of this chapter form a single chain: the accountability/responsibility distinction establishes what a human must hold; the disposal right names where it sits; competence and oversight supply the preconditions; and the audit trail makes the chain demonstrable. Break any link — an unnamed owner, an unexplained output, an unrecorded review — and accountability degrades into the diffuse "the system decided" posture that the examination requires candidates to recognise and reject.

Candidates are therefore expected to demonstrate that accountability structures survive contact with AI: that decision logs name humans rather than systems; that competence requirements are defined for anyone disposing of AI recommendations; that overrides of model output are recorded and reasoned rather than silent; and that post-incident reviews ask first who owned the decision, not what the tool suggested. An organisation whose delivery decisions cannot each be traced to a named, competent, informed human has not adopted AI — it has surrendered to it.

AI proposes; the professional disposes. Applied to accountability, the doctrine is absolute: AI may draft, forecast, score and recommend, but it can be neither accountable nor responsible in the professional sense. Every AI contribution must be explainable, validated and owned; every decision must carry the name of a competent human who disposed of the recommendation and answers for the outcome. Where no such name exists, the decision is ungoverned — and the failure is human before it is technical.

Key points

- Responsibility (performing work) can be delegated, shared and AI-assisted; accountability (answering for a decision) can never be delegated, shared or transferred to a tool.
- Every decision traces to one named, competent human: the "A" in any assignment matrix is a single identifiable person, never a system, vendor or committee.
- Every AI output is a proposal that must pass three tests — explainable, validated and owned — before it informs a decision; the disposal (accept, amend, reject) is always a human act.
- Validation is proportionate to consequence and includes independent re-performance and challenge of the governing assumption; an unexplained acceptance counts as no review at all.
- The audit trail is built at the moment of decision and records provenance, validation, disposition, rationale and the named individual — for rejections and amendments as rigorously as for acceptances.
- Automation bias (rubber-stamping) is the most heavily examined failure mode; where a professional cannot explain an output, the obligation is escalation, not signature.
- Governance and AI-assurance frameworks — ISO 21502, ISO/IEC 42001, ISO/IEC 23894, the NIST AI RMF, the EU AI Act — converge on the human in the loop as the locus of accountability; consequences attach to the named human, never the tool.

Practice questions

Question 1. During a post-incident review, a delivery director explains a failed baseline change by stating that "the forecasting model recommended the revised dates and the steering board endorsed them." According to the Institute's doctrine, what is the fundamental defect in this position?

A. The steering board lacked the technical competence to endorse a model output B. The model's recommendation was not validated against independent evidence C. No single named human held accountability for the decision — it was diffused between a

tool and a committee D. The decision log failed to record the model version and data cut-off

Correct answer: C. Accountability can never rest with a tool or be diffused across a committee; B and D may also be failings, but the fundamental defect is the absence of a named accountable individual.

Question 2. In the worked example, what is the net benefit of the project manager's amended recovery decision, and why does it differ from the AI's claimed £110,000?

A. £52,000 — the critical-path reforecast shows only 4 weeks of true slippage, so £128,000 of crashing recovers £180,000 of liquidated damages B. £110,000 — the amendment changes the actions but not the net benefit C. £180,000 — the full liquidated damages saved by the amended package D. £450,000 — the value of the 10 weeks the AI's package claimed to recover

Correct answer: A. Net benefit = $(4 \times £45,000) - £128,000 = £52,000$; the AI's figure relied on SPI extrapolation that treated £890,000 of floating, non-critical variance as critical-path delay.

Question 3. A project manager records in the decision log: "AI cost forecast reviewed and accepted." No independent calculation, input check or assumption challenge is documented. Which pitfall does this best illustrate?

A. Retrospective reconstruction of the audit trail B. Rubber-stamping driven by automation bias C. Selective recording of acceptances only D. Competence mismatch in the accountability assignment

Correct answer: B. A recorded review with no independent re-performance or challenge is rubber-stamping, the automation-bias failure mode; A, C and D concern when, what or by whom the record is made, none of which is evidenced here.

Integrated Case Study: The Halcyon Operations Centre

Purpose of this capstone

This closing chapter draws the competencies of the PCI PML-AI Body of Knowledge together into a single, realistic delivery. It introduces no new syllabus content; its purpose is integrative — to show how initiation, business-case discipline, governance, scope and requirements, planning and execution, stakeholder leadership, procurement, contracting and commercial delivery interlock on one live project, and how artificial intelligence is applied at every stage under the Institute's governing doctrine: **AI proposes; the professional disposes**. Every AI output is explainable, validated and owned by a competent, accountable human. Candidates should pause at each decision point and form a view before reading on: the examination routinely tests this kind of cross-competency synthesis, and while the figures are illustrative, the reasoning is the examinable substance.

The scenario

Marston Vale Water, a fictional regional utility, operates three ageing control rooms whose technology is approaching end of support. A new resilience standard issued by its regulator must be met within four years. The executive board mandates **Project Halcyon**: consolidation of the three sites into a single operations centre — a new building on utility-owned land, and an integrated control platform configured from commercially available software. The Director of Operations is appointed sponsor; a project delivery leader is mandated to develop the business case. The delivery approach is hybrid: predictive for the civil and fit-out works, iterative for the platform configuration and migration.

Stage one — initiation and the business case

The delivery leader's first act is not planning but justification. Four options are appraised: do minimum (remediate the worst site), refurbish all three, consolidate into a leased facility, or into a new build. Whole-life costs, benefits, risks and regulatory compliance are assessed for each. The new build carries the highest capital cost — **USD 78 million** against USD 52 million for the lease — but the strongest benefits case: consolidation savings, resilience compliance with margin, and an asset with a forty-year life. The business case recommends the new build, states its benefits in measurable terms (a 22.0% reduction in operating cost per site-year; incident-response performance at the regulator's threshold) and names benefit owners who will outlive the project.

Applied AI. An AI assistant drafts the options framework and generates comparative whole-life cost profiles from the utility's asset data and anonymised benchmark

projects. The delivery leader finds one benchmark set unverifiable and excludes it, re-runs the comparison, and documents both the exclusion and its immaterial effect. The investment committee is told which figures are benchmark-derived and which are engineering estimates; the recommendation is the leader's, made explainable to the board without reference to any tool.

Decision point 1. A steering-group member urges skipping straight to design to protect the regulatory deadline. The sponsor declines: an unjustified project cannot be governed, because nothing has been agreed against which to judge change. The business case is approved at the first gate and becomes the standing test — is this investment still worth making? — applied at every later decision.

Stage two — governance, scope and requirements

The sponsor chairs a project board comprising the senior user (head of network operations), senior supplier representation and an independent assurance voice. Tolerances are set — cost $\pm 5.0\%$, schedule ± 2 months, and zero tolerance on the regulatory compliance date — with escalation obligatory the moment a forecast breaches them.

Requirements are elicited from control-room staff, engineering, the regulator's standard and the emergency services with which the centre must interoperate. They are analysed, de-duplicated and prioritised: MoSCoW prioritisation for the platform, a baselined specification for the building. The delivery leader resists the senior user's wish to class every requirement "must have": a prioritisation in which nothing may drop is not a prioritisation but a hope.

Applied AI. Language-model tooling clusters 1,140 requirement statements into candidate themes and flags 63 apparent duplicates and 17 contradictions. The business analyst confirms 41 duplicates, rejects the remainder as genuinely distinct, and resolves the contradictions in facilitated workshops — the machine sorted; the humans decided. Traceability from every baselined requirement to a named stakeholder and an acceptance test is owned by the analyst, not the tool.

Decision point 2. Should the platform's "could have" requirements be contracted now or deferred? The board defers them to a post-migration release, keeping release one small enough to protect the compliance date — scope discipline exercised at the moment of commitment rather than discovered in delivery.

Stage three — procurement, contracts and commercial structure

The procurement strategy splits the work into two lots. The building is let as a design-and-build contract under a **target-cost arrangement with a pain/gain share**, because ground conditions and fit-out interfaces carry genuine two-way uncertainty that a fixed price would merely convert into claims. The platform is let as a **fixed-price implementation with output-based acceptance criteria** and a rate-carded change

mechanism, awarded through competitive dialogue. Evaluation criteria and weightings are published before tenders are opened; the panel scores independently and moderates openly.

Applied AI. During evaluation, AI tooling extracts each bidder's stated assumptions, exclusions and departures into a comparison matrix and flags one bid whose programme logic is internally inconsistent. The panel verifies the flag against the bid documents — it is real — and raises it in clarification. No score is generated by the machine: scoring is the panel's judgement, recorded with reasons, because a procurement decision that cannot be defended to an unsuccessful bidder or an auditor is not a decision but a liability.

Decision point 3. The lowest platform bid sits 18.0% below the next and below the utility's own should-cost estimate. Rather than banking the saving, the panel treats abnormally low pricing as a delivery risk, tests the bidder's resourcing assumptions in dialogue, and uncovers a scope misreading. The bid is corrected upward and loses; the contract is awarded on evaluated value for money, and the record states why.

Stage four — planning, execution and commercial delivery

An integrated schedule links the civil works, fit-out, platform sprints and migration rehearsals, with the critical path running through fit-out into systems integration. Progress is measured through earned value for the works and validated, releasable increments for the platform. In month eleven, ground contamination is found: the contractor gives early warning, the compensation mechanism is applied as written, and the pain/gain share does what it was designed to do — the event costs money, not the relationship. Forecast cost moves to +4.6% of baseline: within tolerance, so reported, not escalated. When a later design change threatens +6.1%, escalation to the board is immediate, and the board funds it from contingency against a re-confirmed business case.

Applied AI. Run monthly, a schedule risk analysis tool proposes a probabilistic completion range and identifies the migration-rehearsal window as the emerging driver. The planner audits the model's logic links and duration distributions before any output reaches the board pack, and the delivery leader presents the forecast as a range with stated confidence — owning it as the project's forecast, not the software's.

Decision point 4. With the compliance date under pressure, the board weighs three recovery options: compress commissioning (a quality risk), migrate one control room at a time (slower benefits, protected compliance), or seek a short regulatory extension (an uncertain precedent). It selects phased migration — a deliberate, recorded trade of benefit timing for delivery certainty.

Stage five — stakeholder leadership under pressure

Consolidation means relocation for 210 staff across three sites, and the recognised union signals formal opposition. The delivery leader treats this as leadership work, not communications work: the union is engaged as a stakeholder with legitimate interests, options for shift patterns and travel support are co-developed, and the senior user — not the project office — fronts the change with operational staff. An AI-drafted engagement pack is reviewed line by line before release; one sentence overstating job security is caught and corrected by its human owner — a small moment that illustrates the entire doctrine. Relocation terms are agreed two months before migration, and the first shift in the new centre is fully crewed.

Stage six — closure and benefits

The centre enters service ahead of the compliance date. The deferred "could have" scope passes to a governed release-two decision, not an assumed one. The project closes formally: contracts are settled, lessons are recorded while memory is fresh, and the benefit owners begin measurement against the baselines the business case fixed. Twelve months on, the operating-cost benefit is tracking at 19.0% against the 22.0% forecast — reported honestly, with a recovery plan, because benefits realisation is a discipline, not a press release.

Integrative discussion questions

1. At Decision point 1 the sponsor refused to begin design before the business case was approved. Which specific governance capabilities would have been lost had the project proceeded, and how would each later decision point have been weakened?
2. The case allocates ground-condition risk through a target-cost contract and platform delivery risk through a fixed price. Justify each allocation from first principles — then describe a change of circumstances that would reverse one of them.
3. Identify every point in the case where an AI output was corrected, excluded or overridden. For each, state the competence the responsible human needed to catch the flaw, and what would have failed had it been absent.
4. Decision point 4 traded benefit timing for delivery certainty. Reconstruct the decision as the board should have documented it: options, evaluation against the business case, tolerance implications and named ownership of the residual risk.
5. The union engagement succeeded because leadership, not process, was applied. Analyse what would probably have happened had the delivery leader relied solely on the AI-drafted communications pack — and who would have been accountable for the outcome.

PCI PML-AI Body of Knowledge — Glossary of Terms

This glossary consolidates the key terms of the PCI Project Management Leader – AI (PCI PML-AI) Body of Knowledge, Chapters 1-24. Each term is defined once here and carries the same meaning wherever it appears in the Body of Knowledge, the examination blueprint and the Institute's study materials. Candidates preparing for the PCI PML-AI examination should treat these definitions as authoritative.

Agile delivery — An iterative, incremental approach in which requirements and solutions evolve through short, time-boxed cycles of planning, delivery and review. Value is released progressively and priorities are reordered as learning emerges.

AI-enabled project management — The application of artificial intelligence techniques — such as forecasting, classification and natural-language analysis — to support project planning, monitoring, reporting and decision-making. The Institute's doctrine applies throughout: AI proposes; the professional disposes.

AI governance — The policies, controls and oversight arrangements through which an organisation directs and constrains its use of artificial intelligence in delivery. Effective AI governance ensures every AI output is explainable, validated and owned by a competent human.

"AI proposes; the professional disposes" — The Institute's governing doctrine for artificial intelligence in project delivery: AI may draft, forecast, flag and recommend, but a competent, accountable human examines, validates and decides. No AI output is acted upon unexplained, unvalidated or unowned.

Algorithmic bias — Systematic distortion in an AI system's outputs, typically arising from unrepresentative training data or flawed design, that produces unfair or misleading results. Detecting and mitigating bias is a core obligation of responsible AI practice.

Assurance — Independent, evidence-based confirmation, provided to governance bodies, that a project is being managed appropriately and remains likely to achieve its objectives. Assurance is distinct from, and must not be captured by, the delivery team it examines.

Automation — The use of technology to execute defined, repeatable delivery tasks — such as data capture, reporting and workflow routing — without manual effort. Automation removes toil from routine work; accountability for the automated outcome remains with the professional.

Baseline — The approved, version-controlled statement of scope, schedule or cost against which performance is measured. A baseline may be altered only through formal change control.

Benefit — A measurable improvement arising from project outcomes that stakeholders regard as advantageous and that contributes to organisational objectives.

Benefits realisation — The structured identification, planning, tracking and embedding of the benefits for which a project was undertaken, continuing beyond project closure under named business ownership.

Business case — The documented justification for a project, presenting the options appraised together with expected benefits, costs, risks and timescales. It is maintained as a living document and retested at each governance gate.

Change control — The formal process by which proposed departures from a baseline are captured, assessed for impact, approved or rejected at the correct level of authority, and recorded.

Configuration management — The discipline of identifying, version-controlling and auditing project deliverables and their documentation so that all parties work from the current approved version.

Contingency — Time or money held in reserve against identified risks, released under defined governance rules rather than absorbed silently into estimates.

Contract — A legally binding agreement between parties defining the goods, services or works to be delivered, the price and payment mechanism, and each party's rights, obligations and remedies.

Cost-benefit analysis — An appraisal technique that compares the whole-life costs of an option with the value of its expected benefits, so that alternatives can be ranked on a consistent economic basis.

Cost-reimbursable contract — A contract form in which the supplier is paid allowable actual costs plus an agreed fee, placing cost risk largely with the buyer. It suits work whose scope cannot yet be defined tightly enough for a fixed price.

Critical path — The longest sequence of dependent activities through the schedule network, which determines the shortest achievable project duration. Any delay to a critical-path activity delays completion unless recovered.

Decision intelligence — The discipline of improving decision-making by combining data, analytics and AI with structured human judgement, making each decision's framing, options, assumptions and consequences explicit and reviewable.

Decision rights — The explicit allocation of authority stating who may take which decisions, at what thresholds and by what process — including which decisions may never be delegated to an automated system.

Digital delivery — The conduct of project work through connected digital tools, shared data standards and collaborative platforms, so that plans, models and records exist as structured, reusable data rather than static documents.

Digital twin — A data-driven digital representation of a physical asset, system or process, kept current with its real-world counterpart and used to simulate, monitor and optimise performance.

Earned value management (EVM) — An integrated method for measuring project performance that compares the value of work performed with the work scheduled and its actual cost, yielding objective schedule and cost indices and forecasts.

Estimate at completion (EAC) — A forecast of the total cost of the project at its finish, derived from actual costs to date plus an estimate of the remaining work informed by performance so far.

Explainability — The degree to which the reasoning behind an AI output can be understood and articulated by the accountable professional. An output that cannot be explained cannot responsibly be relied upon.

Fixed-price contract — A contract form in which the supplier delivers defined scope for an agreed sum, carrying the majority of cost risk. It suits well-specified requirements in stable conditions.

Governance gate — A formal decision point between stages at which the governing body reviews evidence, retests the business case and decides whether the project should proceed, change direction or stop.

Hallucination — Plausible but false or fabricated content produced by a generative AI system. The possibility of hallucination is a principal reason every AI output must be validated by a competent human before use.

Handover — The controlled transfer of deliverables, documentation, knowledge and operational responsibility from the project to its business or operational owner, with acceptance formally recorded.

Human accountability — The principle that a named, competent person remains answerable for every decision, deliverable and outcome, however it was produced. Accountability may be supported by tools but can never be transferred to them.

Human-in-the-loop — An operating pattern in which a competent professional reviews, validates and approves an AI-generated proposal before it takes effect, preserving human ownership of consequential actions.

Hybrid delivery — A deliberately tailored approach combining predictive and agile elements within one project — for example, a predictive governance wrapper around iterative build cycles — selected to fit the certainty and cadence of the work.

Invitation to tender (ITT) — A formal procurement document inviting prospective suppliers to submit priced, compliant offers against a defined specification and published evaluation criteria.

Iteration — A short, fixed-length delivery cycle (in Scrum, a sprint) that produces a reviewable increment of the product and closes with reflection that informs the next cycle.

Large language model (LLM) — A class of generative AI trained on large text corpora to produce fluent language outputs such as drafts, summaries and analyses. Its outputs are probabilistic and must be verified before professional use.

Lessons learned — Knowledge derived from project experience, captured and shared so that successes are repeated and failures are not. Capture is consolidated at closure but should occur throughout delivery.

Machine learning — A family of AI techniques in which models learn patterns from data in order to make predictions or classifications without being explicitly programmed for each case.

Milestone — A significant, zero-duration point in the schedule marking completion of a defined achievement, used for progress measurement and reporting.

Portfolio — The totality of an organisation's projects, programmes and related change investments, managed collectively to balance risk and resources against strategic objectives.

Predictive delivery — A plan-driven approach in which scope, schedule and cost are defined and baselined up front and delivery proceeds through sequential phases. It is best suited to stable, well-understood requirements.

Procurement — The process of identifying the need for externally supplied goods, services or works and acquiring them — from make-or-buy analysis and market engagement through supplier selection to contract award.

Product backlog — A single, ordered list of everything that may be needed in the product, continually refined and reprioritised by the product owner as understanding grows.

Product owner — The role accountable for maximising the value of the product, owning and ordering the backlog and acting as the authoritative voice of the business and its customers.

Professional judgement — The considered application of competence, experience and ethical standards to a decision, particularly where evidence is incomplete or automated recommendations conflict. It is the human faculty no tool can replace.

Programme — A group of related projects and change activities coordinated to deliver outcomes and benefits that could not be achieved by managing them independently.

Project — A temporary endeavour with defined objectives, undertaken to create a unique output, outcome or benefit within agreed constraints of time, cost and quality.

Project charter — The document that formally authorises a project, appointing the project manager, granting authority to apply resources and recording the objectives, constraints and success criteria agreed at initiation.

Project closure — The formal end of a project, at which deliverables are accepted, contracts are closed, resources are released, performance is evaluated and lessons are recorded.

Project governance — The framework of authority, accountability and decision-making — sponsor, board, gates and reporting — through which an organisation directs and controls its projects.

Project leadership — The practice of setting direction, building and motivating teams and creating the conditions in which people deliver; it is distinguished from, and complementary to, the management of plans and controls.

Project life cycle — The sequence of phases a project passes through from initiation to closure. Its shape — predictive, agile or hybrid — is chosen to suit the nature of the work.

Project management — The application of knowledge, skills, methods and judgement to lead defined work from initiation to closure so that objectives are achieved within agreed constraints.

Project sponsor — The senior accountable owner of the business case, who chairs governance, secures resources and remains answerable for the realisation of benefits.

Quality assurance — The planned, systematic activities that give confidence that the processes producing deliverables are appropriate and being followed. Assurance addresses the process rather than the individual output.

Quality control — The inspection, testing and measurement of deliverables against defined acceptance criteria to verify that each output meets its specification.

Quantitative risk analysis — The numerical modelling of combined risk and uncertainty — commonly by simulation across integrated schedule and cost models — to express outcomes as ranges and confidence levels rather than single-point figures.

Requirement — A documented, verifiable statement of a need or condition that the project's outputs must satisfy, agreed with stakeholders and used as the basis of acceptance.

Requirements traceability — The ability to follow each requirement forward to design, build, test and acceptance, and backward to its origin, so that coverage and change impact can be demonstrated.

Responsible AI — The design, selection and use of artificial intelligence in ways that are lawful, ethical, transparent, secure and subject to human control, with risks such as bias, error and data misuse actively governed.

Risk register — The living record of identified threats and opportunities, recording for each its assessment, owner, planned responses and current status.

Robotic process automation (RPA) — Software that executes rule-based, repetitive tasks across existing systems — such as data transfer and report assembly — exactly as configured. It follows rules and exercises no judgement.

Rolling-wave planning — Progressive elaboration in which near-term work is planned in detail while later work is held at a higher level until better information becomes available.

Scope — The totality of outputs, outcomes and work required to deliver a project, expressed in terms of what is included and, just as importantly, what is excluded.

Scope creep — The uncontrolled growth of scope through incremental, unapproved additions that bypass change control, eroding schedule, cost and quality.

Stakeholder — Any individual, group or organisation that can affect, is affected by, or perceives itself to be affected by the project.

Stakeholder engagement — The deliberate, planned work of identifying and analysing stakeholders and of communicating, consulting and building the relationships needed for delivery — an activity to be led, not merely administered.

Sustainability in project delivery — The management of a project's environmental, social and economic impacts across the whole life of its outputs, embedding considerations such as carbon, resource use and community effect into decisions from options appraisal onward.

Tailoring — The deliberate adaptation of methods, governance and controls to fit a specific project's size, risk, complexity and context — the judgement at the heart of hybrid delivery.

Variation — A formally instructed change to the scope, conditions or price of a contract, processed under the contract's change mechanism and priced and agreed before or as the changed work proceeds.

Velocity — An empirical measure of the amount of backlog work a team completes per iteration, used to inform forecasting. It is a planning aid for the team, not a productivity target for comparison across teams.

Whole-life cost — The total cost of an asset or output over its entire life — acquisition, operation, maintenance and disposal — used in appraisal so that decisions are not driven by capital cost alone.

Work breakdown structure (WBS) — The hierarchical decomposition of the total scope into progressively smaller elements down to work packages, providing the framework for estimating, assignment and control.

Work package — The lowest level of the work breakdown structure at which work is estimated, scheduled, assigned to a single owner and controlled.